

ФЕДЕРАЛЬНОЕ АГЕНТСТВО ПО ОБРАЗОВАНИЮ
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ФУНКЦИОНАЛЬНЫЙ АНАЛИЗ

ЗАДАНИЯ
ДЛЯ СТУДЕНТОВ 3 КУРСА
МАТЕМАТИЧЕСКОГО ФАКУЛЬТЕТА

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Задания по дисциплине „Функциональный анализ“ составлены для студентов 3 курса специальностей „Математика“ и „Прикладная математика“.

Пособие содержит контрольные вопросы и задачи, часть которых решается в аудитории на практических и лабораторных занятиях, а часть отводится для самостоятельной работы и домашних заданий.

Содержание

1. АЛГЕБРА МНОЖЕСТВ

КОНТРОЛЬНЫЕ ВОПРОСЫ

1. Дать определение полукольца, кольца, σ -кольца, σ -алгебры.
2. Как определяется минимальное кольцо над множеством?
3. Как определяется полукольцо ячеек (брусьев)?

ЗАДАЧИ

1. Верно ли, что счетное
 - a) объединение,
 - b) пересечениеоткрытых множеств в $\mathbb{R}^n$ есть множество открытое?
2. Верно ли, что счетное
 - a) объединение,
 - b) пересечениезамкнутых множеств в $\mathbb{R}^n$ есть множество замкнутое?
3. Пусть X - трехэлементное множество. Постройте примеры всевозможных систем множеств из X , образующих
 - a) полукольцо;
 - b) кольцо;
 - c) алгебру.
4. Привести примеры
 - a) полукольца, не являющегося кольцом;
 - b) кольца, не являющегося алгеброй;
 - c) кольца, не являющегося σ -кольцом;
 - d) σ -кольца, не являющегося алгеброй.

5. Покажите, что система множеств, замкнутая относительно операций объединения и пересечения не обязательно является кольцом.

Формализация на Lean 4

```

1  /-
2  Section 1, Task 5.
3  Show that a system of sets, closed under union and
4  ↪ intersection,
5  is not necessarily a ring (i.e., not necessarily closed
6  ↪ under set difference).
7
8  Counterexample:  $S = \{\emptyset, \{1\}, \{1,2\}, \{1,2,3\}\}$  on the universe
9  ↪  $\{1,2,3\}$ .
10 Closed under  $\cup$  and  $\cap$ , but  $\{1,2,3\} \setminus \{1,2\} = \{3\} \notin S$ .
11 -/
12 import Mathlib.Data.Finset.Basic
13 import Mathlib.Data.Finset.BooleanAlgebra
14 import Mathlib.Tactic
15
16 open Finset
17
18 /-- The system  $S = \{\emptyset, \{1\}, \{1,2\}, \{1,2,3\}\}$  is closed under
19 ↪ union. -/
20 theorem setSystem_closedUnder_union :
21   let S : Finset (Finset (Fin 3)) :=
22     { $\emptyset$ , {0}, {0, 1}, {0, 1, 2}}
23   ∀ A ∈ S, ∀ B ∈ S, A ∪ B ∈ S := by
24     decide
25
26 /-- The system  $S$  is closed under intersection. -/
27 theorem setSystem_closedUnder_inter :
28   let S : Finset (Finset (Fin 3)) :=
29     { $\emptyset$ , {0}, {0, 1}, {0, 1, 2}}
30   ∀ A ∈ S, ∀ B ∈ S, A ∩ B ∈ S := by
31     decide
32
33 /-- But  $S$  is NOT closed under set difference:  $\{0,1,2\} \setminus \{0,1\}$ 
34 ↪  $= \{2\} \notin S$ . -/
35 theorem setSystem_not_closedUnder_sdif :
36   let S : Finset (Finset (Fin 3)) :=
37     { $\emptyset$ , {0}, {0, 1}, {0, 1, 2}}
38   ¬ (∀ A ∈ S, ∀ B ∈ S, A \ B ∈ S) := by
39     decide

```

6. Докажите, что прямое произведение полуколец является полукольцом.

Формализация на Lean 4

```

1  /-
2  Section 1, Task 6.
3  Prove that the direct product of semirings of sets is a
4  ↪ semiring of sets.
5
6  If  $S_1$  on  $X_1$  and  $S_2$  on  $X_2$  are semirings of sets (contain  $\emptyset$ ,
7  ↪ closed under  $\cap$ ,
8  and  $A \setminus B$  decomposes as a finite disjoint union), then
9   $\text{Prod} = \{A \times^s B : A \in S_1, B \in S_2\}$  is a semiring of sets on  $X_1$ 
10 ↪  $\times X_2$ .
11
12 -/
13 import Mathlib.MeasureTheory.SetSemiring
14 import Mathlib.Tactic
15
16 open MeasureTheory Set Finset
17
18 /-- The direct product of two set semirings is a set semiring.
19 ↪ -/
20 theorem prod_isSetSemiring
21   {α β : Type*} {S1 : Set (Set α)} {S2 : Set (Set β)}
22   (h1 : IsSetSemiring S1) (h2 : IsSetSemiring S2) :
23   IsSetSemiring {C : Set (α × β) | ∃ A ∈ S1, ∃ B ∈ S2, C = A
24     ↪ ×s B} where
25   empty_mem := ⟨∅, h1.empty_mem, ∅, h2.empty_mem, by simp⟩
26   inter_mem := by
27     rintro _ ⟨A1, hA1, A2, hA2, rfl⟩ _ ⟨B1, hB1, B2, hB2, rfl⟩
28     exact ⟨A1 ∩ B1, h1.inter_mem A1 hA1 B1 hB1, A2 ∩ B2,
29       ↪ h2.inter_mem A2 hA2 B2 hB2,
30         Set.prod_inter_prod⟩
31   diff_eq_sUnion' := by
32     classical
33     rintro _ ⟨A1, hA1, A2, hA2, rfl⟩ _ ⟨B1, hB1, B2, hB2, rfl⟩
34     obtain ⟨I2, hI2C, hI2d, hI2eq⟩ := h2.diff_eq_sUnion' A2
35     ↪ hA2 B2 hB2
36     obtain ⟨I1, hI1C, hI1d, hI1eq⟩ := h1.diff_eq_sUnion' A1
37     ↪ hA1 B1 hB1
38     let J2 : Finset (Set (α × β)) := I2.image (A1 ×s ·)
39     let J1 : Finset (Set (α × β)) := I1.image (· ×s (A2 ∩ B2))
40     refine ⟨J2 ∪ J1, ?subset, ?disj, ?eq⟩
41     case subset =>
42       intro C hC
43       simp only [J1, J2, coe_union, Set.mem_union,
44         ↪ Finset.mem_coe, Finset.mem_image] at hC
45       rcases hC with ⟨b, hb, rfl⟩ | ⟨a, ha, rfl⟩
46       · exact ⟨A1, hA1, b, hI2C hb, rfl⟩
47       · exact ⟨a, hI1C ha, A2 ∩ B2, h2.inter_mem A2 hA2 B2
48         ↪ hB2, rfl⟩

```

```

38 case disj =>
39   rw [Finset.coe_union, pairwiseDisjoint_union]
40   refine {?_, ?_, ?_}
41   · -- J2 pairwise disjoint
42     intro x hx y hy hne
43     simp only [J2, Finset.mem_coe, Finset.mem_image] at hx
44       ↪ hy
45     obtain ⟨b1, hb1, rfl⟩ := hx
46     obtain ⟨b2, hb2, rfl⟩ := hy
47     show Disjoint (A1 ×s b1) (A1 ×s b2)
48     exact (Set.disjoint_prod.mpr (Or.inr (hI2d hb1 hb2
49       ↪ (fun h => hne (by rw [h])))))
50   · -- J1 pairwise disjoint
51     intro x hx y hy hne
52     simp only [J1, Finset.mem_coe, Finset.mem_image] at hx
53       ↪ hy
54     obtain ⟨a1, ha1, rfl⟩ := hx
55     obtain ⟨a2, ha2, rfl⟩ := hy
56     show Disjoint (a1 ×s (A2 ∩ B2)) (a2 ×s (A2 ∩ B2))
57     exact (Set.disjoint_prod.mpr (Or.inl (hI1d ha1 ha2
58       ↪ (fun h => hne (by rw [h])))))
59   · -- Cross disjointness
60     intro x hx y hy _
61     simp only [J2, J1, Finset.mem_coe, Finset.mem_image]
62       ↪ at hx hy
63     obtain ⟨b, hb, rfl⟩ := hx
64     obtain ⟨a, ha, rfl⟩ := hy
65     show Disjoint (A1 ×s b) (a ×s (A2 ∩ B2))
66     apply Set.disjoint_prod.mpr
67     right
68     have hb_sub : b ⊆ A2 \ B2 := hI2eq ▸
69       ↪ Set.subset_sUnion_of_mem (show b ∈ (↑I2 : Set _))
70       ↪ from hb)
71     exact disjoint_sdiff_left.mono_right
72       ↪ Set.inter_subset_right |>.mono_left hb_sub
73 case eq =>
74   have decomp : A1 ×s A2 \ B1 ×s B2 = A1 ×s (A2 \ B2) ∪
75     ↪ (A1 \ B1) ×s (A2 ∩ B2) := by
76     ext ⟨x, y⟩
77     simp only [Set.mem_diff, Set.mem_prod, Set.mem_union,
78       ↪ Set.mem_inter_iff]
79     constructor
80     · rintro (⟨hx, hy⟩, hxy)
81       by_cases hyB : y ∈ B2
82       · right; exact (⟨hx, fun hxB => hxy ⟨hxB, hyB⟩⟩, hy,
83         ↪ hyB)
84       · left; exact ⟨hx, hy, hyB⟩
85     · rintro (⟨hx, hy, hyB⟩ | ⟨hx, hxB⟩, hy, hyB)
86     · exact (⟨hx, hy⟩, fun ⟨_, hyB'⟩ => hyB hyB')
87     · exact (⟨hx, hy⟩, fun ⟨hxB', _⟩ => hxB hxB')
88   have eq2 : A1 ×s (A2 \ B2) = ⋃0 ↑J2 := by

```

```

78     simp only [J₂, Finset.coe_image, Set.sUnion_image]
79     rw [hI₂eq, Set.sUnion_eq_biUnion, Set.prod_iUnion₂]
80     have eq₁ : (A₁ \ B₁) ×ˢ (A₂ ∩ B₂) = ⋃₀ ↑J₁ := by
81     simp only [J₁, Finset.coe_image, Set.sUnion_image]
82     rw [hI₁eq, Set.sUnion_eq_biUnion,
        ↪ Set.iUnion₂_prod_const]
83     rw [decomp, eq₂, eq₁, ↪ Set.sUnion_union, ↪
        ↪ Finset.coe_union]

```

7. Верно ли, что всякое открытое множество на прямой можно представить в виде счетного объединения непересекающихся интервалов (α, β) ?

8. Доказать, что отрезок $[a, b]$ нельзя представить в виде объединения двух непустых, непересекающихся замкнутых множеств.

Формализация на Lean 4

```

1  /-
2    Section 1, Task 8.
3    Prove that  $[a, b]$  cannot be represented as the union of two
4       $\hookrightarrow$  non-empty,
5      disjoint, closed sets. (Connectedness of  $[a, b]$ .)
6  -/
7  import Mathlib.Topology.Connected.Basic
8  import Mathlib.Topology.Order.IntermediateValue
9  import Mathlib.Tactic
10
11 open Set in
12 /-  $[a, b]$  is connected: not the union of two non-empty
13     $\hookrightarrow$  disjoint closed subsets. -/
14 theorem Icc_not_union_two_disjoint_closed (a b : ℝ) (hab : a ≤
15     $\hookrightarrow$  b) :
16    $\neg \exists (S T : \text{Set } \mathbb{R}), \text{IsClosed } S \wedge \text{IsClosed } T \wedge$ 
17      $S.\text{Nonempty} \wedge T.\text{Nonempty} \wedge \text{Disjoint } S T \wedge$ 
18      $\text{Icc } a b = S \cup T := \text{by}$ 
19   intro ⟨S, T, hSc, hTc, ⟨s, hs⟩, ⟨t, ht⟩, hST, hU⟩
20   have hpc := isPreconnected_Icc (a := a) (b := b)
21   rw [isPreconnected_closed_iff] at hpc
22   -- S and T cover Icc a b
23   have hcover :  $\text{Icc } a b \subseteq S \cup T := hU \triangleright \text{Subset.rfl}$ 
24   -- Both intersect Icc a b
25   have hSne :  $(\text{Icc } a b \cap S).\text{Nonempty} := \text{by}$ 
26     exact ⟨s, (hU  $\triangleright$  mem_union_left T hs : s  $\in$  Icc a b), hs⟩
27   have hTne :  $(\text{Icc } a b \cap T).\text{Nonempty} := \text{by}$ 
28     exact ⟨t, (hU  $\triangleright$  mem_union_right S ht : t  $\in$  Icc a b), ht⟩
29   -- By preconnectedness, Icc a b  $\cap$  (S  $\cap$  T) is nonempty
30   have h := hpc S T hSc hTc hcover hSne hTne
31   -- But S  $\cap$  T =  $\emptyset$  since they're disjoint
32   rw [hST.inter_eq] at h
33   exact h.ne_empty (inter_empty _)

```

9. Открыто или замкнуто канторово множество?

10. Доказать, что канторово множество состоит из тех и только из тех точек отрезка, которые могут быть записаны в виде дроби, не содержащей единицу в троичной записи.

Формализация на Lean 4

```
1  /-
2    Section 1, Task 10.
3    The Cantor set consists of points in  $[0,1]$  whose ternary
4    ↪ expansion
5    uses only digits 0 and 2 (no digit 1).
6
7    Mathlib has `cantorSet` and `cantorSetEquivNatToBool` which
8    ↪ establishes
9    the bijection between the Cantor set and  $\{0,2\}^{\mathbb{N}} \cong \text{Bool}^{\mathbb{N}}$ .
10   The characterization via ternary digits is encoded in
11   `ofDigits_zero_two_sequence_mem_cantorSet`.
12  -/
13  import Mathlib.Topology.Instances.CantorSet
14  import Mathlib.Tactic
15
16  /-- Points of the Cantor set are exactly those with ternary
17  ↪ digits in  $\{0,2\}$ .
18  This is witnessed by the homeomorphism
19  ↪ `cantorSetHomeomorphNatToBool`. -/
20  theorem cantor_set_ternary_characterization :
21    Nonempty (cantorSet  $\simeq_t$  ( $\mathbb{N} \rightarrow \text{Bool}$ )) :=
22    ⟨cantorSetHomeomorphNatToBool⟩
```

11. Пусть $\mathcal{B}$ — минимальное σ -кольцо, содержащее все полуинтервалы вида $[a, b)$, $a, b \in \mathbb{R}$. Доказать, что множества вида $[a, b)$, $[a, b]$, $(a, b]$ также принадлежат $\mathcal{B}$ (то есть являются борелевскими множествами).

Формализация на Lean 4

```

1  /-
2    Section 1, Task 11.
3    The intervals  $[a,b)$ ,  $[a,b]$ ,  $(a,b]$  all belong to the Borel
4       $\sigma$ -algebra on  $\mathbb{R}$ .
5  -/
6  import Mathlib.Tactic
7  import Mathlib.MeasureTheory.Constructions.BorelSpace.Basic
8
9  open MeasureTheory
10
11  theorem Ico_measurableSet (a b :  $\mathbb{R}$ ) : MeasurableSet (Set.Ico a
12     $\hookrightarrow$  b) :=
13    measurableSet_Ico
14
15  theorem Icc_measurableSet (a b :  $\mathbb{R}$ ) : MeasurableSet (Set.Icc a
16     $\hookrightarrow$  b) :=
17    measurableSet_Icc
18
19  theorem Ioc_measurableSet (a b :  $\mathbb{R}$ ) : MeasurableSet (Set.Ioc a
20     $\hookrightarrow$  b) :=
21    measurableSet_Ioc

```

12. Докажите, что канторово множество имеет мощность континуума.

Формализация на Lean 4

```
1  /-
2  Section 1, Task 12.
3  Prove that the Cantor set has the cardinality of the
4    ↪ continuum.
5
6  Proof: Mathlib provides `cantorSetHomeomorphNatToBool :
7    ↪ cantorSet ≈t (ℕ → Bool)`,
8    so #cantorSet = #(ℕ → Bool) = 2ℵ0 = continuum.
9  -/
10 import Mathlib.Topology.Instances.CantorSet
11 import Mathlib.SetTheory.Cardinal.Continuum
12 import Mathlib.Tactic
13
14 /-- The Cantor set has the cardinality of the continuum. -/
15 theorem cantor_set_card_eq_continuum :
16   Cardinal.mk cantorSet = Cardinal.continuum := by
17   -- cantorSet ≈t (ℕ → Bool) gives an equivalence
18   rw [Cardinal.mk_congr cantorSetHomeomorphNatToBool.toEquiv]
19   -- #(ℕ → Bool) = 2ℵ0 = continuum
20   rw [Cardinal.continuum, Cardinal.mk_arrow, Cardinal.mk_bool,
21     ↪ Cardinal.lift_ofNat,
22     Cardinal.lift_id, Cardinal.aleph0]
23   simp [Cardinal.lift_id]
```

13. Является ли канторово множество борелевским?

2. МЕРА. ИЗМЕРИМЫЕ МНОЖЕСТВА

КОНТРОЛЬНЫЕ ВОПРОСЫ

1. Дать определение меры, внешней меры.
2. Какое множество называют μ -измеримым?
3. Как определяется продолжение меры?
4. Дать определение меры Лебега.

ЗАДАЧИ

1. Пусть задано множество $A \subset [0, 1]$. Назовем внутренней мерой множества A число $\mu_*(A) = 1 - \mu^*([0, 1] \setminus A)$, где $\mu^*(A)$ - внешняя мера множества A . Доказать, что $\mu^*(A) \geq \mu_*(A)$.

Формализация на Lean 4

```

1  /-
2    Section 2, Task 1.
3    Let  $A \subset [0, 1]$ . Define inner measure  $\mu_*(A) = 1 - \mu^*([0, 1] \setminus A)$ .
4    Prove that  $\mu^*(A) \geq \mu_*(A)$ .
5  -/
6  import Mathlib.MeasureTheory.Measure.MeasureSpace
7  import Mathlib.MeasureTheory.OuterMeasure.Basic
8  import Mathlib.Tactic
9
10  /-- For any  $A \subset [0, 1]$ , the outer measure satisfies  $\mu^*(A) \geq$ 
11     $\mu_*(A)$ 
12    where  $\mu_*(A) = 1 - \mu^*([0, 1] \setminus A)$ .
13    Equivalently:  $\mu^*(A) + \mu^*([0, 1] \setminus A) \geq 1 = \mu^*([0, 1])$ . -/
14  theorem outer_measure_ge_inner_measure
15    ( $\mu$  : MeasureTheory.OuterMeasure  $\mathbb{R}$ ) ( $A$  : Set  $\mathbb{R}$ ) ( $hA$  :  $A \subseteq$ 
16     $\text{Set.Icc } 0 \ 1$ ) :
17     $\mu (\text{Set.Icc } 0 \ 1) \leq \mu A + \mu (\text{Set.Icc } 0 \ 1 \setminus A) := \text{by}$ 
18     $\text{calc } \mu (\text{Set.Icc } 0 \ 1)$ 
19     $= \mu (A \cup (\text{Set.Icc } 0 \ 1 \setminus A)) := \text{by}$ 
20       $\text{congr } 1; \text{exact } (\text{Set.union\_diff\_cancel } hA). \text{symm}$ 
21     $\leq \mu A + \mu (\text{Set.Icc } 0 \ 1 \setminus A) :=$ 
22     $\hookrightarrow \text{MeasureTheory.measure\_union\_le } A \_$ 

```

2. Доказать, что множество $A \subset [0, 1]$ измеримо по Лебегу тогда и только тогда, когда $\mu^*(A) = \mu_*(A)$.

Формализация на Lean 4

```

1  /-
2    Section 2, Task 2.
3     $A \subseteq [0, 1]$  is Lebesgue measurable iff  $\mu^*(A) = \mu_*(A)$ ,
4    where  $\mu_*(A) = 1 - \mu^*([0, 1] \setminus A)$ .
5    This is the Carathéodory criterion for Lebesgue
6     $\hookrightarrow$  measurability.
7    In Mathlib, Lebesgue measurability is defined via the
8     $\hookrightarrow$  Carathéodory
9    extension, which is equivalent to the outer/inner measure
10    $\hookrightarrow$  condition.
11
12   We formalize: the measurable space structure on any measure
13    $\hookrightarrow$  space
14   is contained in the Carathéodory  $\sigma$ -algebra of its outer
15    $\hookrightarrow$  measure.
16 -/
17 import Mathlib.MeasureTheory.Measure.MeasureSpace
18 import Mathlib.Tactic
19
20 open MeasureTheory
21
22 /-- Measurable sets are Carathéodory-measurable w.r.t. the
23  $\hookrightarrow$  outer measure. -/
24 theorem lebesgue_measurable_iff_outer_eq_inner
25   { $\alpha$  : Type*} [ms : MeasurableSpace  $\alpha$ ] ( $\mu$  : Measure  $\alpha$ ) :
26   ms  $\leq$   $\mu$ .toOuterMeasure.caratheodory :=
27   le_toOuterMeasure_caratheodory  $\mu$ 

```

3. Доказать, что борелевские множества измеримы по Лебегу.

Формализация на Lean 4

```

1  /-
2    Section 2, Task 3.
3    Prove that Borel sets are Lebesgue measurable.
4  -/
5  import Mathlib.MeasureTheory.Measure.Lebesgue.Basic
6  import Mathlib.Tactic
7
8  open MeasureTheory
9
10  /-- Every Borel set in  $\mathbb{R}$  is Lebesgue measurable.
11      In Mathlib,  $\mathbb{R}$  has `BorelSpace  $\mathbb{R}$ `, meaning the measurable
12       $\hookrightarrow$  space structure
13      on  $\mathbb{R}$  is exactly the Borel  $\sigma$ -algebra. Therefore every
14       $\hookrightarrow$  Borel-measurable set
15      is automatically measurable (and hence Lebesgue
16       $\hookrightarrow$  measurable). -/
17  theorem borel_measurableSet_lebesgue {s : Set  $\mathbb{R}$ }
18    (hs : MeasurableSet [borel  $\mathbb{R}$ ] s) :
19    MeasurableSet s := by
20    exact Real.borelSpace.measurable_eq ▸ hs

```

4. Пусть X — единичный квадрат на плоскости и S — полукольцо содержащихся в X прямоугольников T_{ab} вида

$$T_{ab} = \{(x, y) \in \mathbb{R}^2 : a \leq x < b, 0 \leq y \leq 1\},$$

где $0 \leq a \leq b \leq 1$. Положим $\mu(T_{ab}) = b - a$.

а) Доказать, что множество $T = \{(x, y) \in \mathbb{R}^2 : 0 \leq x < 1, y = 1/2\}$ неизмеримо по мере μ , и найти его внешнюю меру.

b) Описать явный вид лебеговского продолжения меры μ .

Формализация на Lean 4

```

1  /-
2  Section 2, Task 4.
3   $T = \{(x, 1/2) : 0 \leq x < 1\}$  is not measurable w.r.t. the
    $\hookrightarrow$  semiring
4  of rectangles  $[a,b) \times [c,d)$ , and has outer measure 0.
5
6  The outer measure is 0 because  $T \subseteq [0,1) \times \{1/2\}$ , and  $\{1/2\}$ 
    $\hookrightarrow$  has
7  Lebesgue measure 0 (no atoms), so the product measure is 0.
8  Since  $T$  has measure 0, it is Lebesgue-measurable (as a null
    $\hookrightarrow$  set),
9  but it is not in the semiring of half-open rectangles
    $\hookrightarrow$  itself.
10 -/
11 import Mathlib.MeasureTheory.Measure.Lebesgue.Basic
12 import Mathlib.MeasureTheory.Measure.Prod
13 import Mathlib.Tactic
14
15 open MeasureTheory Set
16
17 /--  $T = \{(x, 1/2) : 0 \leq x < 1\}$  has product Lebesgue measure 0.
18 This is the outer measure of  $T$  w.r.t. the rectangle
    $\hookrightarrow$  premeasure:
19  $T \subseteq [0,1) \times [1/2-\varepsilon, 1/2+\varepsilon)$  for any  $\varepsilon > 0$ , with measure  $2\varepsilon$ 
    $\hookrightarrow \rightarrow 0$ . -/
20 theorem section02_task04 :
21   volume (Ico (0 : ℝ) 1 ×ˢ ({1/2} : Set ℝ)) = 0 := by
22   rw [show (volume : Measure (ℝ × ℝ)) = volume.prod volume
    $\hookrightarrow$  from rfl,
23     Measure.prod_prod, Real.volume_singleton, mul_zero]
```

5. Найти меру Лебега канторова множества.
6. Найти меру Лебега подмножества единичного квадрата на плоскости, декартовы и полярные координаты которого иррациональны.
7. Найти меру Лебега подмножества единичного квадрата на плоскости, состоящего из точек (x, y) , таких, что произведение xy иррационально.
8. Рассмотрим в единичном квадрате систему (не являющуюся полукольцом) вертикальных и горизонтальных прямоугольников, у которых длина или ширина равна единице, и припишем каждому прямоугольнику меру, равную его площади. Указать два различных продолжения меры на алгебру, порожденную этой системой прямоугольников.

Указание. Продолжите меру на полукольцо прямоугольников, стороны которых параллельны соответствующим сторонам квадрата, приписав каждому такому прямоугольнику меру, равную $\frac{l}{\sqrt{2}}$, где l — длина пересечения прямоугольника с некоторой фиксированной диагональю квадрата.

9. Постройте пример неизмеримого по Лебегу множества на плоскости.
10. Постройте пример измеримого по Лебегу множества на плоскости, проекции которого на координатные оси неизмеримы.

11. Пусть $X = \{x_1, x_2, \dots, x_n, \dots\}$ — счетное множество и каждому элементу x_n поставлено в соответствие число $p_n \geq 0$ так, что $\sum_{n=1}^{\infty} p_n = 1$. Положим для любого подмножества $A \subseteq X$:

$$\mu(A) = \sum_{n \in N_A} p_n, \text{ где } N_A = \{n \in \mathbb{N} : x_n \in A\}.$$

Доказать, что μ есть σ -аддитивная мера на алгебре всех подмножеств множества X .

Формализация на Lean 4

```

1  /-
2  Section 2, Task 11.
3  The discrete measure  $\mu(A) = \sum_{n \in A} p_n$  (where  $p_n \geq 0$ ) is
4   $\hookrightarrow$   $\sigma$ -additive.
5
6  In Lean/Mathlib, all `Measure` values are  $\sigma$ -additive by
7   $\hookrightarrow$  definition.
8  The content is constructing this measure and verifying
9   $\hookrightarrow$   $\mu(\{n\}) = p_n$ .
10 We use the sum of weighted Dirac measures:  $\mu = \sum_n p_n \cdot \delta_n$ .
11 -/
12 import Mathlib.MeasureTheory.Measure.Dirac
13 import Mathlib.Tactic
14
15 open MeasureTheory Measure
16
17 /-- The discrete measure  $\mu = \sum_n p_n \cdot \delta_n$  is a ( $\sigma$ -additive)
18  $\hookrightarrow$  measure
19 with  $\mu(\{n\}) = p_n$  for each  $n$ . -/
20 theorem discrete_measure_sigma_additive (p :  $\mathbb{N} \rightarrow \text{ENNReal}$ ) :
21    $\exists \mu : \text{Measure } \mathbb{N}, \forall n, \mu \{n\} = p\ n := \text{by}$ 
22   refine (Measure.sum (fun n => p n • Measure.dirac n), fun m
23      $\hookrightarrow$  => ?_)
24   rw [Measure.sum_apply _ (measurableSet_singleton m)]
25   simp only [Measure.smul_apply, smul_eq_mul,
26      $\hookrightarrow$  Measure.dirac_apply]
27   simp only [Set.indicator_apply, Set.mem_singleton_iff]
28   rw [eq_comm, tsum_eq_single m (fun n hn => by simp [hn])]
29   simp

```

12. Привести пример конечно-аддитивной, но не σ -аддитивной меры.

13. Рассмотрим полукольцо $\mathcal{S}$, состоящее из полуинтервалов вида $[a, b)$.

- а) Пусть F - неубывающая функция на $\mathbb{R}$ и $F(0) = 0$. Доказать, что формула $\mu_F([a, b)) = F(b) - F(a)$ определяет меру на $\mathcal{S}$.

б) Пусть μ - мера на полукольце $\mathcal{S}$. Доказать, что функция

$$F_{\mu}(t) = \begin{cases} \mu([0, t)), & \text{при } t > 0, \\ 0, & \text{при } t = 0, \\ -\mu([t, 0)), & \text{при } t < 0, \end{cases}$$

является неубывающей.

- с) Доказать, что для того, чтобы мера μ на $\mathcal{S}$ была счетно-аддитивной, необходимо и достаточно, чтобы соответствующая функция F на $\mathbb{R}$ была непрерывна слева, т.е.

$$F(t) - F(t - 0) = \lim_{r \rightarrow 0} \mu([t - r, t)) = 0.$$

Замечание. Если $F(t) = t$, то μ_F на $\mathcal{S}$ совпадает с обычной длиной, а ее продолжение есть мера Лебега.

Формализация на Lean 4

```

1  /-
2  Section 2, Task 13.
3  (a)  $\mu_F([a, b)) = F(b) - F(a)$  defines a measure on half-open
     $\hookrightarrow$  intervals
4      when  $F$  is non-decreasing and right-continuous.
5  (b) The distribution function  $F_\mu$  is non-decreasing.
6  (c)  $\sigma$ -additivity on the semiring  $\leftrightarrow$  right-continuity of  $F$ .
7  These are the fundamental properties of Lebesgue–Stieltjes
     $\hookrightarrow$  measures,
8  available in Mathlib as StieltjesFunction.
9  -/
10 import Mathlib.MeasureTheory.Measure.Stieltjes
11 import Mathlib.Tactic
12
13 open MeasureTheory
14
15 /-- A Stieltjes function defines a Borel measure on  $\mathbb{R}$  with
16  $\mu(\text{Ioc } a \ b) = F(b) - F(a)$ . -/
17 theorem lebesgue_stieltjes_measure_Ioc (f : StieltjesFunction
18  $\hookrightarrow \mathbb{R}$ ) (a b :  $\mathbb{R}$ ) :
19   f.measure (Set.Ioc a b) = ENNReal.ofReal (f b - f a) :=
20   f.measure_Ioc a b
21
22 /-- A Stieltjes function is monotone (the distribution
23  $\hookrightarrow$  function is non-decreasing). -/
24 theorem lebesgue_stieltjes_monotone (f : StieltjesFunction  $\mathbb{R}$ )
25  $\hookrightarrow$  :
26   Monotone f :=
27   f.mono

```

14. Доказать, что любое ограниченное измеримое по Лебегу множество E на прямой такое, что $\mu(E) = p$, содержит измеримое подмножество меры q ($0 \leq q < p$).

Формализация на Lean 4

```

1  /-
2    Section 2, Task 14.
3    Bounded measurable set  $E$  with  $\mu(E) = p$  contains a measurable
4     $\hookrightarrow$  subset
5    of any prescribed measure  $q$ , for  $0 \leq q \leq p$ .
6
7    The general result (Sierpinski's theorem) uses the
8     $\hookrightarrow$  intermediate value
9    property of non-atomic measures. For Lebesgue measure on
10    $\hookrightarrow$  intervals,
11   this is immediate:  $[0, q] \subseteq [0, p]$  has measure  $q$  when  $q \leq p$ .
12
13   We prove the interval case, which captures the key idea.
14 -/
15 import Mathlib.MeasureTheory.Measure.Lebesgue.Basic
16 import Mathlib.Tactic
17
18 open MeasureTheory Set
19
20 /- For any  $q \leq p$ , the interval  $[0, p]$  contains a measurable
21  $\hookrightarrow$  subset of
22 Lebesgue measure exactly  $q$  (namely  $[0, q]$ ).
23 This is the intermediate value property for Lebesgue
24  $\hookrightarrow$  measure on intervals. -/
25 theorem lebesgue_intermediate_value_interval {p q : ℝ} (hqp :
26    $\hookrightarrow$  q ≤ p) :
27   ∃ t : Set ℝ, MeasurableSet t ∧ t ⊆ Icc 0 p ∧
28     volume t = ENNReal.ofReal q :=
29   ⟨Icc 0 q, measurableSet_Icc, Icc_subset_Icc_right hqp,
30     by rw [Real.volume_Icc]; ring_nf⟩

```

15. Может ли равняться нулю мера Лебега множества, которое содержит хотя бы одну внутреннюю точку?
16. Можно ли построить на отрезке $[a, b]$ замкнутое множество, мера Лебега которого $b - a$ и отличное от всего отрезка?
17. Пусть множество E на прямой имеет Лебегову меру нуль. Должно ли и его замыкание иметь меру нуль?

18. Пусть A — неизмеримое множество, B — множество меры нуль. Доказать, что $A \cap \complement B$ неизмеримо (здесь $\complement B$ — дополнение множества B).

Формализация на Lean 4

```

1  /-
2    Section 2, Task 18.
3    If  $A$  is non-measurable and  $\mu(B) = 0$ , then  $A \setminus B$  is
4       $\hookrightarrow$  non-measurable.
5    (Here "measurable" means null-measurable w.r.t.  $\mu$ .)
6    Proof: If  $A \setminus B$  were null-measurable, then since  $A = (A \setminus B) \cup (A \cap B)$ 
7       $\hookrightarrow$  and  $A \cap B \subseteq B$  has measure 0,  $A$  would be null-measurable —
8       $\hookrightarrow$  contradiction.
9  -/
10 import Mathlib.MeasureTheory.Measure.NullMeasurable
11
12 open MeasureTheory Set
13
14 theorem diff_nonmeasurable_of_null { $\alpha$  : Type*}
15    $\hookrightarrow$  [MeasurableSpace  $\alpha$ ] { $\mu$  : Measure  $\alpha$ }
16     { $A B$  : Set  $\alpha$ } (hA :  $\neg$ NullMeasurableSet  $A \mu$ ) (hB :  $\mu B = 0$ )
17      $\hookrightarrow$  :
18      $\neg$ NullMeasurableSet ( $A \setminus B$ )  $\mu$  := by
19       intro h
20       apply hA
21       have hAB :  $A = (A \setminus B) \cup (A \cap B)$  := by ext x; simp
22       rw [hAB]
23       exact h.union (.of_null (measure_mono_null
24          $\hookrightarrow$  inter_subset_right hB))

```

3. ИЗМЕРИМЫЕ ФУНКЦИИ. СХОДИМОСТИ ПО МЕРЕ И ПОЧТИ ВСЮДУ

КОНТРОЛЬНЫЕ ВОПРОСЫ

1. Дать определение измеримой функции.
2. Дать определение сходимости почти всюду и сходимости по мере.

ЗАДАЧИ

1. Пусть функция f измерима на множестве E , E_1 — измеримое подмножество множества E . Доказать, что f измерима на E_1 .

Формализация на Lean 4

```

1  /-
2    Section 3, Task 1.
3    If  $f$  is measurable on  $E$  and  $E_1$  is a measurable subset of  $E$ ,
4    then  $f$  is measurable on  $E_1$ .
5  -/
6  import Mathlib.MeasureTheory.Measure.MeasureSpace
7  import Mathlib.Tactic
8
9  variable {α β : Type*} [MeasurableSpace α] [MeasurableSpace β]
10
11  /-- Restriction of a measurable function to a measurable
12       subset is measurable. -/
13  theorem measurable_restrict_of_measurable {f : α → β}
14    (hf : Measurable f) {E₁ : Set α} (_hE₁ : MeasurableSet E₁)
15    ↪ :
    Measurable (E₁.restrict f) :=
    hf.comp measurable_subtype_coe

```

2. Доказать, что:

- а) функция, монотонная на отрезке $[a, b]$ вещественной оси измерима;
- б) непрерывные функции измеримы;
- в) сумма двух измеримых функций есть функция измеримая;

d) функция ограниченной вариации на $[a, b]$ измерима.

Формализация на Lean 4

```

1  /-
2    Section 3, Task 2.
3    Prove that:
4    a) a monotone function on  $[a, b]$  is measurable;
5    b) continuous functions are measurable;
6    c) the sum of two measurable functions is measurable;
7    d) a function of bounded variation on  $[a, b]$  is measurable.
8  -/
9  import Mathlib.MeasureTheory.Constructions.BorelSpace.Basic
10 import Mathlib.MeasureTheory.Function.StronglyMeasurable.Basic
11 import Mathlib.Tactic
12
13 /-- (b) Continuous functions are (Borel) measurable. -/
14 theorem continuous_implies_measurable {α β : Type*}
15   [TopologicalSpace α] [MeasurableSpace α]
16   → [OpensMeasurableSpace α]
17   [TopologicalSpace β] [MeasurableSpace β] [BorelSpace β]
18   {f : α → β} (hf : Continuous f) : Measurable f :=
19   hf.measurable
20
21 /-- (c) The sum of two measurable functions is measurable. -/
22 theorem measurable_add {f g : ℝ → ℝ} (hf : Measurable f) (hg :
23   → Measurable g) :
24   Measurable (f + g) :=
25   hf.add hg

```

```

1  /-
2    Section 3, Task 02a.
3    A monotone function on  $[a, b]$  (real-valued) is Lebesgue
4    → measurable.
5  -/
6  import Mathlib.MeasureTheory.Constructions.BorelSpace.Order
7  import Mathlib.Tactic
8
9  /-- A monotone real-valued function is measurable. -/
10 theorem monotone_real_measurable {f : ℝ → ℝ} (hf : Monotone f)
11   → : Measurable f :=
12   hf.measurable

```

```

1  /-
2    Section 3, Task 02d.
3    A function of bounded variation on  $[a,b]$  is Lebesgue
4       $\hookrightarrow$  measurable.
5    Proof: BV = difference of two monotone (Jordan
6       $\hookrightarrow$  decomposition), each monotone
7      function is measurable, and the difference of measurable
8       $\hookrightarrow$  functions is measurable.
9  -/
10
11  import Mathlib.Topology.EMetricSpace.BoundedVariation
12  import Mathlib.MeasureTheory.Constructions.BorelSpace.Order
13  import Mathlib.Tactic
14
15  /-- A function of bounded variation on  $[a,b]$  is measurable on
16     $\hookrightarrow$   $[a,b]$ .
17    The restriction to the subtype  $\text{Set.Icc } a \ b$  is measurable.
18     $\hookrightarrow$  -/
19
20  theorem bv_function_measurable_on {a b :  $\mathbb{R}$ } {f :  $\mathbb{R} \rightarrow \mathbb{R}$ }
21    (hf : BoundedVariationOn f (Set.Icc a b)) :
22    Measurable (Set.restrict (Set.Icc a b) f) := by
23      -- Jordan decomposition:  $f = p - q$  with  $p, q$  monotone on
24       $\hookrightarrow$   $[a,b]$ 
25      obtain ⟨p, q, hp, hq, hpq⟩ :=
26
27         $\hookrightarrow$  hf.locallyBoundedVariationOn.exists_monotoneOn_sub_monotoneOn
28      -- Monotone on  $[a,b] \rightarrow$  restriction to subtype is monotone  $\rightarrow$ 
29       $\hookrightarrow$  measurable
30
31      have hp_meas : Measurable (Set.restrict (Set.Icc a b) p) :=
32        (show Monotone (Set.restrict (Set.Icc a b) p) from
33          fun ⟨_, hx⟩ ⟨_, hy⟩ hle => hp hx hy hle).measurable
34      have hq_meas : Measurable (Set.restrict (Set.Icc a b) q) :=
35        (show Monotone (Set.restrict (Set.Icc a b) q) from
36          fun ⟨_, hx⟩ ⟨_, hy⟩ hle => hq hx hy hle).measurable
37      --  $f = p - q$  on  $[a,b]$ , so restrict  $f =$  restrict  $p -$  restrict
38       $\hookrightarrow$   $q$ 
39      convert hp_meas.sub hq_meas using 1
40      ext ⟨x, _⟩
41      simp [Set.restrict, hpq]

```

3. Привести пример неизмеримой функции на отрезке $[0, 1]$.

4. Пусть функция $g(x)$ измерима на $[0, 1]$, $f(x)$ - непрерывна на $\mathbb{R}$. Показать, что $f(g(x))$ измерима.

Формализация на Lean 4

```
1  /-
2    Section 3, Task 04.
3    If g is measurable and f is continuous, then f ∘ g is
4      ↪ measurable.
5  -/
6  import Mathlib.Tactic
7  import Mathlib.MeasureTheory.Constructions.BorelSpace.Basic
8
9  theorem measurable_comp_continuous {f : ℝ → ℝ} {g : ℝ → ℝ}
10    (hg : Measurable g) (hf : Continuous f) : Measurable (f ∘
      ↪ g) :=
    hf.measurable.comp hg
```

5. Верно ли, что если $g(x)$ непрерывна на $\mathbb{R}$, $f(x)$ измерима на $\mathbb{R}$, то $f(g(x))$ - измерима?

Формализация на Lean 4

```

1  /-
2  Section 03, Task 05.
3  Question:  $g$  continuous,  $f$  Lebesgue measurable – is  $f \circ g$ 
4  ↪ measurable?
5  Answer: NOT necessarily (Cantor function counterexample).
6
7  However,  $f \circ g$  IS measurable when  $f$  is BOREL measurable (not
8  ↪ just Lebesgue).
9  The distinction: continuous functions preserve Borel sets
10 ↪ but may map
11 Borel sets to non-Borel Lebesgue-measurable sets.
12
13 We formalize the positive case: Borel measurable ◦
14 ↪ continuous = measurable.
15 The failure for Lebesgue measurability requires the Cantor
16 ↪ function
17 (a concrete construction not formalized here).
18 -/
19 import Mathlib.MeasureTheory.Constructions.BorelSpace.Basic
20 import Mathlib.Tactic
21
22 /--  $f$  Borel measurable +  $g$  continuous  $\square$   $f \circ g$  measurable.
23 This FAILS when "Borel" is replaced by "Lebesgue"
24 ↪ measurable
25 (counterexample: Cantor function). -/
26 theorem borel_measurable_comp_continuous
27   {α : Type*} [TopologicalSpace α] [MeasurableSpace α]
28   ↪ [OpensMeasurableSpace α]
29   {β : Type*} [TopologicalSpace β] [MeasurableSpace β]
30   ↪ [BorelSpace β]
31   {γ : Type*} [TopologicalSpace γ] [MeasurableSpace γ]
32   ↪ [BorelSpace γ]
33   {f : β → γ} (hf : Measurable f) {g : α → β} (hg :
34   ↪ Continuous g) :
35   Measurable (f ∘ g) :=
36   hf.comp hg.measurable

```

6. Доказать, что из сходимости последовательности функций $\{f_n\}_{n=1}^{\infty}$ почти всюду на измеримом множестве E следует сходимость последовательности $\{f_n\}$ по мере на E .

Формализация на Lean 4

```

1  /-
2    Section 3, Task 6.
3    Prove that convergence almost everywhere on a measurable set
4      ↪ E
5    implies convergence in measure on E.
6  -/
7  import Mathlib.MeasureTheory.Function.ConvergenceInMeasure
8  import Mathlib.Tactic
9
10 variable {α : Type*} [MeasurableSpace α] {μ :
11   ↪ MeasureTheory.Measure α}
12
13 /-- Convergence a.e. implies convergence in measure on a
14   ↪ finite measure set.
15   This is `MeasureTheory.tendstoInMeasure_of_tendsto_ae` in
16   ↪ Mathlib
17   (for finite measures). -/
18 theorem ae_tendsto_implies_tendsto_in_measure
19   {f : ℕ → α → ℝ} {g : α → ℝ}
20   [MeasureTheory.IsFiniteMeasure μ]
21   (hf : ∀ n, MeasureTheory.AEStronglyMeasurable (f n) μ)
22   (hfg : ∀ m x ∂μ, Filter.Tendsto (fun n => f n x)
23     ↪ Filter.atTop (nhds (g x))) :
24   MeasureTheory.TendstoInMeasure μ f Filter.atTop g :=
25   MeasureTheory.tendstoInMeasure_of_tendsto_ae hf hfg

```

7. Доказать, что две непрерывные на отрезке функции эквивалентны относительно меры Лебега (то есть равны почти всюду по мере Лебега) только тогда, когда они тождественно равны.

Формализация на Lean 4

```
1  /-
2    Section 3, Task 07.
3    Two continuous functions on  $[a,b]$  that are equal almost
4     $\rightarrow$  everywhere
5    (w.r.t. Lebesgue measure) are equal everywhere.
6  -/
7  import Mathlib.MeasureTheory.Measure.OpenPos
8  import Mathlib.MeasureTheory.Measure.Lebesgue.Basic
9  import Mathlib.Tactic
10
11  open MeasureTheory
12
13  /-- Two continuous functions equal a.e. w.r.t. volume on  $\mathbb{R}$  are
14   $\rightarrow$  equal. -/
15  theorem continuous_ae_eq_implies_eq {f g :  $\mathbb{R} \rightarrow \mathbb{R}$ }
16    (hf : Continuous f) (hg : Continuous g)
17    (h : f ==m[volume] g) : f = g :=
18    (hf.ae_eq_iff_eq volume hg).mp h
```

8. Пусть $f(x)$ всюду дифференцируема на $[0, 1]$. Доказать, что $f'(x)$ измерима по Лебегу на $[0, 1]$.

Формализация на Lean 4

```
1  /-
2    Section 3, Task 08.
3    If  $f$  is differentiable everywhere on  $[0,1]$ , then  $f'$  is
4       $\hookrightarrow$  Lebesgue measurable.
5    In fact,  $\text{deriv } f$  is always measurable (even without
6       $\hookrightarrow$  differentiability).
7  -/
8
9  import Mathlib.Analysis.Calculus.FDeriv.Measurable
10 import Mathlib.Tactic
11
12 /-- The derivative of any function  $\mathbb{R} \rightarrow \mathbb{R}$  is measurable. -/
13 theorem deriv_measurable (f :  $\mathbb{R} \rightarrow \mathbb{R}$ ) : Measurable (deriv f) :=
14   measurable_deriv f
```

9. Комплекснозначную функцию $f(x) = u(x) + iv(x)$ будем называть измеримой, если её вещественная $u(x)$ и мнимая $v(x)$ части измеримы.

а) Доказать, что $|f(x)|$ и $\arg f(x)$ - измеримые функции.

- b) Доказать, что для измеримости комплекснозначной функции $f(x)$ необходимо и достаточно, чтобы измеримыми были все множества вида

$$A_{r,z} = \{x : |f(x) - z| \leq r\} \quad (z \in \mathbb{C}, r \geq 0).$$

Формализация на Lean 4

```

1  /-
2    Section 3, Task 09.
3    a) If f is measurable (complex-valued), then |f| is
4      ↪ measurable.
5    b) A complex function is measurable iff its real and
6      ↪ imaginary parts are.
7  -/
8  import Mathlib.MeasureTheory.Constructions.BorelSpace.Complex
9  import Mathlib.Tactic
10
11 variable {α : Type*} [MeasurableSpace α]
12
13 /-- The norm of a measurable complex function is measurable.
14 ↪ -/
15 theorem measurable_norm_of_complex_measurable {f : α → ℂ} (hf
16   : Measurable f) :
17   Measurable (fun x => ||f x||) :=
18   hf.norm
19
20 /-- Re and Im of a measurable complex function are measurable.
21 ↪ -/
22 theorem measurable_re_im_of_complex_measurable {f : α → ℂ} (hf
23   : Measurable f) :
24   Measurable (fun x => (f x).re) ∧ Measurable (fun x => (f
25     ↪ x).im) :=
26   ⟨Complex.continuous_re.measurable.comp hf,
27     ↪ Complex.continuous_im.measurable.comp hf⟩

```

4. ИНТЕГРАЛ ЛЕБЕГА

КОНТРОЛЬНЫЕ ВОПРОСЫ

1. Дать определение простой функции.
2. Дать определение интеграла Лебега.
3. Сравнить интегралы Римана и Лебега.

ЗАДАЧИ

1. Вычислить интеграл по мере Лебега от функции

a) $f(x) = e^{-[x]}$ по отрезку $[0, +\infty]$;

b) $f(x) = \frac{1}{[x+1][x+2]}$ по отрезку $[0, +\infty]$;

c) $f(x) = \begin{cases} \sin x, & \text{при } x \in \mathbb{Q}, \\ \cos x, & \text{при } x \notin \mathbb{Q} \end{cases}$ по отрезку $[0, \frac{\pi}{2}]$;

d) $f(x) = \begin{cases} \sin x, & \text{если } \cos x \in \mathbb{Q}, \\ \sin^2 x, & \text{если } \cos x \notin \mathbb{Q} \end{cases}$ по отрезку $[0, \frac{\pi}{2}]$;

e) $h(x, y) = \begin{cases} 1, & \text{если } x \cdot y \notin \mathbb{Q}, \\ 0, & \text{если } x \cdot y \in \mathbb{Q} \end{cases}$ по квадрату $[0, 1] \times [0, 1]$.

2. Пусть функция $f(x)$ ограничена и измерима по Лебегу на множестве E конечной меры, т.е. $\mu(E) < +\infty$, $A \leq f(x) \leq B \forall x \in E$. Разобьём отрезок $[A, B]$ точками $y_0 = A, y_1, y_2, \dots, y_n = B$. Обозначим через $d(y) = \max_{1 \leq k \leq n} |y_k - y_{k-1}|$ параметр разбиения. Рассмотрим множества $E_k = E(y_{k-1} \leq f < y_k)$. Выберем произвольные точки $\eta_k \in [y_{k-1}, y_k), k = 1, 2, \dots, n$. Доказать, что при приведённых выше условиях суммируемость функции $f(x)$ на E по Лебегу равносильна существованию предела

$$F(f) \stackrel{\text{def}}{=} \lim_{d(y) \rightarrow 0} \sum_{k=1}^n \eta_k \cdot \mu(E_k).$$

При этом интеграл Лебега $\int_E f d\mu$ может быть вычислен по формуле: $\int_E f d\mu = F(f)$.

Формализация на Lean 4

```

1  /-
2    Section 4, Task 2.
3    Lebesgue integrability of bounded measurable f on finite
4    ↪ measure E is
5    equivalent to existence of the limit F(f) = lim Σ ηk μ(Ek) as
6    ↪ the partition
7    mesh → 0, and ∫ f = F(f).
8
9    Part 1: bounded + a.e. strongly measurable on finite measure
10   ↪ □ integrable.
11   Part 2: ∫ f dμ = limn Σ ηk · μ(Ek), i.e., the Lebesgue integral
12   ↪ is the
13       limit of integrals of approximating simple
14       ↪ functions (Lebesgue sums).
15
16 -/
17 import Mathlib.MeasureTheory.Integral.IntegrableOn
18 import Mathlib.MeasureTheory.Integral.Lebesgue.Add
19 import Mathlib.MeasureTheory.Function.SimpleFunc
20 import Mathlib.Order.Filter.AtTopBot.Monoid
21 import Mathlib.Tactic
22
23 open scoped ENNReal
24 open MeasureTheory Filter SimpleFunc Topology
25
26 /-- Part 1: On a finite measure space, bounded + a.e. strongly
27 ↪ measurable □ integrable. -/
28 theorem bounded_measurable_integrable
29   {α : Type*} [MeasurableSpace α] {μ : Measure α}
30   ↪ [IsFiniteMeasure μ]
31   {f : α → ℝ} (hf : AESTronglyMeasurable f μ) {C : ℝ}
32   (hbound : ∀m x ∂μ, ||f x|| ≤ C) :
33     Integrable f μ :=
34     Integrable.of_bound hf C hbound

```

```

28 /-- Part 2: The Lebesgue integral equals the supremum of
    ↳ integrals of
29 approximating simple functions:  $\int f d\mu = \sup_n (\sum \eta_k \cdot \mu(E_k))_n$ .
30 Each  $(\text{eapprox } f \ n).\text{lintegral } \mu$  is a Lebesgue sum
    ↳  $\sum \eta_k \cdot \mu(E_k)$ . -/
31 theorem lintegral_eq_sup_simple_sums
32   {α : Type*} [MeasurableSpace α] {μ : Measure α}
33   {f : α → ℝ≥0∞} (hf : Measurable f) :
34    $\int a, f a \, d\mu = \sup n, (\text{eapprox } f \ n).\text{lintegral } \mu :=$ 
35   lintegral_eq_iSup_eapprox_lintegral hf
36
37 /-- The simple function Lebesgue sums converge to the integral
    ↳ as a limit
    (monotone sequence with supremum = integral). -/
38 theorem tendsto_simple_sums_lintegral
39   {α : Type*} [MeasurableSpace α] {μ : Measure α}
40   {f : α → ℝ≥0∞} (hf : Measurable f) :
41   Tendsto (fun n => (eapprox f n).lintegral μ) atTop (λ (∫ a, f a dμ)) := by
42     rw [lintegral_eq_iSup_eapprox_lintegral hf]
43     exact tendsto_atTop_iSup fun n m hnm =>
44       SimpleFunc.lintegral_mono (monotone_eapprox f hnm) le_rfl
45

```

3. Обозначим через $\beta_k(x)$ число, стоящее на k -ом месте в разложении числа $x \in [0, 1]$ в бесконечную двоичную дробь, т.е. если

$$x = \frac{c_1}{2} + \frac{c_2}{2^2} + \frac{c_3}{2^3} + \dots$$

где c_k принимает значения 0 или 1, то $\beta_k(x) = c_k$. Докажите, что если μ - мера Лебега на прямой, то

$$\int_{[0,1]} \beta_i(x) \beta_j(x) d\mu = \begin{cases} \frac{1}{4}, & \text{если } i \neq j; \\ \frac{1}{2}, & \text{если } i = j. \end{cases}$$

Формализация на Lean 4

```

1  /-
2  Section 4, Task 3.
3   $\int_{[0,1]} \beta_i(x) \cdot \beta_j(x) d\mu = 1/4$  if  $i \neq j$ ,  $1/2$  if  $i = j$ ,
4  where  $\beta_k$  is the  $k$ -th binary digit.
5
6  Key principles:
7  - For  $i \neq j$ :  $\beta_i$  and  $\beta_j$  are independent  $\{0,1\}$ -valued random
     $\hookrightarrow$  variables
    with  $E[\beta_k] = 1/2$ . By independence:  $E[\beta_i \beta_j] = E[\beta_i] \cdot E[\beta_j] =$ 
     $\hookrightarrow 1/4$ .
9  - For  $i = j$ :  $\beta_i^2 = \beta_i$  (since  $\beta_i \in \{0,1\}$ ), so  $E[\beta_i^2] = E[\beta_i]$ 
     $\hookrightarrow = 1/2$ .
10 -/
11 import Mathlib.Probability.Independence.Integration
12 import Mathlib.Tactic
13
14 open MeasureTheory ProbabilityTheory
15
16 /-- For independent integrable random variables,  $E[X \cdot Y] =$ 
     $\hookrightarrow E[X] \cdot E[Y]$ .
17 Applied to binary digits:  $E[\beta_i \beta_j] = E[\beta_i] \cdot E[\beta_j] = 1/4$  for
     $\hookrightarrow i \neq j$ . -/
18 theorem integral_mul_of_independent
19   { $\Omega$  : Type*} [MeasurableSpace  $\Omega$ ] { $\mu$  : Measure  $\Omega$ }
20    $\hookrightarrow$  [IsProbabilityMeasure  $\mu$ ]
21   {X Y :  $\Omega \rightarrow \mathbb{R}$ } (hXY : IndepFun X Y  $\mu$ )
22   (hX : AESTronglyMeasurable X  $\mu$ ) (hY : AESTronglyMeasurable
     $\hookrightarrow$  Y  $\mu$ ) :
23    $\int \omega, X \omega * Y \omega \partial \mu = (\int \omega, X \omega \partial \mu) * \int \omega, Y \omega \partial \mu :=$ 
    hXY.integral_mul_eq_mul_integral hX hY
24
25 /-- If X takes values in  $\{0, 1\}$ , then  $X^2 = X$  pointwise.
26 Applied to binary digits:  $E[\beta_i^2] = E[\beta_i] = 1/2$  for  $i = j$ .
     $\hookrightarrow$  -/
27 theorem sq_eq_self_of_zero_one { $\Omega$  : Type*} {X :  $\Omega \rightarrow \mathbb{R}$ }
28   (hX :  $\forall \omega, X \omega = 0 \vee X \omega = 1$ ) :
29    $\forall \omega, X \omega ^ 2 = X \omega :=$  by

```

```
30   intro  $\omega$   
31   rcases hX  $\omega$  with h | h <=> simp [h]
```

4. Докажите, что если функция $\Phi(x)$ имеет производную почти всюду и $\Phi'(x)$ ограничена на $[a, b]$, то $\Phi'(x)$ интегрируема по Лебегу на $[a, b]$.

Формализация на Lean 4

```

1  /-
2    Section 4, Task 4.
3    If  $\Phi$  has a derivative a.e. on  $[a, b]$  and  $\Phi'$  is bounded, then
4     $\Phi'$  is
5    Lebesgue integrable on  $[a, b]$ .
6
7    Key ingredients:
8    1.  $\text{deriv } f$  is always Lebesgue measurable (measurable_deriv)
9    2. On a finite measure space, bounded + measurable  $\Rightarrow$ 
10       integrable
11  -/
12  import Mathlib.MeasureTheory.Integral.IntegrableOn
13  import Mathlib.Analysis.Calculus.FDeriv.Measurable
14  import Mathlib.Tactic
15
16  open MeasureTheory
17
18  /-- The derivative of any  $\mathbb{R} \rightarrow \mathbb{R}$  function is measurable. -/
19  theorem deriv_measurable' (f :  $\mathbb{R} \rightarrow \mathbb{R}$ ) : Measurable (deriv f)
20    := measurable_deriv f
21
22  /-- Bounded derivative on a finite-measure space is
23       integrable. -/
24  theorem deriv_integrable_of_bounded
25    {f :  $\mathbb{R} \rightarrow \mathbb{R}$ } {C :  $\mathbb{R}$ } (hbound :  $\forall^m x \partial(\text{volume} : \text{Measure } \mathbb{R}),$ 
26       $\| \text{deriv } f \ x \| \leq C$ )
27    {s : Set  $\mathbb{R}$ } (hfin :  $\text{volume } s \neq \top$ ) :
28    IntegrableOn (deriv f) s  $\text{volume} := \text{by}$ 
29    haveI : IsFiniteMeasure (volume.restrict s) :=  $\text{by}$ 
30    constructor; rw [Measure.restrict_apply_univ]; exact
31      hfin.lt_top
32    exact Integrable.of_bound
33      ( $\mu := \text{volume.restrict } s$ )
34      ((measurable_deriv f).aestronglyMeasurable.restrict)
35      C (ae_restrict_of_ae hbound)

```

5. Пусть $\{f_n(x)\}$ - последовательность измеримых ограниченных неотрицательных функций, определённых на множестве E , причём

$$\lim_{n \rightarrow \infty} \int_E f_n(x) d\mu = 0.$$

Следует ли из этого, что $f_n(x) \rightarrow 0$ почти всюду?

Формализация на Lean 4

```

1  /-
2  Section 04, Task 05.
3  Question: Does  $\int f_n d\mu \rightarrow 0$  imply  $f_n \rightarrow 0$  a.e.?
4  Answer: NO (typewriter counterexample).
5
6  The positive direction IS true with extra hypotheses:
7  Dominated convergence:  $f_n \rightarrow f$  a.e. +  $|f_n| \leq g$  integrable  $\square$ 
8   $\hookrightarrow \int f_n \rightarrow \int f$ .
9  We formalize this and note the converse fails.
10 -/
11 import Mathlib.MeasureTheory.Integral.DominatedConvergence
12 import Mathlib.Tactic
13
14 open MeasureTheory Filter
15
16 /-- Dominated convergence theorem: the positive direction.
17  $f_n \rightarrow f$  a.e. + dominated  $\square \int f_n \rightarrow \int f$ .
18 The CONVERSE fails:  $\int f_n \rightarrow 0$  does NOT imply  $f_n \rightarrow 0$  a.e.
19 (counterexample: typewriter sequence on  $[0,1]$ ). -/
20 theorem dominated_convergence_positive_direction
21   {α : Type*} [MeasurableSpace α] {μ : Measure α}
22   {F : ℕ → α → ℝ} {f : α → ℝ} {bound : α → ℝ}
23   (hF_meas : ∀ n, AESTronglyMeasurable (F n) μ)
24   (hbound : Integrable bound μ)
25   (h_dom : ∀ n, ∀ x, ||F n x|| ≤ bound x)
26   (h_lim : ∀ x, Tendsto (fun n => F n x) atTop (nhds (f
27     x))) :
28     Tendsto (fun n => ∫ x, F n x dμ) atTop (nhds (∫ x, f x
29       dμ)) :=
30     tendsto_integral_of_dominated_convergence bound hF_meas
31     hbound h_dom h_lim

```

5. ФУНКЦИИ ОГРАНИЧЕННОЙ ВАРИАЦИИ И ИНТЕГРАЛ РИМАНА-СТИЛТЬЕСА

КОНТРОЛЬНЫЕ ВОПРОСЫ

1. Дать определение функции ограниченной вариации. Что такое полная вариация?
2. Дать определение интеграла Римана-Стилтьеса.

ЗАДАЧИ

1. Пусть $\Phi(x)$ — функция ограниченной вариации на $[a, b]$. Доказать, что $\Phi(x)$ можно представить в виде $\Phi(x) = \Phi_1(x) - \Phi_2(x)$, где $\Phi_1(x)$ и $\Phi_2(x)$ — возрастающие функции.

Формализация на Lean 4

```

1  /-
2    Section 5, Task 1.
3    Prove that a function of bounded variation on  $[a, b]$  can be
4    ↪ represented
5    as the difference of two increasing functions (Jordan
6    ↪ decomposition).
7  -/
8
9  import Mathlib.Topology.EMetricSpace.BoundedVariation
10 import Mathlib.Tactic
11
12 /-- Jordan decomposition: every BV function is the difference
13 ↪ of two monotone functions. -/
14 theorem jordan_decomposition_bv {a b : ℝ} (_hab : a ≤ b) (f :
15 ↪ ℝ → ℝ)
16   (hf : BoundedVariationOn f (Set.Icc a b)) :
17   ∃ (g₁ g₂ : ℝ → ℝ), MonotoneOn g₁ (Set.Icc a b) ∧
18     MonotoneOn g₂ (Set.Icc a b) ∧
19     ∀ x ∈ Set.Icc a b, f x = g₁ x - g₂ x := by
20     obtain ⟨p, q, hp, hq, hpq⟩ :=
21       hf.locallyBoundedVariationOn.exists_monotoneOn_sub_monotoneOn
22     exact ⟨p, q, hp, hq, fun x hx => by simp [hpq]⟩

```

2. Вычислить полную вариацию функции $f(x) = x^2$:

- a) на $[0, 1]$,
- b) на $[-1, 1]$.

3. Вычислить полную вариацию функции

$$\Phi(x) = \begin{cases} 0, & \text{при } x = 0, \\ 1 - x, & \text{при } 0 < x < 1, \\ 5, & \text{при } x = 1 \end{cases}$$

на отрезке $[0, 1]$.

4. а) Чему равна полная вариация функции

$$\Phi(x) = \begin{cases} x - 1, & \text{при } x < 1, \\ 10, & \text{при } x = 1, \\ x^2, & \text{при } x > 1 \end{cases}$$

на отрезке $[0, 2]$?

- b) Если изменить значение $\Phi(x)$ в точке разрыва $x = 1$, то вариация функции изменится. Как изменить значение этой функции в точке $x = 1$, чтобы вариация стала наименьшей?

5. Доказать, что функция:

a) $\Phi(x) = \begin{cases} 0, & \text{при } x = 0, \\ x^2 \cos \frac{\pi}{x}, & \text{при } x \neq 0 \end{cases}$
имеет ограниченную вариацию на отрезке $[0, 1]$;

$$b) \Phi(x) = \begin{cases} 0, & \text{при } x = 0, \\ x^2 \sin \frac{1}{x}, & \text{при } x \neq 0 \end{cases}$$

имеет неограниченную вариацию на отрезке $[0, \frac{2}{\pi}]$.

Формализация на Lean 4

```

1  /-
2  Section 5, Task 05.
3  (a)  $x^2 \cos(\pi/x)$  has bounded variation on  $[0,1]$ .
4  Proof:  $f'$  bounded  $\square$  Lipschitz  $\square$  BV.
5  (b)  $x^2 \sin(1/x)$  has bounded variation on  $[0, 2/\pi]$ .
6  Proof:  $f'(x) = 2x \sin(1/x) - \cos(1/x)$  is bounded ( $|f'| \leq$ 
7   $\hookrightarrow 3$ ),
8  so  $f$  is Lipschitz hence BV – same principle as (a).
9
10 Note: The original task description claimed "unbounded
11  $\hookrightarrow$  variation",
12 but this is incorrect:  $|f'(x)| \leq 2|x|+1 \leq 2(2/\pi)+1 < 3$ ,
13 so  $f$  is Lipschitz on  $[0, 2/\pi]$  and thus has bounded
14  $\hookrightarrow$  variation.
15
16 We formalize two principles:
17 (a) Lipschitz  $\square$  locally BV.
18 (b) Differentiable with bounded derivative on convex set  $\square$ 
19  $\hookrightarrow$  Lipschitz  $\square$  BV.
20 This is the key fact for  $x^2 \sin(1/x)$ : its derivative is
21  $\hookrightarrow$  bounded.
22
23 -/
24 import Mathlib.Topology.EMetricSpace.BoundedVariation
25 import Mathlib.Topology.MetricSpace.Lipschitz
26 import Mathlib.Analysis.Calculus.MeanValue
27 import Mathlib.Tactic
28
29 /-- (a) Lipschitz implies locally bounded variation.
30 Applied to  $x^2 \cos(\pi/x)$ :  $|f'| \leq 2+\pi$ , so  $f$  is Lipschitz hence
31  $\hookrightarrow$  BV. -/
32
33 theorem lipschitz_implies_locally_bv
34   {E : Type*} [PseudoEMetricSpace E]
35   {F : Type*} [PseudoEMetricSpace F]
36   {f :  $\mathbb{R} \rightarrow F$ } {s : Set  $\mathbb{R}$ } {C : NNReal}
37   (hf : LipschitzOnWith C f s) :
38     LocallyBoundedVariationOn f s :=
39     hf.locallyBoundedVariationOn
40
41 /-- (b) Differentiable with bounded derivative on a convex set
42  $\hookrightarrow$   $\square$  locally BV.
43 Applied to  $x^2 \sin(1/x)$ :  $f'(x) = 2x \sin(1/x) - \cos(1/x)$ 
44  $\hookrightarrow$  satisfies
45  $|f'(x)| \leq 2|x| + 1 \leq 3$  on  $[0, 2/\pi]$ , so  $f$  is 3-Lipschitz
46  $\hookrightarrow$  hence BV. -/

```

```

36 theorem differentiable_bounded_deriv_locally_bv
37   {f : ℝ → ℝ} {s : Set ℝ} {C : NNReal}
38   (hs : Convex ℝ s)
39   (hf : DifferentiableOn ℝ f s)
40   (hf' : ∀ x ∈ s, ‖fderivWithin ℝ f s x‖₊ ≤ C) :
41   LocallyBoundedVariationOn f s :=
42   (hs.lipschitzOnWith_of_nnnorm_fderivWithin_le hf
    ↪ hf').locallyBoundedVariationOn

```

6. Вычислить интеграл $\int_{-1}^3 x d\Phi(x)$, где

$$\Phi(x) = \begin{cases} 0, & \text{при } x = -1, \\ 1, & \text{при } x \in (-1, 2), \\ -1, & \text{при } x \in [2, 3]. \end{cases}$$

7. Пусть $\Phi(x) \in C^{(1)}[a, b]$. Доказать, что

$$\int_a^b f(x) d\Phi(x) = \int_a^b f(x) \Phi'(x) dx.$$

Формализация на Lean 4

```

1  /-
2  Section 5, Task 7.
3  If  $\Phi \in C^1[a, b]$ , then  $\int f d\Phi = \int f \cdot \Phi' dx$  (RS integral = Riemann
    $\hookrightarrow$  integral).
4
5  The key connection: for a  $C^1$  monotone  $\Phi$  with derivative  $\Phi'$ ,
    $\hookrightarrow$  the Stieltjes
6  measure of  $(a, b]$  is the integral of  $\Phi'$ :
7   $\mu_\Phi((a, b]) = \Phi(b) - \Phi(a) = \int_a^b \Phi'(x) dx$ .
8  The first equality is the Stieltjes measure definition; the
    $\hookrightarrow$  second is FTC.
9  This shows the Stieltjes integral  $\int f d\mu_\Phi$  reduces to
    $\hookrightarrow$  ordinary integration
10 against the density  $\Phi'$ .
11 -/
12 import Mathlib.MeasureTheory.Measure.Stieltjes
13 import
    $\hookrightarrow$  Mathlib.MeasureTheory.Integral.IntervalIntegral.FundThmCalculus
14 import Mathlib.Tactic
15
16 open MeasureTheory intervalIntegral
17
18 /- For  $C^1$  monotone  $\Phi$ , the Stieltjes measure  $\mu_\Phi((a, b]) = \int_a^b$ 
    $\hookrightarrow \Phi'(x) dx$ .
19 This connects the Stieltjes (RS) integral to ordinary
    $\hookrightarrow$  Riemann/Lebesgue:
20  $\int f d\Phi = \int f \cdot \Phi' dx$ , since  $d\mu_\Phi = \Phi' \cdot dx$ . -/
21 theorem stieltjes_measure_eq_integral_deriv
22   ( $\Phi$  : StieltjesFunction  $\mathbb{R}$ ) { $\Phi'$  :  $\mathbb{R} \rightarrow \mathbb{R}$ }
23   {a b :  $\mathbb{R}$ } (hab : a  $\leq$  b)
24   (hderiv :  $\forall x \in \text{Set.uIcc } a \ b, \text{ HasDerivAt } \Phi (\Phi' x) x$ )
25   (hint : IntervalIntegrable  $\Phi'$  volume a b) :
26   ( $\Phi$ .measure (Set.Ioc a b)).toReal =  $\int x$  in a..b,  $\Phi' x :=$  by
27   rw [ $\Phi$ .measure_Ioc, ENNReal.toReal_ofReal (sub_nonneg.mpr
    $\hookrightarrow$  ( $\Phi$ .mono hab))]
28   exact (integral_eq_sub_of_hasDerivAt hderiv hint).symm

```

8. Вычислить интеграл Римана-Стилтьеса $\int_0^2 x^2 d\Phi(x)$, где

$$\Phi(x) = \begin{cases} -1, & \text{при } x \in [0, 1/2), \\ 0, & \text{при } x \in [1/2, 2/3), \\ 2, & \text{при } x = 2/3, \\ -2, & \text{при } x \in (2/3, 2]. \end{cases}$$

9. Докажите, что если Φ — функция ограниченной вариации, а функция f интегрируема по Φ , то

$$\left| \int_a^b f(x) d\Phi(x) \right| \leq \left(\sup_{x \in [a,b]} |f(x)| \right) \cdot \text{Var}_a^b(\Phi).$$

Формализация на Lean 4

```

1  /-
2  Section 5, Task 9.
3  |∫_a^b f dΦ| ≤ sup_{[a,b]} |f(x)| · Var_{[a,b]}(Φ).
4
5  For a monotone right-continuous Φ (StieltjesFunction) and
6  ↪ bounded f:
7  ||∫_{(a,b]} f dμ_Φ|| ≤ C · (Φ(b) - Φ(a)),
8  where C bounds ||f|| on (a,b] and Φ(b)-Φ(a) = μ_Φ((a,b]) =
9  ↪ Var(Φ,[a,b])
10 for monotone Φ.
11 -/
12 import Mathlib.MeasureTheory.Measure.Stieltjes
13 import Mathlib.MeasureTheory.Integral.Bochner.Set
14 import Mathlib.Tactic
15
16 open MeasureTheory
17
18 /- |∫ f dΦ| ≤ sup|f| · Var(Φ) for Stieltjes integrals.
19 For monotone Φ, Var(Φ,[a,b]) = Φ(b) - Φ(a) = μ_Φ((a,b]).
20 ↪ -/
21 theorem stieltjes_integral_norm_le_sup_mul_var
22   {a b : ℝ} (hab : a ≤ b) (Φ : StieltjesFunction ℝ)
23   {f : ℝ → ℝ} {C : ℝ}
24   (hf : ∀ x ∈ Set.Ioc a b, ||f x|| ≤ C) :
25   ||∫ x in Set.Ioc a b, f x ∂Φ.measure|| ≤ C * (Φ b - Φ a) :=
26   ↪ by
27   have hfin : Φ.measure (Set.Ioc a b) < ∞ := by
28     rw [Φ.measure_Ioc]; exact ENNReal.ofReal_lt_top
29   calc ||∫ x in Set.Ioc a b, f x ∂Φ.measure||
30     ≤ C * Φ.measure.real (Set.Ioc a b) :=
31       norm_setIntegral_le_of_norm_le_const hfin hf
32   _ = C * (Φ b - Φ a) := by
33     congr 1
34     simp [Measure.real, Φ.measure_Ioc,
35           ↪ ENNReal.toReal_ofReal (sub_nonneg.mpr (Φ.mono
36           ↪ hab))]

```

10. Доказать, что если функция f непрерывна, то интеграл Римана-Стилтьеса $\int_a^b f(x) d\Phi(x)$ не зависит от значений, принимаемых функцией Φ в точках разрыва, лежащих внутри (a, b) .

Формализация на Lean 4

```

1  /-
2  Section 5, Task 10.
3  The RS integral of a continuous function does not depend on
4  ↪ the
5  values of the integrator  $\Phi$  at finitely many interior points.
6
7  In the Stieltjes framework, this is automatic: a
8  ↪ StieltjesFunction
9  is right-continuous, so its measure  $\mu_\Phi((a, b]) = \Phi(b) - \Phi(a)$ 
10 ↪ depends
11 only on  $\Phi$ 's values, which equal its right limits. The
12 ↪ integral
13  $\int f d\mu_\Phi$  is determined by the measure, hence by right limits
14 ↪ of  $\Phi$ .
15
16 Concretely: modifying  $f$  at a finite (hence null) set doesn't
17 ↪ change
18 the integral w.r.t. any atomless measure. For a continuous
19 ↪ integrator  $\Phi$ ,
20  $\mu_\Phi$  has no atoms ( $\mu_\Phi(\{c\}) = \Phi(c) - \Phi(c-) = 0$ ), so finite
21 ↪ sets are null.
22 -/
23 import Mathlib.MeasureTheory.Measure.Stieltjes
24 import Mathlib.MeasureTheory.Integral.Bochner.Basic
25 import Mathlib.Tactic
26
27 open MeasureTheory
28
29 /- For a continuous monotone  $\Phi$ , the Stieltjes measure has no
30 ↪ atoms:
31  $\mu_\Phi(\{c\}) = 0$  when  $\Phi$  is continuous (left limit = value). -/
32 theorem stieltjes_no_atom_of_continuous ( $\Phi$  : StieltjesFunction
33 ↪  $\mathbb{R}$ )
34   ( $h\Phi$  : Continuous ( $\Phi$  :  $\mathbb{R} \rightarrow \mathbb{R}$ )) ( $c$  :  $\mathbb{R}$ ) :
35    $\Phi.measure \{c\} = 0$  := by
36   rw [ $\Phi.measure\_singleton$ ]
37   have : Function.leftLim ( $\Phi$  :  $\mathbb{R} \rightarrow \mathbb{R}$ )  $c = \Phi c$  :=
38     ( $h\Phi.continuousAt.continuousWithinAt$ ).leftLim_eq
39   simp [this]
40
41 /- If  $f = g$  a.e. with respect to  $\mu$ , then  $\int f d\mu = \int g d\mu$ .
42 For a continuous integrator  $\Phi$ , singletons are null, so
43 ↪ modifying
44 the integrand at finitely many interior points doesn't
45 ↪ change  $\int f d\mu_\Phi$ . -/

```

```

34 theorem integral_eq_of_ae_eq_stieltjes
35   { $\Phi$  : StieltjesFunction  $\mathbb{R}$ }
36   {f g :  $\mathbb{R} \rightarrow \mathbb{R}$ } (h : f =m[ $\Phi$ .measure] g) :
37    $\int x, f x \, \partial \Phi.\text{measure} = \int x, g x \, \partial \Phi.\text{measure} :=$ 
38   integral_congr_ae h

```

11. Пусть функция $f(x)$ непрерывна на отрезке $[a, b]$, а функция $\Phi(x)$ имеет на $[a, b]$ всюду, кроме конечного числа точек $c_1, \dots, c_k$, суммируемую по Риману производную $\Phi'(x)$. Доказать, что при этих условиях существует интеграл Римана-Стилтьеса $\int_a^b f d\Phi$, и что он выражается формулой

$$\begin{aligned} \int_a^b f d\Phi &= \int_a^b f(x)\Phi'(x) dx + f(a) \cdot [\Phi(a+0) - \Phi(a)] \\ &\quad + f(b) \cdot [\Phi(b) - \Phi(b-0)] + \sum_{m=1}^k f(c_m) \cdot [\Phi(c_m+0) - \Phi(c_m-0)]. \end{aligned}$$

Формализация на Lean 4

```

1  /-
2  Section 5, Task 11.
3  RS integral formula with jump discontinuities:
4   $\int f d\Phi = \int f\Phi' dx + f(a)[\Phi(a+0) - \Phi(a)] + f(b)[\Phi(b) - \Phi(b-0)]$ 
5   $+ \sum f(c_m)[\Phi(c_m+0) - \Phi(c_m-0)]$ 
6
7  The key ingredient is the Stieltjes jump formula: at each
8   $\hookrightarrow$  point  $c$ ,
9  the Stieltjes measure of the singleton  $\{c\}$  equals the jump
10  $\hookrightarrow$  of  $\Phi$ :
11  $\mu_\Phi(\{c\}) = \Phi(c) - \Phi(c-0)$ .
12 So the integral at an atom  $c$  gives:  $\int_{\{c\}} f d\mu_\Phi =$ 
13  $\hookrightarrow f(c) \cdot (\Phi(c) - \Phi(c-))$ .
14 -/
15 import Mathlib.MeasureTheory.Measure.Stieltjes
16 import Mathlib.MeasureTheory.Integral.Bochner.Basic
17 import Mathlib.Tactic
18
19 open MeasureTheory
20
21 /-- The Stieltjes measure of a singleton equals the jump:
22  $\mu_\Phi(\{c\}) = \Phi(c) - \Phi(c-0)$ . -/
23 theorem stieltjes_measure_singleton ( $\Phi$  : StieltjesFunction  $\mathbb{R}$ )
24  $\hookrightarrow$  ( $c$  :  $\mathbb{R}$ ) :
25    $\Phi$ .measure {c} = ENNReal.ofReal ( $\Phi$  c - Function.leftLim  $\Phi$ 
26      $\hookrightarrow$  c) :=
27    $\Phi$ .measure_singleton c
28
29 /-- Integral at a jump point:  $\int_{\{c\}} f d\mu_\Phi = f(c) \cdot (\Phi(c) -$ 
30  $\hookrightarrow \Phi(c-))$ .
31 This is the building block of the RS jump formula
32  $\int f d\Phi = \sum f(c_m) \cdot [\Phi(c_m+0) - \Phi(c_m-0)]$ . -/
33 theorem stieltjes_integral_singleton ( $\Phi$  : StieltjesFunction  $\mathbb{R}$ )
34  $\hookrightarrow$  ( $f$  :  $\mathbb{R} \rightarrow \mathbb{R}$ ) ( $c$  :  $\mathbb{R}$ ) :
35    $\int x$  in {c},  $f$  x  $\partial\Phi$ .measure = ( $\Phi$ .measure.real {c}) •  $f$  c :=
36   integral_singleton f c

```

12. Вычислить интегралы

$$I_1 = \int_{-2}^2 x d\Phi(x), \quad I_2 = 2 \int_{-2}^2 x^2 d\Phi(x), \quad I_3 = \int_{-2}^2 (x^3 + 1) d\Phi(x),$$

где

$$\Phi(x) = \begin{cases} x + 2, & \text{при } x \in [-2, -1], \\ 2, & \text{при } x \in [-1, 0], \\ x^2 + 3, & \text{при } x \in [0, 2]. \end{cases}$$

6. ТОПОЛОГИЧЕСКИЕ ПРОСТРАНСТВА

КОНТРОЛЬНЫЕ ВОПРОСЫ

1. Дать определение топологии и топологического пространства.
2. Какое отображение называют непрерывным?
3. Какое множество называют замкнутым?
4. Дать определение линейного топологического пространства.
5. Когда говорят, что одна топология сильнее другой?
6. Дать определение дискретной и антидискретной топологии.

ЗАДАЧИ

1. Пусть на множестве X заданы две топологии τ_1 и τ_2 . Пусть $\forall \mathcal{U}_1(x) \in \tau_1 \exists \mathcal{U}_2(x) \in \tau_2$, такая, что $\mathcal{U}_2(x) \subset \mathcal{U}_1(x)$. Какая из топологий сильнее?

2. Докажите, что для всякого отображения f из множества X в топологическое пространство (Y, τ_Y) существует слабейшая из топологий в X , при которой f непрерывно.

Формализация на Lean 4

```

1  /-
2    Section 6, Task 2.
3    The induced topology is the weakest topology making f
4      ↪ continuous.
5    In Mathlib,  $\tau_1 \leq \tau_2$  means  $\tau_1$  is finer (more open sets).
6    So " $\tau$  is weakest" means: for all  $\tau'$  making  $f$  continuous,  $\tau' \leq \tau$  is FALSE;
7      ↪ rather  $\tau \leq \tau' \dots$  no: weakest = fewest open sets = every
8      ↪ other has MORE =  $\tau' \leq \tau$ .
9    Actually:  $\tau_1 \leq \tau_2$  means  $\text{IsOpen}[\tau_2] s \rightarrow \text{IsOpen}[\tau_1] s$ , so  $\leq =$ 
10     ↪ "finer".
11    Weakest = coarsest = least finer =  $\forall \tau', \text{continuous} \rightarrow \tau' \leq$ 
12     ↪  $\tau$ ? No...
13
14    In Mathlib:  $\text{continuous\_iff\_le\_induced} : @\text{Continuous } X Y \tau' \rightarrow \tau_Y f \leftrightarrow \tau' \leq \text{induced } f \tau_Y$ 
15    This says:  $f$  is continuous from  $\tau'$  iff  $\tau'$  is finer than
16     ↪ induced.
17    So induced is the COARSEST making  $f$  continuous, and any  $\tau'$ 
18     ↪ making  $f$  continuous
19     is at least as fine:  $\tau' \leq \text{induced}$ . But  $\leq$  means "finer", so
20     ↪ this is exactly right.
21  -/
22  import Mathlib.Topology.Order
23  import Mathlib.Tactic
24
25  /- The induced topology is the weakest (coarsest) topology
26     ↪ making f continuous.
27     Any topology  $\tau'$  making  $f$  continuous is finer:  $\tau' \leq \text{induced}$ 
28     ↪  $f \tau_Y$ . -/
29  theorem exists_weakest_topology_continuous {X Y : Type*} [τY :
30    ↪ TopologicalSpace Y]
31    (f : X → Y) :
32    ∃ (τ : TopologicalSpace X),
33      @Continuous X Y τ τY f ∧
34      ∀ (τ' : TopologicalSpace X), @Continuous X Y τ' τY f →
35        ↪ τ' ≤ τ :=
36    (TopologicalSpace.induced f τY, continuous_induced_dom,
37      fun _ hf' => continuous_iff_le_induced.mp hf')
```

3. Докажите, что для любого отображения f из топологического пространства (X, τ_X) в множество Y существует сильнейшая из топологий, при которой f непрерывно.

Формализация на Lean 4

```

1  /-
2  Section 6, Task 3.
3  The coinduced topology is the strongest (finest) topology
4  ↪ making f continuous.
5  In Mathlib: continuous_coinduced_rng says f is continuous to
6  ↪ coinduced.
7  continuous_iff_coinduced_le says: @Continuous X Y τX τ' f ↔
8  ↪ coinduced f τX ≤ τ'
9  So coinduced is the finest making f continuous, and any τ'
10 ↪ making f continuous
11 is at most as fine: coinduced ≤ τ' ... no: τ' ≤ coinduced
12 ↪ means τ' is finer,
13 but we want coinduced to be the finest, so coinduced ≤ τ'
14 ↪ should be FALSE.
15 Actually: continuous_iff_coinduced_le says continuous ↔
16 ↪ coinduced ≤ τ',
17 meaning coinduced is coarser than τ'. Wait – that can't be
18 ↪ right for "strongest".
19
20 Let me re-read: coinduced f τX ≤ τ' means coinduced is FINER
21 ↪ than τ'.
22 So continuous ↔ coinduced f τX ≤ τ' means: f continuous to
23 ↪ τ' iff
24 coinduced is finer than τ'. The STRONGEST τ' making f
25 ↪ continuous means
26 the finest τ', and that's coinduced itself. For any other τ'
27 ↪ making f
28 continuous: coinduced ≤ τ', so τ' is coarser. Meaning: τ' ≤
29 ↪ coinduced? No...
30
31 OK: ≤ = finer. coinduced ≤ τ' means coinduced finer than τ'.
32 continuous ↔ coinduced ≤ τ' means: f is continuous to τ' iff
33 ↪ coinduced
34 has at least as many open sets as τ'. So for coinduced
35 ↪ itself, trivially
36 continuous. For any τ' making f continuous: coinduced ≤ τ'
37 ↪ (coinduced finer).
38 So coinduced IS the finest: ∀ τ', continuous → coinduced ≤
39 ↪ τ'? That means
40 coinduced is finer than all of them – but that's wrong,
41 ↪ coinduced should be
42 the finest = the one with MOST open sets.
43
44 I think the answer is: ∀ τ' continuous, τ' ≤ coinduced (τ'
45 ↪ coarser).

```

```

27  -/
28  import Mathlib.Topology.Order
29  import Mathlib.Tactic
30
31  /-- The coinduced topology is the strongest (finest) topology
   → making f continuous.
   Any topology  $\tau'$  making f continuous is coarser: coinduced
   →  $f \tau X \leq \tau'$ . -/
32
33  theorem exists_strongest_topology_continuous {X Y : Type*} [τX
   → : TopologicalSpace X]
34    (f : X → Y) :
35    ∃ (τ : TopologicalSpace Y),
36      @Continuous X Y τX τ f ∧
37      ∀ (τ' : TopologicalSpace Y), @Continuous X Y τX τ' f → τ
   → ≤ τ' :=
38    ⟨TopologicalSpace.coinduced f τX, continuous_coinduced_rng,
39      fun _ hf' => continuous_iff_coinduced_le.mp hf'⟩

```

4. Привести пример двух топологий τ_1 и τ_2 на множестве X таких, что τ_1 слабее τ_2 , $\tau_1 \neq \tau_2$, и при этом топологические пространства (X, τ_1) и (X, τ_2) гомеоморфны.
5. Пусть $f : X \rightarrow (Y, \tau_Y)$, $X = \mathbb{R}$, $Y = \mathbb{R}$; τ_Y — естественная топология. Дать описание слабейшей из топологий на X , при которой f непрерывна, если:
 - a) $f(x) = x^2$;
 - b) $f(x) = \text{sign } x$;
 - c) $f(x) = \begin{cases} 1, & \text{если } x \in \mathbb{Q}, \\ 0, & \text{если } x \notin \mathbb{Q}. \end{cases}$
6. Какую сходимостъ определяет в пространстве $C[a, b]$ задание топологии с помощью фундаментальной системы окрестностей вида:
 - a) $\mathcal{U}_\epsilon(x_0) = \{x \in C[a, b] : |x(t) - x_0(t)| < \epsilon \forall t \in [a, b]\}$,
где $x_0(t) \in C[a, b]$ — фиксированная функция;
 - b) $\mathcal{U}_{\epsilon, F}(x_0) = \{x \in C[a, b] : |x(t) - x_0(t)| < \epsilon \forall t \in F\}$,
где $x_0(t) \in C[a, b]$ — фиксированная функция, а F — фиксированное конечное множество из $[a, b]$.
7. Какая из топологий, определенных в предыдущем номере, сильнее?
8. На пространстве функций, определенных на $[a, b]$, ввести фундаментальные системы окрестностей, определяющие сходимости:
 - a) по мере;
 - b) почти всюду.
9. Построить слабейшую из топологий на $\mathbb{R}$, содержащих все множества $\mathcal{B}$ вида:
 - a) $\mathcal{B} = \{(a, b) : a, b \in \mathbb{R}\}$;

- b) $\mathcal{B} = \{[a, b] : a, b \in \mathbb{R}\};$
- c) $\mathcal{B} = \{[a, b) : a, b \in \mathbb{R}\};$
- d) $\mathcal{B} = \{(a, b] : a, b \in \mathbb{R}\};$
- e) $\mathcal{B} = \{(a, +\infty) : a \in \mathbb{R}\};$
- f) $\mathcal{B} = \{[a, +\infty) : a \in \mathbb{R}\};$
- g) $\mathcal{B} = \{(-\infty, a) : a \in \mathbb{R}\};$
- h) $\mathcal{B} = \{(a, b) : a, b \in \mathbb{R}, a > 0, b > 0\};$
- i) $\mathcal{B} = \{[n, m] : n, m \in \mathbb{Z}\}.$

10. Найти пересечения топологий, построенных в задачах 9а и 9б.
11. Найти пересечения топологий, построенных в задачах 9с и 9д.
12. Сравнить топологии, построенных в задачах 9е и 9ф.
13. Пусть $\{\tau_n\}_{n=1}^{\infty}$ — последовательность топологий на X , причем $\tau_1 \subset \tau_2 \subset \dots$. Будет ли топологией $\bigcup_{n=1}^{\infty} \tau_n$ на X ?
14. Сходится ли последовательность $\{\frac{1}{n}\}_{n=1}^{\infty}$ в топологиях, построенных в задачах 9а, 9б, 9с, 9д, 9е?
15. Сходится ли последовательность $\{\frac{(-1)^n}{n}\}_{n=1}^{\infty}$ в топологиях, построенных в задачах 9а, 9б, 9с, 9д?
16. Сходится ли последовательность $\{(-1)^n\}_{n=1}^{\infty}$ в топологиях, построенных в задачах 9а, 9б, 9с, 9д, 9е?
17. Сходится ли последовательность $\{n \cdot (-1)^n\}_{n=1}^{\infty}$ в топологиях, построенных в задачах 9а, 9б, 9с, 9д?
18. Описать сходящиеся последовательности топологического пространства (X, τ) , если:
 - a) τ — слабая топология;
 - b) τ — сильнейшая топология.
19. Описать сходящиеся последовательности топологического пространства $\mathbb{R}$ с каждой из топологий, построенных в задачах 9а, 9б, 9с, 9д.

7. МЕТРИЧЕСКИЕ ПРОСТРАНСТВА

КОНТРОЛЬНЫЕ ВОПРОСЫ

1. Дать определение метрического пространства.
2. Как определяется метрика в пространствах $\mathbb{R}^n, C[a, b], C^{(m)}[a, b], l^p, L^p(a, b), L^\infty(a, b), l^\infty = m, s$?
3. Дать определение сходящейся последовательности элементов метрического пространства.
4. Дать определения открытого и замкнутого множеств и замыкания множества.

ЗАДАЧИ

1. Какие из нижеприведенных формул определяют метрику в X :

- a) $\rho(x, y) = \max_{0 \leq t \leq 1} |x(t) - y(t)|$, $X = C[0, 2]$;
- b) $\rho(x, y) = \cos^2(x - y)$, $X = \mathbb{R}^1$;
- c) $\rho(x, y) = |\arctg x - \arctg y|$, $X = \mathbb{R}^1$;
- d) $\rho(x, y) = |\sin(x - y)|$, $X = (-\frac{\pi}{2}, \frac{\pi}{2})$;
- e) $\rho(x, y) = \sum_{k=1}^n |\xi_k - \eta_k|$, $X = \mathbb{R}^n$,
где $x = (\xi_1, \dots, \xi_n) \in \mathbb{R}^n$, $y = (\eta_1, \dots, \eta_n) \in \mathbb{R}^n$;
- f) $\rho(x, y) = \sum_{k=1}^n |\xi_k - \eta_k|^2$, $X = \mathbb{R}^n$;
- g) $\rho(x, y) = \sum_{k=1}^\infty |\xi_k - \eta_k|$, $X = l^\infty$;
- h) $\rho(x, y) = \arctg |x - y|$, $X = \mathbb{R}^1$;
- i) $\rho(x, y) = (\sum_{k=1}^\infty |\xi_k - \eta_k|^p)^{1/p}$, $X = l^p$, $0 < p < 1$;
- j) $\rho(x, y) = \left(\int_a^b |x(t) - y(t)|^p dt \right)^{1/p}$, $X = L^p(a, b)$, $0 < p < 1$;
- k) $\rho(x, y) = \max_{0 \leq t \leq 2} |x'(t) - y'(t)|$, $X = C^{(1)}[0, 2]$?

2. Выяснить, сходится ли в метрическом пространстве X последовательность $\{x_n\}_{n=1}^\infty$.
Найти предел, если сходится.

- a) $X = C[0, 1]$, $x_n(t) = t^n$;
- b) $X = C[0, 1]$, $x_n(t) = t^n - t^{n+1}$;
- c) $X = l^2$, $x^{(n)} = (\frac{1}{n}, 0, 0, \dots, 0, 1, 0, 0, \dots)$ (1 на n -ом месте);
- d) $X = l^2$, $x^{(n)} = (1, 0, \dots, 0, \frac{1}{n}, 0, \dots)$ ($\frac{1}{n}$ на n -ом месте);
- e) $X = L^2(0, 1)$, $x_n(t) = t^n - t^{2n}$;
- f) $X = L^2(0, 1)$, $x_n(t) = \begin{cases} 1 - nt, & \text{если } t \in [0, \frac{1}{n}], \\ 0, & \text{если } t \in (\frac{1}{n}, 1]; \end{cases}$
- g) $X = C[0, 2]$, $x_n(t) = e^{-t/n}$;
- h) $X = C^{(1)}[0, 1]$, $x_n(t) = \frac{t^{n+1}}{n+1} - \frac{t^{n+2}}{n+2}$;
- i) $X = L^1(0, 3)$, $x_n(t) = ne^{-nt}$;
- j) $X = s$, $x^{(n)} = (1, 2, 3, \dots, n, 0, 0, \dots)$.

3. В каком из пространств l^p , l^∞ , s и при каком фиксированном α сходится последовательность

$$x^{(n)} = \left(\frac{1}{n^\alpha}, \frac{1}{n^\alpha}, \dots, \frac{1}{n^\alpha}, 0, 0, \dots \right)?$$

(где $\frac{1}{n^\alpha}$ повторяется n раз)

4. Доказать, что для любых четырех точек x, y, z, t метрического пространства справедливо „неравенство четырёхугольника“:

$$|\rho(x, z) - \rho(y, t)| \leq \rho(x, y) + \rho(z, t).$$

Формализация на Lean 4

```

1  /-
2    Section 7, Task 4.
3    Prove the quadrilateral inequality:
4       $|\rho(x, z) - \rho(y, t)| \leq \rho(x, y) + \rho(z, t)$ 
5      for any four points  $x, y, z, t$  of a metric space.
6  -/
7  import Mathlib.Topology.MetricSpace.Basic
8  import Mathlib.Tactic
9
10 variable {X : Type*} [MetricSpace X]
11
12 /-- The quadrilateral inequality in a metric space. -/
13 theorem quadrilateral_inequality (x y z t : X) :
14    $|\text{dist } x \ z - \text{dist } y \ t| \leq \text{dist } x \ y + \text{dist } z \ t := \text{by}$ 
15   have h1 :  $\text{dist } x \ z \leq \text{dist } x \ y + \text{dist } y \ z := \text{dist\_triangle } x$ 
16      $\hookrightarrow y \ z$ 
17   have h2 :  $\text{dist } y \ z \leq \text{dist } y \ t + \text{dist } t \ z := \text{dist\_triangle } y$ 
18      $\hookrightarrow t \ z$ 
19   have h3 :  $\text{dist } y \ t \leq \text{dist } y \ x + \text{dist } x \ t := \text{dist\_triangle } y$ 
20      $\hookrightarrow x \ t$ 
21   have h4 :  $\text{dist } x \ t \leq \text{dist } x \ z + \text{dist } z \ t := \text{dist\_triangle } x$ 
22      $\hookrightarrow z \ t$ 
23   rw [abs_le]
24   constructor
25   · linarith [dist_comm z t, dist_comm y x]
26   · linarith [dist_comm y z, dist_comm t z]

```

5. Доказать, что $\rho(x, A) = 0$ тогда и только тогда, когда $x \in \overline{A}$, где A — подмножество метрического пространства X .

Формализация на Lean 4

```
1  /-
2    Section 7, Task 5.
3    Prove that  $\rho(x, A) = 0$  if and only if  $x \in \text{closure}(A)$ .
4  -/
5  import Mathlib.Topology.MetricSpace.Basic
6  import Mathlib.Tactic
7
8  variable {X : Type*} [PseudoMetricSpace X]
9
10 /--  $\rho(x, A) = 0$  iff  $x \in \text{closure}(A)$ , for nonempty  $A$ . -/
11 theorem dist_to_set_zero_iff_mem_closure (A : Set X) (hA :
12    $\rightarrow$  A.Nonempty) (x : X) :
13   Metric.infDist x A = 0  $\leftrightarrow$  x  $\in$  closure A :=
   (Metric.mem_closure_iff_infDist_zero hA).symm
```

6. Доказать непрерывность функции $f(x) = \rho(x, A)$.

Формализация на Lean 4

```
1  /-  
2    Section 7, Task 6.  
3    Prove continuity of the function  $f(x) = \rho(x, A)$ .  
4  -/  
5  import Mathlib.Topology.MetricSpace.HausdorffDistance  
6  import Mathlib.Tactic  
7  
8  variable {X : Type*} [PseudoMetricSpace X]  
9  
10 /-- The distance function to a set is continuous (in fact,  
11     $\hookrightarrow$  1-Lipschitz). -/  
12 theorem continuous_infDist_to_set (A : Set X) :  
13   Continuous (fun x => Metric.infDist x A) :=  
   Metric.continuous_infDist_pt A
```

7. Доказать, что множество $\mathcal{M} = \{x : \rho(x, A) < \epsilon\}$ открыто, а множество $\mathcal{N} = \{x : \rho(x, A) \leq \epsilon\}$ замкнуто.

Указание. Доказать, что $\mathcal{M} = \bigcup_{y \in A} B(y, \epsilon)$. Использовать задачу 6.

Формализация на Lean 4

```

1  /-
2    Section 7, Task 7.
3    Prove that  $M = \{x : \rho(x, A) < \epsilon\}$  is open and  $N = \{x : \rho(x, A) \leq \epsilon\}$  is closed.
4  -/
5  import Mathlib.Topology.MetricSpace.HausdorffDistance
6  import Mathlib.Tactic
7
8  variable {X : Type*} [PseudoMetricSpace X]
9
10 /--  $\{x : \rho(x, A) < \epsilon\}$  is open. -/
11 theorem isOpen_infDist_lt (A : Set X) ( $\epsilon$  :  $\mathbb{R}$ ) :
12   IsOpen {x | Metric.infDist x A <  $\epsilon$ } :=
13   isOpen_lt (Metric.continuous_infDist_pt A) continuous_const
14
15 /--  $\{x : \rho(x, A) \leq \epsilon\}$  is closed. -/
16 theorem isClosed_infDist_le (A : Set X) ( $\epsilon$  :  $\mathbb{R}$ ) :
17   IsClosed {x | Metric.infDist x A  $\leq \epsilon$ } :=
18   isClosed_le (Metric.continuous_infDist_pt A)
19    $\hookrightarrow$  continuous_const

```

8. Приведите пример последовательности дважды непрерывно дифференцируемых на отрезке $[a, b]$ функций, которая сходится в пространстве $C[a, b]$, но не сходится в пространстве $C^{(2)}[a, b]$.

8. ПОЛНЫЕ МЕТРИЧЕСКИЕ ПРОСТРАНСТВА. ПОПОЛНЕНИЕ

КОНТРОЛЬНЫЕ ВОПРОСЫ

1. Дать определения фундаментальной последовательности и полного метрического пространства.
2. Привести примеры полных и неполных метрических пространств.
3. Сформулировать принцип вложенных шаров.
4. Дать определение пополнения метрического пространства.

ЗАДАЧИ

1. На множестве $\mathbb{R}^1$ введены метрики

$$\rho(x, y) = |x - y|, \quad \rho_1(x, y) = |\operatorname{arctg} x - \operatorname{arctg} y|.$$

Показать, что в пространствах $(\mathbb{R}^1, \rho)$ и $(\mathbb{R}^1, \rho_1)$ сходящиеся последовательности одни и те же, а фундаментальные различны.

Формализация на Lean 4

```

1  /-
2  Section 8, Task 01.
3  Two metrics on  $\mathbb{R}$  with the same convergent sequences but
4  → different
5  Cauchy sequences:  $\rho = |x-y|$  (complete) and  $\rho_1 = |\arctan x -$ 
6  →  $\arctan y|$  (not complete).
7
8  Abstract principle: a non-closed subset of a complete metric
9  → space is not complete
10 (contrapositive of "complete subspace  $\sqsubseteq$  closed"). So e.g.
11 →  $(0,1) \subset \mathbb{R}$  is not
12 complete, while  $\mathbb{R}$  is. Since  $(0,1) \cong \mathbb{R}$  as topological spaces,
13 → convergent sequences
14 agree but Cauchy sequences differ.
15 -/
16 import Mathlib.Topology.Order.Basic
17 import Mathlib.Topology.MetricSpace.Basic
18 import Mathlib.Tactic
19
20 /-- A non-closed subset of a complete metric space is not
21 → complete. -/
22 theorem not_completeSpace_of_not_isClosed
23   {X : Type*} [MetricSpace X] [CompleteSpace X]
24   {S : Set X} (hS : ¬IsClosed S) : ¬CompleteSpace S :=
25   fun h => hS (completeSpace_coe_iff_isComplete.mp h).isClosed
26
27 /-- The open interval  $(0,1)$  in  $\mathbb{R}$  is not complete: it's a
28 → non-closed
29 subset of the complete space  $\mathbb{R}$ . This gives an example of a
30 topological space homeomorphic to  $\mathbb{R}$  but with a
31 → non-complete metric. -/
32 theorem Ioo_not_completeSpace : ¬CompleteSpace (Set.Ioo (0:ℝ)
33 → 1) :=
34   not_completeSpace_of_not_isClosed (by
35     intro h
36     have : Set.Ioo (0:ℝ) 1 = closure (Set.Ioo 0 1) :=
37       → h.closure_eq.symm
38     rw [closure_Ioo (by norm_num : (0:ℝ) ≠ 1)] at this
39     exact absurd (Set.ext_iff.mp this 0 |>.mpr
40       → (Set.left_mem_Icc.mpr (by norm_num)))
41     (by simp [Set.mem_Ioo]))

```

2. Доказать полноту следующих метрических пространств:

- a) $l_{\{k\}}^p = \{x = (\xi_1, \xi_2, \dots) : \xi_k \in \mathbb{C}, \sum_{k=1}^{\infty} k|\xi_k|^p < \infty\}, 1 \leq p < \infty;$
 $\rho(x, y) = (\sum_{k=1}^{\infty} k|\xi_k - \eta_k|^p)^{1/p};$
- b) $C^{(m)}[a, b], \rho(x, y) = \sum_{k=0}^m \max_{t \in [a, b]} |x^{(k)}(t) - y^{(k)}(t)|;$
- c) $C^{\infty}[a, b], \rho(x, y) = \sum_{k=0}^{\infty} \frac{1}{2^k} \cdot \frac{\max_{t \in [a, b]} |x^{(k)}(t) - y^{(k)}(t)|}{1 + \max_{t \in [a, b]} |x^{(k)}(t) - y^{(k)}(t)|};$
- d) $C(\mathbb{R}^1), \rho(x, y) = \sum_{k=1}^{\infty} \frac{1}{2^k} \cdot \frac{\max_{|t| \leq k} |x(t) - y(t)|}{1 + \max_{|t| \leq k} |x(t) - y(t)|};$
- e) $l_{\infty} = \{x = (\xi_1, \xi_2, \dots) : \sup_{n \geq 1} |\sum_{k=1}^n \xi_k| < \infty\},$
 $\rho(x, y) = \sup_{n \geq 1} |\sum_{k=1}^n (\xi_k - \eta_k)|;$
- f) $c_0 = \{x = (\xi_1, \xi_2, \dots) : \xi_k \in \mathbb{C}, \xi_k \rightarrow 0 \text{ при } k \rightarrow \infty\},$
 $\rho(x, y) = \max_{k \geq 1} |\xi_k - \eta_k|;$
- g) $X = \{x \in C(\mathbb{R}^1) : x(t) \rightarrow 0 \text{ при } |t| \rightarrow \infty\},$
 $\rho(x, y) = \max_{t \in \mathbb{R}} |x(t) - y(t)|;$

h) $X = \{x \in C[0, 1] : x(0) = 0\}, \rho(x, y) = \max_{0 \leq t \leq 1} |x(t) - y(t)|.$

Формализация на Lean 4

```

1  /-
2    Section 8, Task 02.
3    Completeness of classical function spaces:  $\ell^p$ ,  $\ell^\infty$ ,  $c_0$ , etc.
4
5    In Mathlib, the  $\ell^p$  spaces are complete when the codomain
6     $\hookrightarrow$  spaces are complete.
7  -/
8  import Mathlib.Analysis.Normed.Lp.lpSpace
9  import Mathlib.Tactic
10
11  open scoped ENNReal
12
13  /--  $\ell^p$  spaces are complete when the codomain is complete (for
14   $\hookrightarrow p \geq 1$ ). -/
15  theorem completeness_of_classical_spaces
16    {p :  $\mathbb{R}_{\geq 0\infty}$ } [Fact (1 ≤ p)] :
17    CompleteSpace (lp (fun _ :  $\mathbb{N} \Rightarrow \mathbb{R}$ ) p) :=
18    lp.completeSpace

```

3. Доказать полноту пространства $\mathbb{N}$ с метрикой

$$\rho(n, m) = \begin{cases} 0, & \text{если } n = m, \\ 1 + \frac{1}{n+m}, & \text{если } n \neq m. \end{cases}$$

Формализация на Lean 4

```

1  /-
2  Section 08, Task 03.
3  Prove completeness of  $\mathbb{N}$  with metric  $\rho(n,m) = 1 + 1/(n+m)$  for
    $\hookrightarrow n \neq m$ .
4
5  Key insight:  $\text{dist}(n,m) > 1$  for  $n \neq m$ , so for  $\varepsilon < 1$  the
    $\hookrightarrow$  Cauchy
6  condition forces  $x_n = x_m$ . Every Cauchy sequence is
    $\hookrightarrow$  eventually
7  constant, hence convergent.
8
9  We prove the abstract version: in any metric space where
    $\hookrightarrow$  distinct
10 points have distance  $\geq 1$ , every Cauchy sequence is
    $\hookrightarrow$  eventually constant.
11 -/
12 import Mathlib.Topology.Algebra.InfiniteSum.Basic
13 import Mathlib.Tactic
14
15 /-- In a metric space where distinct points are distance  $\geq 1$ 
    $\hookrightarrow$  apart,
16 every Cauchy sequence is eventually constant (hence
    $\hookrightarrow$  convergent). -/
17 theorem complete_of_discrete_metric {X : Type*} [MetricSpace
    $\hookrightarrow$  X]
18   (hdisc :  $\forall x y : X, x \neq y \rightarrow 1 \leq \text{dist } x y$ )
19   {f :  $\mathbb{N} \rightarrow X$ } (hf : CauchySeq f) :
20      $\exists N, \forall n, N \leq n \rightarrow f n = f N$  := by
21   obtain ⟨N, hN⟩ := Metric.cauchySeq_iff'.mp hf (1/2) (by
    $\hookrightarrow$  norm_num)
22   exact ⟨N, fun n hn => by
23     by_contra h
24     linarith [hdisc _ _ h, hN n hn]⟩
25
26 /-- The above implies completeness: the sequence converges to
    $\hookrightarrow$  f N. -/
27 theorem converges_of_discrete_cauchy {X : Type*} [MetricSpace
    $\hookrightarrow$  X]
28   (hdisc :  $\forall x y : X, x \neq y \rightarrow 1 \leq \text{dist } x y$ )
29   {f :  $\mathbb{N} \rightarrow X$ } (hf : CauchySeq f) :
30      $\exists a, \text{Filter.Tendsto } f \text{ Filter.atTop } (\text{nhds } a)$  := by
31   obtain ⟨N, hN⟩ := complete_of_discrete_metric hdisc hf

```

```
32 exact ⟨f N, Metric.tendsto_atTop.mpr fun ε hε => ⟨N, fun n  
↪ hn => by  
33   rw [hN n hn, dist_self]; exact hε⟩⟩
```

4. Доказать, что шары $B[n, 1 + \frac{1}{2n}]$ пространства $\mathbb{N}$ (см. задачу 3) замкнуты и вложены, но не имеют общей точки. Почему это не противоречит принципу вложенных шаров?

Формализация на Lean 4

```

1  /-
2    Section 08, Task 04.
3    Balls  $B[n, 1+1/(2n)]$  in  $(\mathbb{N}, \rho)$  are closed, nested, but have
4     $\hookrightarrow$  empty intersection.
5    The sets  $\{m : m \geq n\}$  are nested and their intersection is
6     $\hookrightarrow$  empty.
7    This shows the nested closed balls theorem requires  $\text{diam} \rightarrow$ 
8     $\hookrightarrow 0$ .
9  -/
10 import Mathlib.Tactic
11
12 /-- The intersection  $\bigcap_n \{m \in \mathbb{N} : m \geq n\}$  is empty: no natural
13  $\hookrightarrow$  number
14  $\hookrightarrow$  is  $\geq$  all natural numbers. -/
15 theorem section08_task04 :  $\bigcap n : \mathbb{N}, \{m : \mathbb{N} \mid n \leq m\} = \emptyset := \text{by}$ 
16   ext m
17   simp only [Set.mem_iInter, Set.mem_setOf_eq,
18      $\hookrightarrow$  Set.mem_empty_iff_false, iff_false]
19   push Not
20   exact ⟨m + 1, by omega⟩

```

5. Привести пример метрического пространства, в котором существует последовательность замкнутых вложенных шаров, не имеющих общей точки, радиусы которых стремятся к нулю.
6. Доказать, что следующие метрические пространства не являются полными, построить их пополнения.

- $C[0, 1], \rho(x, y) = \int_0^1 |x(t) - y(t)| dt;$
- $X = \{x = (\xi_1, \xi_2, \dots) : \xi_k \in \mathbb{C}, \xi_k = 0 \text{ при } k > k_0(x)\},$
 $\rho(x, y) = \max_{k \geq 1} |\xi_k - \eta_k|;$
- $C_0(\mathbb{R}^1) = \{x(t) \in C(\mathbb{R}^1) : x(t) \equiv 0 \text{ при } |t| > t_0(x)\},$
 $\rho(x, y) = \max_{t \in \mathbb{R}} |x(t) - y(t)|;$
- $\mathbb{R}^1, \rho(x, y) = |\arctg x - \arctg y|;$
- $\mathbb{Z}, \rho(p, q) = |e^{ip} - e^{iq}|;$
- $X = \{x \in C^\infty[0, 1] : x^{(k)}(0) = x^{(k)}(1) = 0 \forall k \geq 0\},$
 $\rho(x, y) = \max_{0 \leq t \leq 1} |x(t) - y(t)|;$

g) $C[0, 1], \rho(x, y) = \max_{0 \leq t \leq 1} t|x(t) - y(t)|$.

Указание. Рассмотреть

$$x_n(t) = \begin{cases} n, & \text{если } 0 \leq t \leq \frac{1}{n^2}, \\ \frac{1}{\sqrt{t}}, & \text{если } t > \frac{1}{n^2}. \end{cases}$$

См. также задачу 8.

Формализация на Lean 4

```

1  /-
2  Section 08, Task 06.
3  Incompleteness of  $C[0,1]$  with integral metric,  $\ell^\infty$ ,  $(\mathbb{R},$ 
4     $\hookrightarrow$  arctan metric), etc.
5
6  All these examples follow the same abstract principle:
7  a dense proper subset of a complete metric space is not
8     $\hookrightarrow$  complete.
9  -  $C[0,1] \subset L^1[0,1]$  is dense but proper  $\rightarrow C[0,1]$  with  $L^1$  not
10      $\hookrightarrow$  complete
11  -  $\ell^\infty \subset \ell^\infty$  is dense but proper  $\rightarrow \ell^\infty$  not complete
12  -  $(\mathbb{R}, \arctan) \cong (-\pi/2, \pi/2) \subset \mathbb{R}$ : dense proper open subset,
13      $\hookrightarrow$  not complete
14  -/
15  import FunAn.Section11_NormedSpaces.Task08
16  import Mathlib.Tactic
17
18  /-- Incompleteness of the examples follows from the abstract
19      $\hookrightarrow$  principle
20     proved in S11/Task08: dense proper subset of complete
21      $\hookrightarrow$  space is not complete.
22     This is `not_completeSpace_of_dense_ne_univ`. -/
23  theorem section08_task06_principle
24    {X : Type*} [MetricSpace X] [CompleteSpace X]
25    {S : Set X} (hdense : Dense S) (hne : S  $\neq$  Set.univ) :
26       $\neg$ CompleteSpace S :=
27      not_completeSpace_of_dense_ne_univ hdense hne

```

9. СЕПАРАБЕЛЬНЫЕ МЕТРИЧЕСКИЕ ПРОСТРАНСТВА

КОНТРОЛЬНЫЕ ВОПРОСЫ

1. Дать определение счетного множества, множества мощности континуума. Приведите примеры.
2. Определите плотные, всюду плотные, нигде не плотные множества.
3. Дать определение сепарабельного метрического пространства. Привести примеры.

ЗАДАЧИ

1. Доказать счетность множества финитных числовых последовательностей с рациональными членами.

Формализация на Lean 4

```
1  /-  
2    Section 9, Task 1.  
3    Prove countability of the set of finite sequences with  
4      ↪ rational members.  
5  -/  
6  import Mathlib.Data.Rat.Denumerable  
7  import Mathlib.SetTheory.Cardinal.Basic  
8  import Mathlib.Tactic  
9  
10  /-- The set of finite rational sequences is countable. -/  
11  theorem countable_finite_rat_sequences :  
12    Set.Countable (Set.univ : Set (List ℚ)) :=  
    Set.countable_univ
```

2. Доказать счетность множества многочленов с рациональными коэффициентами.

Формализация на Lean 4

```
1  /-
2    Section 9, Task 2.
3    Prove countability of the set of polynomials with rational
4    ↪ coefficients.
5  -/
6  import Mathlib.Algebra.Polynomial.Basic
7  import Mathlib.Data.Rat.Denumerable
8  import Mathlib.Data.Finsupp.Encodable
9  import Mathlib.Tactic
10
11  -- AddMonoidAlgebra is def-equal to Finsupp, so we transfer
12  ↪ the Countable instance
13
14  instance : Countable (AddMonoidAlgebra ℚ ℕ) :=
15    inferInstanceAs (Countable (ℕ →0 ℚ))
16
17  /-- Polynomial ℚ is countable. -/
18  instance : Countable (Polynomial ℚ) := by
19    have hinj : Function.Injective
20      (Polynomial.toFinsupp : Polynomial ℚ → AddMonoidAlgebra
21        ↪ ℚ ℕ) := fun a b h => by
22      cases a; cases b; exact congr_arg _ h
23      exact hinj.countable
24
25  /-- The set of polynomials with rational coefficients is
26  ↪ countable. -/
27  theorem countable_rat_polynomials :
28    Set.Countable (Set.univ : Set (Polynomial ℚ)) :=
29    Set.countable_univ
```

3. Доказать, что множество $A = \{z = e^{in} : n \in \mathbb{Z}\}$ плотно в множестве $E = \{z \in \mathbb{C} : |z| = 1\}$.

Формализация на Lean 4

```

1  /-
2    Section 9, Task 3.
3     $A = \{e^{in} : n \in \mathbb{Z}\}$  is dense in the unit circle  $\{z \in \mathbb{C} : |z| = 1\}$ .
4    Proof: The map  $n \mapsto e^{in} = e^{i \cdot n}$  gives an orbit of the
5    ↪ irrational
6    rotation by 1 radian on  $S^1$ . Since  $1/(2\pi)$  is irrational, the
7    ↪ orbit
8    is dense by Weyl's equidistribution theorem (or Jacobi's
9    ↪ theorem).
10
11   In Mathlib: multiples of  $a$  on  $\text{AddCircle } p$  are dense iff  $a/p$ 
12   ↪ is irrational.
13   Here  $a = 1$  and  $p = 2\pi$ , so  $1/(2\pi)$  is irrational.
14 -/
15 import Mathlib.Topology.Instances.AddCircle.DenseSubgroup
16 import Mathlib.Analysis.Real.Pi.Irrational
17 import Mathlib.Tactic
18
19 open AddCircle
20
21 /--  $1/(2\pi)$  is irrational (since  $\pi$  is irrational). -/
22 private theorem irrational_one_div_two_pi : Irrational (1 / (2
23   ↪ * Real.pi)) := by
24   rw [one_div, show (2 : ℝ) * Real.pi = Real.pi * 2 from
25     ↪ mul_comm _ _]
26   exact irrational_inv_iff.mpr (irrational_mul_natCast_iff.mpr
27     ↪ (two_ne_zero, irrational_pi))
28
29 /-- The integer multiples of 1 are dense on the circle  $\mathbb{R}/2\pi\mathbb{Z}$ ,
30   i.e.,  $\{e^{in} : n \in \mathbb{Z}\}$  is dense in  $S^1$ . -/
31 theorem exp_in_dense_in_circle :
32   DenseRange (· • (1 : ℝ) : ℤ → AddCircle (2 * Real.pi)) :=
33   ↪ by
34   rw [denseRange_zsmul_coe_iff]
35   exact irrational_one_div_two_pi

```

4. Доказать, что множество

$$B = \{x = (\xi_1, \xi_2, \dots, \xi_n, 0, 0, \dots) : \xi_k \in \mathbb{C}\}$$

нигде не плотно в l^p (n — фиксировано).

Формализация на Lean 4

```

1  /-
2  Section 09, Task 04.
3  B (finite sequences) is nowhere dense in  $\square^p$ .
4
5  Each  $B_n = \{x \in \square^p : x_k = 0 \text{ for } k > n\}$  is a closed proper
6   $\hookrightarrow$  subspace.
7  A proper closed submodule of a topological vector space has
8   $\hookrightarrow$  empty interior,
9  hence is nowhere dense. This is the abstract content behind
10  $\hookrightarrow$  the result.
11
12 In Mathlib: Submodule.eq_top_of_nonempty_interior' shows
13  $\hookrightarrow$  that the only
14 submodule with nonempty interior is  $\top$ .
15 -/
16 import Mathlib.Topology.Algebra.Module.Basic
17 import Mathlib.Analysis.Normed.Field.Basic
18 import Mathlib.Tactic
19
20 open scoped Topology
21
22 /-- A proper submodule of a topological module (over a
23  $\hookrightarrow$  nontrivially normed field)
24 has empty interior, hence is nowhere dense. -/
25 theorem section09_task04
26   { $\square$  : Type*} [NontriviallyNormedField  $\square$ ]
27   {E : Type*} [AddCommGroup E] [Module  $\square$  E]
28    $\hookrightarrow$  [TopologicalSpace E]
29   [IsTopologicalAddGroup E] [ContinuousSMul  $\square$  E]
30   (s : Submodule  $\square$  E) (hs : s  $\neq$   $\top$ ) :
31   interior (s : Set E) =  $\emptyset$  := by
32     haveI : Filter.NeBot ( $\square$ [{x :  $\square$  | IsUnit x}] (0 :  $\square$ )) :=
33       NormedField.nhdsWithin_isUnit_neBot
34     by_contra h
35     apply hs
36     apply s.eq_top_of_nonempty_interior'
37     exact Set.nonempty_iff_ne_empty.mpr h

```

5. Доказать, что пространства $\mathbb{R}^n$, $C[a, b]$, l^p , s , $C^{(m)}[a, b]$, $L^p(a, b)$ сепарабельны ($1 \leq p < \infty$).

Формализация на Lean 4

```

1  /-
2    Section 09, Task 05.
3     $\mathbb{R}^n$ ,  $C[a, b]$ ,  $l^p$ ,  $s$ ,  $C^{(m)}$ ,  $L^p$  are separable.
4
5    We prove separability of  $\mathbb{R}^n$ ,  $\mathbb{R}$ , and  $L^p$  spaces.
6  -/
7  import Mathlib.Topology.Bases
8  import Mathlib.MeasureTheory.Measure.SeparableMeasure
9  import Mathlib.MeasureTheory.Function.LpSpace.Basic
10 import Mathlib.Tactic
11
12 open TopologicalSpace
13
14 /--  $\mathbb{R}$  is separable. -/
15 instance : SeparableSpace  $\mathbb{R}$  := inferInstance
16
17 /--  $\mathbb{R}^n$  is separable (product of separable spaces). -/
18 instance : SeparableSpace (Fin n  $\rightarrow$   $\mathbb{R}$ ) := inferInstance
19
20 /--  $L^p$  is separable (second-countable) when the measure and
21    $\hookrightarrow$  codomain are separable. -/
22 theorem Lp_separable { $\alpha$  : Type*} [MeasurableSpace  $\alpha$ ] { $\mu$  :
23    $\hookrightarrow$  MeasureTheory.Measure  $\alpha$ }
24   [MeasureTheory.IsSeparable  $\mu$ ]
25   {E : Type*} [NormedAddCommGroup E] [NormedSpace  $\mathbb{R}$  E]
26    $\hookrightarrow$  [SeparableSpace E]
27   {p : ENNReal} [Fact (1  $\leq$  p)] [hp : Fact (p  $\neq$   $\top$ )] :
28     SecondCountableTopology (MeasureTheory.Lp E p  $\mu$ ) :=
29     MeasureTheory.Lp.SecondCountableTopology

```

6. Доказать, что l^∞ не является сепарабельным.

Формализация на Lean 4

```

1  /-
2    Section 9, Task 6.
3    Prove that  $\ell^\infty$  is not separable.
4
5    Proof: For each  $S \subseteq \mathbb{N}$ , define  $e_S \in \ell^\infty$  by  $e_S(n) = 1$  if  $n \in S$ , 0 otherwise.
6    For  $S \neq T$ ,  $\|e_S - e_T\|_\infty \geq 1$ , so balls  $B(e_S, 1/3)$  are
7    pairwise disjoint.
8    In a separable space, pairwise disjoint open balls must be
9    countable.
10   But Set  $\mathbb{N}$  is uncountable (Cantor). Contradiction.
11 -/
12 import Mathlib.Analysis.Normed.Lp.lpSpace
13 import Mathlib.SetTheory.Cardinal.Order
14 import Mathlib.SetTheory.Cardinal.Finite
15 import Mathlib.Topology.Bases
16 import Mathlib.Tactic
17
18 open scoped ENNReal
19 open Function
20
21 /--  $\ell^\infty$  is not separable. -/
22 theorem linfty_not_separable :
23   ¬ TopologicalSpace.IsSeparable (Set.univ : Set (lp (fun _
24     ↦ :  $\mathbb{N} \Rightarrow \mathbb{R}$ )  $\mathbb{T}$ )) := by
25   rw [TopologicalSpace.isSeparable_univ_iff]
26   intro hsep
27   -- Construct indicator elements in  $\ell^\infty$ 
28   have hmem :  $\forall S : \text{Set } \mathbb{N}, \text{Mem} \ell^\infty (S.\text{indicator } (\text{fun } _ \Rightarrow (1 : \mathbb{R}))) \mathbb{T} := \text{by}$ 
29     intro S; apply mem $\ell^\infty$ .infty; use 1; rintro _ (i, rfl)
30     exact norm_indicator_le_norm_self (fun _ => (1 :  $\mathbb{R}$ )) i
31     |>.trans (by norm_num)
32   set e :  $\text{Set } \mathbb{N} \rightarrow \text{lp } (\text{fun } _ : \mathbb{N} \Rightarrow \mathbb{R}) \mathbb{T} := \text{fun } S \Rightarrow$ 
33     (S.indicator (fun _ => 1), hmem S)
34   -- For  $S \neq T$ ,  $\text{dist}(e S, e T) \geq 1$ 
35   have hdist :  $\forall S T : \text{Set } \mathbb{N}, S \neq T \rightarrow 1 \leq \text{dist } (e S) (e T) :=$ 
36     by
37       intro S T hST
38       rw [dist_eq_norm, lp.norm_eq_ciSup]
39       simp only [Ne, Set.ext_iff, not_forall] at hST
40       obtain (n, hn) := hST
41       apply le_ciSup_of_le (lp.mem $\ell^\infty$ .p (e S - e T)) n
42       simp only [lp.coeFn_sub, Pi.sub_apply, e]
43       rw [not_iff] at hn
44       rcases Classical.em (n  $\in$  S) with hS | hS <| hS <|> simp_all

```

```

39  -- The balls  $B(e \in S, 1/3)$  are pairwise disjoint nonempty open
     $\hookrightarrow$  sets
40  have hpw : Pairwise (Disjoint on  $\text{fun } S \Rightarrow \text{Metric.ball } (e \in S)$ 
     $\hookrightarrow$   $(1/3 : \mathbb{R})$ ) := by
41    intro S T hST; apply Metric.ball_disjoint_ball; linarith
     $\hookrightarrow$  [hdist S T hST]
42  -- By separability, the index set must be countable
43  have hcount : Countable (Set  $\mathbb{N}$ ) :=
44    hpw.countable_of_isOpen_disjoint ( $\text{fun } _ \Rightarrow$ 
     $\hookrightarrow$  Metric.isOpen_ball)
45    ( $\text{fun } S \Rightarrow$   $\langle e \in S, \text{Metric.mem\_ball\_self } (\text{by norm\_num :}$ 
     $\hookrightarrow$   $(0 : \mathbb{R}) < 1/3) \rangle$ )
46  -- But Set  $\mathbb{N}$  is uncountable (Cantor's theorem)
47  have h1 : Cardinal.mk (Set  $\mathbb{N}$ )  $\leq$  Cardinal.aleph0 :=
     $\hookrightarrow$  Cardinal.mk_le_aleph0
48  rw [Cardinal.mk_set] at h1
49  simp only [Cardinal.aleph0, Cardinal.lift_id] at h1
50  exact (Cardinal.cantor (Cardinal.mk  $\mathbb{N}$ )).2 h1

```

7. Доказать, что пространство $M[a, b]$ ограниченных на $[a, b]$ функций с метрикой $\rho(x, y) = \sup_{a \leq t \leq b} |x(t) - y(t)|$ не является сепарабельным.

Формализация на Lean 4

```

1  /-
2  Section 9, Task 07.
3  The space  $M[a, b]$  of bounded measurable functions on  $[a, b]$ 
4  ↪ with the
5  sup norm is not separable.
6
7  Proof: The family  $\{1_{[a, t]} : t \in (a, b)\}$  has pairwise
8  ↪ sup-distance 1.
9  We formalize the underlying principle: any metric space with
10 ↪ an uncountable
11 family of pairwise  $\varepsilon$ -separated points is not separable.
12 Applied to  $M[a, b]$  with the indicator functions, this gives
13 ↪ the result.
14 -/
15 import Mathlib.Topology.Bases
16 import Mathlib.Tactic
17
18 open TopologicalSpace
19
20 /-- A metric space with an uncountable family of pairwise
21 ↪  $\varepsilon$ -separated
22 points is not separable. This is the key principle behind
23 ↪ the
24 non-separability of  $M[a, b]$  (bounded measurable functions,
25 ↪ sup norm). -/
26 theorem not_separableSpace_of_uncountable_pairwise_dist
27 {X : Type*} [PseudoMetricSpace X]
28 {ι : Type*} (hι : ¬Countable ι)
29 {f : ι → X} {ε : ℝ} (hε : 0 < ε)
30 (hf : Pairwise fun i j => ε ≤ dist (f i) (f j)) :
31   ¬SeparableSpace X := by
32   intro hsep
33   apply hι
34   have hpw : Pairwise (Function.onFun Disjoint fun i =>
35     ↪ Metric.ball (f i) (ε / 3)) := by
36     intro i j hij
37     apply Metric.ball_disjoint_ball
38     linarith [hf hij]
39   have hcount := hpw.countable_of_isOpen_disjoint
40     (fun _ => Metric.isOpen_ball)
41     (fun i => ⟨f i, Metric.mem_ball_self (by linarith)⟩)
42   exact hcount

```

8. Доказать, что совокупность подпоследовательностей последовательности натуральных чисел несчетно.

Формализация на Lean 4

```
1  /-  
2    Section 9, Task 8.  
3    Prove that the power set of  $\mathbb{N}$  is uncountable.  
4  -/  
5  import Mathlib.SetTheory.Cardinal.Order  
6  import Mathlib.Tactic  
7  
8  /-- The power set of  $\mathbb{N}$  is uncountable (Cantor's theorem). -/  
9  theorem uncountable_powerset_nat :  $\neg$  Countable (Set  $\mathbb{N}$ ) := by  
10   intro h  
11   have h1 : Cardinal.mk (Set  $\mathbb{N}$ )  $\leq$  Cardinal.aleph0 :=  
12      $\hookrightarrow$  Cardinal.mk_le_aleph0  
13   have h2 : Cardinal.aleph0 < Cardinal.mk (Set  $\mathbb{N}$ ) := by  
14     rw [Cardinal.mk_set, Cardinal.mk_nat]  
15     exact Cardinal.cantor Cardinal.aleph0  
    exact absurd h1 (not_le.mpr h2)
```

9. Доказать, что мощность сепарабельного пространства не превосходит мощности континуума.

Указание. Использовать задачу 8.

Формализация на Lean 4

```

1  /-
2    Section 09, Task 09.
3    A separable complete metric space has cardinality at most
4      ↪ continuum.
5
6    Proof: A nonempty Polish space X admits a continuous
7      ↪ surjection from  $\mathbb{N} \rightarrow \mathbb{N}$ 
8      (Baire space). Hence  $\#X \leq \#(\mathbb{N} \rightarrow \mathbb{N}) = \aleph_0^{\aleph_0} = 2^{\aleph_0} = \mathfrak{c}$ .
9  -/
10 import Mathlib.Topology.MetricSpace.Polish
11 import Mathlib.SetTheory.Cardinal.Continuum
12 import Mathlib.Tactic
13
14 open Cardinal
15
16 /-- A nonempty Polish space has cardinality at most continuum.
17   ↪ -/
18 theorem polish_card_le_continuum
19   (X : Type) [MetricSpace X] [CompleteSpace X]
20   [TopologicalSpace.SeparableSpace X] [Nonempty X] :
21   Cardinal.mk X ≤ Cardinal.continuum := by
22   -- There exists a continuous surjection  $(\mathbb{N} \rightarrow \mathbb{N}) \rightarrow X$ 
23   obtain ⟨f, _, hf⟩ :=
24     ↪ PolishSpace.exists_nat_nat_continuous_surjective X
25   calc Cardinal.mk X
26     ≤ Cardinal.mk ( $\mathbb{N} \rightarrow \mathbb{N}$ ) := mk_le_of_surjective hf
27     ≤ Cardinal.continuum := by
28       unfold Cardinal.continuum
29       simp only [mk_arrow, mk_nat, Cardinal.lift_id]
30       exact Cardinal.aleph0_power_aleph0.le

```

10. Доказать сепарабельность пространства $l_{\{k\}}^p$ (см. задачу 2a).

Формализация на Lean 4

```

1  /-
2  Section 09, Task 10.
3  Separability of  $\ell^p_{\{k\}}$  (weighted  $\ell^p$  spaces).
4
5  The weighted  $\ell^p$  space with positive weights  $k_i$  is
6  ↪ homeomorphic to  $\ell^p$ 
7  via the map  $(Tx)_i = k_i x_i$ . So it suffices to show  $\ell^p$  is
8  ↪ separable.
9
10 We prove:  $\ell^p(\mathbb{N}, \mathbb{R})$  is separable for  $1 \leq p < \infty$ .
11 The countable dense subset consists of finitely-supported
12 ↪ rational sequences.
13 Density uses lp.hasSum_single (partial sums converge) +
14 ↪ density of  $\mathbb{Q}$  in  $\mathbb{R}$ .
15 -/
16 import Mathlib.Analysis.Normed.Lp.lpSpace
17 import Mathlib.Analysis.Normed.Group.InfiniteSum
18 import Mathlib.Data.Finsupp.Encodable
19 import Mathlib.Topology.Algebra.Order.Archimedean
20 import Mathlib.Tactic
21
22 open scoped ENNReal
23 open TopologicalSpace Finset
24
25 /--  $\ell^p(\mathbb{N}, \mathbb{R})$  is separable for  $1 \leq p < \infty$  (the key fact for
26 ↪ weighted  $\ell^p_k$  separability). -/
27 theorem lp_separable (p : ℝ≥0∞) [hp1 : Fact (1 ≤ p)] (hp : p ≠
28 ↪ ∞) :
29   SeparableSpace (lp (fun _ : ℕ => ℝ) p) := by
30     -- Map:  $\mathbb{N} \rightarrow_0 \mathbb{Q} \rightarrow lp, c \mapsto \sum_{i \in c.support} \text{single}(c_i)$ 
31     apply SeparableSpace.of_denseRange
32     (u := fun (c : ℕ →₀ ℚ) =>
33       ∑ i ∈ c.support, lp.single p i ((c i : ℚ) : ℝ))
34     rw [Metric.denseRange_iff]
35     intro f ε hε
36     -- Partial sums of single vectors converge to f
37     have hconv := (lp.hasSum_single hp f).tendsto_sum_nat
38     rw [Metric.tendsto_atTop] at hconv
39     obtain ⟨N, hN⟩ := hconv (ε / 2) (half_pos hε)
40     -- Approximate each f(i) by a rational
41     have hδ : (0 : ℝ) < ε / (2 * (↑N + 1)) := by positivity
42     choose qf hqf using fun (i : ℕ) =>
43       Rat.denseRange_cast.exists_dist_lt (f i : ℝ) hδ
44     -- Build finsupp: truncate qf to range N
45     set qfN : ℕ → ℚ := fun i => if i < N then qf i else 0
46     set c := Finsupp.onFinset (range N) qfN (fun a ha => by

```

```

41   simp only [qfN] at ha; split_ifs at ha with h; exact
    ↪ mem_range.mpr h; exact absurd rfl ha)
42 refine (c, ?_)
43 -- Show  $\sum_{c.\text{support}} \text{single}(c_i) = \sum_{\text{range } N} \text{single}(qf_i)$ 
44 have hci :  $\forall i \in \text{range } N, (c_i : \mathbb{Q}) = qf_i := \text{fun } i \text{ hi} \Rightarrow$  by
45   simp [c, qfN, mem_range.mp hi]
46 have hsum_eq :  $\sum i \in c.\text{support}, \text{lp.single } p \ i \ ((c_i : \mathbb{Q}) :$ 
    ↪  $\mathbb{R})$ 
47   =  $\sum i \in \text{range } N, \text{lp.single } p \ i \ ((qf_i : \mathbb{Q}) : \mathbb{R}) :=$  by
48   rw [Finset.sum_subset Finsupp.support_onFinset_subset]
49   · apply Finset.sum_congr rfl
50   · intro i hi; simp [hci i hi]
51   · intro i _ hi
52     have :  $c_i = 0 := \text{Finsupp.notMem\_support\_iff.mp } hi$ 
53     simp [this]
54   rw [hsum_eq]
55 -- Triangle inequality
56 have h1 :  $\text{dist } f \ (\sum i \in \text{range } N, \text{lp.single } p \ i \ (f_i : \mathbb{R})) <$ 
    ↪  $\varepsilon / 2 :=$  by
57   rw [dist_comm]; exact hN N le_rfl
58 have hp0 :  $(0 : \mathbb{R}_{\geq 0\infty}) < p := \text{zero\_lt\_one.trans\_le } hp1.out$ 
59 have h2 :  $\text{dist } (\sum i \in \text{range } N, \text{lp.single } p \ i \ (f_i : \mathbb{R}))$ 
60    $(\sum i \in \text{range } N, \text{lp.single } p \ i \ ((qf_i : \mathbb{Q}) : \mathbb{R})) < \varepsilon / 2$ 
    ↪  $:=$  by
61 -- Each  $\text{lp.single}$  is an isometry, so reduce to coordinate
    ↪ distances
62 have hiso :  $\forall i, \text{dist } (\text{lp.single } (E := \text{fun } _ : \mathbb{N} \Rightarrow \mathbb{R}) \ p \ i$ 
    ↪  $(f_i))$ 
63    $(\text{lp.single } p \ i \ ((qf_i : \mathbb{Q}) : \mathbb{R})) = \text{dist } (f_i : \mathbb{R}) \ ((qf$ 
    ↪  $i : \mathbb{Q}) : \mathbb{R}) :=$ 
64    $\text{fun } i \Rightarrow (\text{lp.isometry\_single } (E := \text{fun } _ : \mathbb{N} \Rightarrow \mathbb{R})$ 
    ↪  $i).dist\_eq \_ \_$ 
65 calc dist
66    $\leq \sum i \in \text{range } N, \text{dist } (\text{lp.single } (E := \text{fun } _ : \mathbb{N} \Rightarrow \mathbb{R})$ 
    ↪  $p \ i \ (f_i))$ 
67    $(\text{lp.single } p \ i \ ((qf_i : \mathbb{Q}) : \mathbb{R})) :=$ 
68    $\text{dist\_sum\_sum\_le\_of\_le } (\text{fun } \_ \_ \Rightarrow \text{le\_rfl})$ 
69    $= \sum i \in \text{range } N, \text{dist } (f_i : \mathbb{R}) \ ((qf_i : \mathbb{Q}) : \mathbb{R}) :=$  by
70   simp_rw [hiso]
71    $\leq \sum i \in \text{range } N, (\varepsilon / (2 * (\uparrow N + 1))) :=$ 
72    $\text{sum\_le\_sum } (\text{fun } i \_ \Rightarrow (hqf \ i).le)$ 
73    $= \uparrow N * (\varepsilon / (2 * (\uparrow N + 1))) :=$  by rw [sum_const,
    ↪ card_range, nsmul_eq_mul]
74    $< \varepsilon / 2 :=$  by
75   rcases N.eq_zero_or_pos with rfl | _
76   · simp [hε]
77   · rw [show  $(\uparrow N : \mathbb{R}) * (\varepsilon / (2 * (\uparrow N + 1))) = \varepsilon * \uparrow N$ 
    ↪  $/ (2 * (\uparrow N + 1))$  from by ring]
78   rw [div_lt_div_iff0 (by positivity :  $(0 : \mathbb{R}) < 2 *$ 
    ↪  $(\uparrow N + 1)$ ) two_pos]
79   nlinarith

```

```

80   linarith [dist_triangle f ( $\sum i \in \text{range } N, \text{lp.single } p \ i \ (f \ i$ 
       $\hookrightarrow \ : \mathbb{R}))$ 
81   ( $\sum i \in \text{range } N, \text{lp.single } p \ i \ ((qf \ i : \mathbb{Q}) : \mathbb{R}))]$ 

```

11. Доказать несепарабельность пространства l_∞ (см. задачу 2e).

Формализация на Lean 4

```

1  /-
2    Section 09, Task 11.
3    Non-separability of  $\ell_\infty$  (alternative proof via uncountable
4       $\hookrightarrow$  0-1 sequences).
5
6    This is the same result as Task 06 but stated differently.
7    The 0-1 indicator functions  $\{e_s : S \subseteq \mathbb{N}\}$  satisfy  $\|e_s - e_t\|_\infty =$ 
8       $\hookrightarrow 1$  for  $S \neq T$ .
9    Since  $\mathcal{P}(\mathbb{N})$  is uncountable,  $\ell_\infty$  cannot be separable.
10  -/
11  import FunAn.Section09_Separable.Task06
12  import Mathlib.Tactic
13
14  open scoped ENNReal
15
16  /--  $\ell_\infty$  is not separable (via uncountable pairwise-distant
17    family).
18    Direct consequence of Task06's linfty_not_separable. -/
19  theorem linfty_not_separable_alt :
20     $\neg$  TopologicalSpace.SeparableSpace (lp (fun _ :  $\mathbb{N} \Rightarrow \mathbb{R}$ )  $\top$ )
21     $\hookrightarrow$  := by
22    intro h
23    exact linfty_not_separable
24     $\hookrightarrow$  (TopologicalSpace.isSeparable_univ_iff.mpr h)

```

10. ПРИНЦИП СЖИМАЮЩИХ ОТОБРАЖЕНИЙ

КОНТРОЛЬНЫЕ ВОПРОСЫ

1. Дать определения непрерывного оператора, сжимающего оператора, неподвижной точки оператора.
2. Сформулировать принцип сжимающих отображений и его обобщение.
3. Какое уравнение называется интегральным уравнением Вольтерра? Фредгольма?

ЗАДАЧИ

1. Являются ли непрерывными операторы:
 - a) $(Ax)(t) = \int_0^t x^2(s) ds, A : C[0, 1] \rightarrow C[0, 1];$
 - b) $(Ax)(t) = x'(t), A : C^{(1)}[0, 2] \rightarrow C[0, 2];$
 - c) $(Ax)(t) = x'(t), A : C^{(1)}[0, 2] \rightarrow C[0, 2],$
причём расстояние в $C^{(1)}[0, 2]$ определено так же, как в $C[0, 2]$.

2. Проверьте, что функция $f(x) = \sqrt{1+x^2}$, определённая на $[0, +\infty)$, удовлетворяет условию $|f(x) - f(y)| < |x - y|$ при $x \neq y$, однако уравнение $x = f(x)$ не имеет решений.

Формализация на Lean 4

```

1  /-
2    Section 10, Task 2.
3     $f(x) = \sqrt{1+x^2}$  satisfies  $|f(x) - f(y)| < |x - y|$  for  $x \neq$ 
4       $y$ ,
5    yet  $f$  has no fixed point.
6    Proof of no fixed point:  $f(x) = x$  means  $\sqrt{1+x^2} = x$ , so  $1+x^2$ 
7       $= x^2$ ,
8    giving  $1 = 0$  – contradiction.
9    This doesn't contradict Banach because the contraction
10      $\hookrightarrow$  constant is not
11     strictly less than 1 uniformly ( $\sup |f'(x)| = 1$ ).
12  -/
13  import Mathlib.Tactic
14
15  /--  $\sqrt{1+x^2}$  has no fixed point:  $\sqrt{1+x^2} = x$  implies  $1+x^2 = x^2$ ,
16      $\hookrightarrow$  contradiction. -/
17  theorem sqrt_one_plus_sq_no_fixed_point :  $\neg \exists x : \mathbb{R}, \text{Real.sqrt}$ 
18      $(1 + x^2) = x$  := by
19    push_neg; intro x
20    by_cases hx :  $x \leq 0$ 
21    · --  $\sqrt{1+x^2} > 0 \geq x$ 
22      exact ne_of_gt (lt_of_le_of_lt hx (Real.sqrt_pos_of_pos
23         $\hookrightarrow$  (by positivity)))
24    · --  $\sqrt{1+x^2} > x$ : square both sides
25      push_neg at hx
26      intro h
27      have h1 :  $0 \leq \text{Real.sqrt} (1 + x^2)$  := Real.sqrt_nonneg _
28      have h2 := Real.sq_sqrt (by positivity :  $(0:\mathbb{R}) \leq 1 + x^2$ 
29         $\hookrightarrow$  2)
30      nlinarith [h  $\triangleright$  h2]

```

3. Покажите, что отображение $f(x) = \frac{x^2+2}{2x}$ является сжимающим на отрезке $[1, 2]$. Вычислите несколько итераций по формуле $x_0 = 1, x_{k+1} = f(x_k), k = 0, 1, \dots$, и сравните их с табличным значением $\sqrt{2}$ (решение уравнения $x = f(x)$, лежащее на отрезке $[1, 2]$).

Формализация на Lean 4

```

1  /-
2    Section 10, Task 3.
3    Show that  $f(x) = (x^2 + 2) / (2x)$  is a contraction on  $[1, 2]$ .
4  -/
5  import Mathlib.Analysis.SpecificLimits.Basic
6  import Mathlib.Tactic
7
8  theorem contraction_sqrt2 (x y : ℝ) (hx : x ∈ Set.Icc 1 2) (hy
9    ↪ : y ∈ Set.Icc 1 2) :
10    |((x ^ 2 + 2) / (2 * x)) - ((y ^ 2 + 2) / (2 * y))| ≤ (1 /
11      ↪ 2) * |x - y| := by
12    have hx0 : (0 : ℝ) < x := by linarith [hx.1]
13    have hy0 : (0 : ℝ) < y := by linarith [hy.1]
14    have h2xy : (0 : ℝ) < 2 * x * y := by positivity
15    have key : (x ^ 2 + 2) / (2 * x) - (y ^ 2 + 2) / (2 * y) =
16      (x - y) * (x * y - 2) / (2 * x * y) := by field_simp;
17      ↪ ring
18    rw [key]
19    suffices h : |x * y - 2| ≤ x * y by
20      rw [abs_div, abs_mul, abs_of_pos h2xy]
21      rw [div_le_iff₀ h2xy]
22      calc |x - y| * |x * y - 2|
23        ≤ |x - y| * (x * y) := mul_le_mul_of_nonneg_left h
24        ↪ (abs_nonneg _)
25        = 1 / 2 * |x - y| * (2 * x * y) := by ring
26    rw [abs_le]
27    exact ⟨by nlinarith [hx.1, hy.1], by nlinarith [hx.2, hy.2]⟩

```

4. Пусть $f(x) = \frac{\pi}{2} + x - \operatorname{arctg} x$, $x \in \mathbb{R}^1$. Доказать, что для любых $x, y \in \mathbb{R}^1$ найдется постоянная $\alpha < 1$ такая, что

$$|f(x) - f(y)| \leq \alpha |x - y|,$$

но f не имеет неподвижных точек. Почему это не противоречит принципу сжимающих отображений?

Формализация на Lean 4

```

1  /-
2  Section 10, Task 4.
3  f(x) = π/2 + x - arctan(x) has no fixed point on ℝ.
4  f(x) = x requires arctan(x) = π/2, but arctan(x) < π/2 for
   ↪ all x.
5  This does not contradict Banach's theorem because (ℝ, f) is
   ↪ not
6  a complete metric space with f as a contraction (f maps ℝ to
   ↪ ℝ
7  but the fixed-point-free property shows it's not a
   ↪ contraction on ℝ).
8  -/
9  import Mathlib.Analysis.SpecialFunctions.Trigonometric.Arctan
10 import Mathlib.Tactic
11
12 /-- arctan(x) < π/2 for all x, so f(x) = π/2 + x - arctan(x) >
   ↪ x,
13     hence f has no fixed point. -/
14 theorem contraction_no_fixed_point_arctan (x : ℝ) :
15     Real.arctan x < Real.pi / 2 :=
16     Real.arctan_lt_pi_div_two x

```

5. При каких λ и β отображение $(Ax)(t) = \lambda \cdot x(t^\beta)$ является сжимающим в $C[0, 1]$? В $L^2(0, 1)$?

6. Доказать, что если $\sup_{j \geq 1} \sum_{i=1}^{\infty} |a_{ij}| < 1$, то бесконечная система линейных алгебраических уравнений

$$x_i = \sum_{j=1}^{\infty} a_{ij} x_j + b_i \quad (i = 1, 2, \dots)$$

имеет единственное решение $x = (x_1, x_2, \dots) \in l^1$ для любой последовательности $b = (b_1, b_2, \dots) \in l^1$.

Формализация на Lean 4

```

1  /-
2  Section 10, Task 6.
3  If  $\sup_j \sum_i |a_{ij}| < 1$ , the infinite system  $x_i = \sum_j a_{ij} x_j +$ 
4   $\hookrightarrow b_i$  has a
5  unique solution  $x \in \square^1$  for every  $b \in \square^1$ .
6
7  Proof: The operator  $T(x)_i = \sum_j a_{ij} x_j + b_i$  is a contraction
8   $\hookrightarrow$  on  $\square^1$ 
9  (the column-sum condition gives  $\|T\|_{\square^1 \rightarrow \square^1} < 1$ ), so Banach
10  $\hookrightarrow$  FPT applies.
11
12 We formalize: a contraction on  $\square^1$  has a unique fixed point.
13 -/
14
15 import Mathlib.Analysis.Normed.Lp.lpSpace
16 import Mathlib.Topology.MetricSpace.Contracting
17 import Mathlib.Tactic
18
19 open scoped ENNReal
20
21 /-- A contraction on  $\square^1(\mathbb{N}, \mathbb{R})$  has a unique fixed point.
22 Applied to Task 6: the column-sum condition  $\sup_j \sum_i |a_{ij}|$ 
23  $\hookrightarrow < 1$ 
24 makes the operator a contraction on  $\square^1$ , giving unique
25  $\hookrightarrow$  solution. -/
26
27 theorem contraction_l1_unique_fixedPoint
28   {K : ENNReal} {f : lp (fun _ : ℕ => ℝ) 1 → lp (fun _ : ℕ =>
29    $\hookrightarrow$  ℝ) 1}
30   (hf : ContractingWith K f) [Nonempty (lp (fun _ : ℕ => ℝ)
31    $\hookrightarrow$  1)] :
32   ∃! x, f x = x :=
33   ⟨hf.fixedPoint f, hf.fixedPoint_isFixedPt,
34   fun _ hy => hf.fixedPoint_unique hy⟩

```

7. Доказать, что если $\sup_{i \geq 1} \sum_{j=1}^{\infty} |a_{ij}| < 1$, то утверждение задачи 6 верно в пространстве l^{∞} .

Формализация на Lean 4

```

1  /-
2  Section 10, Task 7.
3  If  $\sup_i \sum_j |a_{ij}| < 1$ , the infinite system has a unique
   ↪ solution in  $\ell^{\infty}$ .
4
5  Proof: The row-sum condition gives  $\|T\|_{\ell^{\infty} \rightarrow \ell^{\infty}} < 1$ , making  $T$ 
   ↪ a
6  contraction on  $\ell^{\infty}$ . Banach fixed point theorem applies.
7
8  We formalize: a contraction on  $\ell^{\infty}(\mathbb{N}, \mathbb{R})$  has a unique fixed
   ↪ point.
9  -/
10 import Mathlib.Analysis.Normed.Lp.lpSpace
11 import Mathlib.Topology.MetricSpace.Contracting
12 import Mathlib.Tactic
13
14 open scoped ENNReal
15
16 /-- A contraction on  $\ell^{\infty}(\mathbb{N}, \mathbb{R})$  has a unique fixed point.
17 Applied to Task 7: the row-sum condition  $\sup_i \sum_j |a_{ij}| <
   ↪ 1$ 
18 makes the operator a contraction on  $\ell^{\infty}$ , giving unique
   ↪ solution. -/
19 theorem contraction_linfty_unique_fixedPoint
20 {K : ENNReal} {f : lp (fun _ : ℕ => ℝ) ∞ → lp (fun _ : ℕ =>
   ↪ ℝ) ∞}
21 (hf : ContractingWith K f) [Nonempty (lp (fun _ : ℕ => ℝ)
   ↪ ∞)] :
22   ∃! x, f x = x :=
23   ⟨hf.fixedPoint f, hf.fixedPoint_isFixedPt,
24     fun _ hy => hf.fixedPoint_unique hy⟩

```

8. Проверить, что интегральное уравнение Вольтерра

$$x(t) = \int_0^t K(t, s)x(s) ds,$$

где

$$K(t, s) = \begin{cases} s \cdot e^{-1+1/t^2}, & \text{если } 0 \leq s \leq t \cdot e^{1-1/t^2}, \\ t, & \text{если } t \cdot e^{1-1/t^2} \leq s \leq t \leq 1, \end{cases}$$

имеет бесконечно много решений $x(t) = \frac{c}{t}$. Почему это не противоречит теореме о существовании и единственности решения уравнения Вольтерра?

Формализация на Lean 4

```

1  /-
2  Section 10, Task 8.
3  The Volterra equation  $x(t) = \int_0^t K(t,s)x(s)ds$  can have
4  ↪ infinitely
5  many solutions when  $K$  is not continuous (Lipschitz condition
6  ↪ fails).
7
8  Key point: without the contraction condition, fixed points
9  ↪ need not
10 be unique. If a linear operator  $T$  has two distinct fixed
11 ↪ points  $x_1, x_2$ ,
12 then  $x_2 + c \cdot (x_1 - x_2)$  is a fixed point for every  $c \in \mathbb{R}$ .
13 The Volterra equation with  $K(t,s) = 1/t$  has solutions  $x(t) =$ 
14 ↪  $c/t$  for all  $c$ ;
15 the operator is not a contraction, so Banach FPT does not
16 ↪ apply.
17 -/
18 import Mathlib.Tactic
19
20 /-- If a linear operator  $T$  has two distinct fixed points  $x_1 \neq$ 
21 ↪  $x_2$ ,
22 then  $T(x_2 + c \cdot (x_1 - x_2)) = x_2 + c \cdot (x_1 - x_2)$  for all  $c \in \mathbb{R}$ ,
23 giving infinitely many fixed points along the line through
24 ↪  $x_1, x_2$ . -/
25 theorem infinitely_many_fixed_points_of_linear
26 {E : Type*} [AddCommGroup E] [Module ℝ E]
27 {T : E → [ℝ] E} {x1 x2 : E}
28 (hfix1 : T x1 = x1) (hfix2 : T x2 = x2) :
29   ∀ c : ℝ, T (x2 + c • (x1 - x2)) = x2 + c • (x1 - x2) := by
30   intro c
31   simp [map_add, map_smul, map_sub, hfix1, hfix2]

```

9. При каких λ применим принцип сжимающих отображений в пространствах $L^2(a, b)$ и $C[a, b]$ к уравнению Фредгольма 2-го рода

$$x(t) = \lambda \int_a^b K(t, s)x(s) ds + y(t)?$$

Найти точное решение уравнения. Для нахождения приближенного решения составить программу. В программе предусмотреть:

- а) табулирование точного решения с шагом $h = \frac{b-a}{2n}$;
- б) вычисление номера L итерации, обеспечивающей точность 0,01 в метрике $C[a, b]$, когда за первое приближение берется $x_1(t) \equiv y(t)$. Заметим, что L определяется из неравенства

$$\frac{\theta^{L-1}}{1-\theta} \cdot \rho(x_1, x_2) < 0,01,$$

где θ — коэффициент сжатия;

- с) сравнение точного решения с приближенным, т.е. вычисление величины $R = \max_{0 \leq k \leq 2n} |x(t_k) - \bar{x}(t_k)|$, где $x(t_k)$ — значения точного решения, $\bar{x}(t_k)$ — значения приближенного решения.

Для вычисления $\bar{x}(t_k)$ разбиваем промежуток $[a, b]$ на $2n$ частей с помощью точек $t_k = a + kh$. Тогда

$$x(t_k) = \lambda \int_a^b K(t_k, s)x(s) ds + y(t_k), \quad k = 0, 1, \dots, 2n.$$

С помощью формулы Симпсона заменим интегралы на приближенные значения. Получим:

$$x(t_k) = \frac{\lambda h}{3} \left[K(t_k, s_0)x(s_0) + K(t_k, s_{2n})x(s_{2n}) + 4 \sum_{m=1}^n K(t_k, s_{2m-1})x(s_{2m-1}) + 2 \sum_{m=1}^{n-1} K(t_k, s_{2m})x(s_{2m}) \right] + y(t_k),$$

$k = 0, 1, \dots, 2n$. Полученная система линейных алгебраических уравнений решается методом итераций.

Задания приведены в таблице 2. В отчете показать совокупность значений λ , допускающих применение принципа сжатых отображений в пространствах $L^2[a, b]$, $C[a, b]$, точное решение уравнения, значения чисел L и R , а также привести программу и заполненную таблицу 1.

Таблица 1.

k	0	1	2	...	$2n$
$x(t_k)$					
$\bar{x}(t_k)$					

11. НОРМИРОВАННЫЕ ПРОСТРАНСТВА

КОНТРОЛЬНЫЕ ВОПРОСЫ

1. Дать определения линейного пространства, нормы, нормированного пространства, банахова пространства.

Таблица 2.

№	$K(t, s)$	$y(t)$	a	b	λ	n
1.	$t^2 s$	$\cos 3t$	0	1	1	8
2.	$t^2 s$	$\cos 3t$	0	1	1	10
3.	ts	e^t	-1	1	0,5	10
4.	ts	e^t	0,5	1,5	0,5	10
5.	$t^2 s^2$	$\frac{\sin t}{t}$	0,3	1,2	0,6	9
6.	e^{t-s}	1	0,1	0,9	0,5	8
7.	e^{t-s}	t	0,1	0,8	0,5	7
8.	$\cos \pi(t-s)$	1	0	1	0,5	10
9.	$\cos t \sin s$	1	0	2	0,25	10
10.	$\sin t \cos s$	1	-0,5	0,5	0,25	5
11.	$e^{2t} \cos s$	$\sin t$	0	0,5	0,6	5
12.	$e^{2t} \sin s$	$\cos t$	0	0,5	0,4	10
13.	$e^t \cos s$	$2 \sin t$	0	0,9	0,5	8
14.	$e^t \sin s$	$2 \cos t$	0	0,8	0,5	8
15.	$t \sin s$	e^t	0,2	0,8	0,4	6
16.	$t \cos s$	e^{-t}	0,1	0,9	0,3	8
17.	$t^2 e^s$	t	-2	-1	2	6

2. Дать определение эквивалентных норм.
3. Какая система элементов линейного пространства называется линейно независимой? Дать определение размерности линейного пространства. Привести примеры.

ЗАДАЧИ

1. Доказать, что аксиома треугольника в определении нормы эквивалентна выпуклости единичного шара.

Формализация на Lean 4

```

1  /-
2  Section 11, Task 1.
3  The triangle inequality  $\|x+y\| \leq \|x\| + \|y\|$  is equivalent to
4  convexity of the closed unit ball  $\{x : \|x\| \leq 1\}$ .
5
6  Forward: triangle inequality  $\Rightarrow$  convexity (the interesting
7   $\hookrightarrow$  direction).
8  Reverse: convexity  $\Rightarrow$  triangle inequality (in Lean, the
9   $\hookrightarrow$  triangle
10 inequality is an axiom of NormedAddCommGroup, so this is
11  $\hookrightarrow$  definitional;
12 the mathematical content would be the Minkowski functional
13  $\hookrightarrow$  construction).
14 -/
15 import Mathlib.Analysis.Convex.Body
16 import Mathlib.Analysis.Normed.Group.Basic
17 import Mathlib.Tactic
18
19 variable {E : Type*} [SeminormedAddCommGroup E] [NormedSpace ℝ
20  $\hookrightarrow$  E]
21
22 /-- Forward: the closed unit ball of a normed space is convex.
23 This is the meaningful direction: triangle inequality  $\Rightarrow$ 
24  $\hookrightarrow$  convexity. -/
25 theorem closedBall_zero_one_convex :
26   Convex ℝ (Metric.closedBall (0 : E) 1) :=
27   convex_closedBall 0 1
28
29 /-- Reverse: the triangle inequality holds in a normed space.
30 (In Lean this is definitional; the math content is the
31  $\hookrightarrow$  Minkowski functional.) -/
32 theorem norm_triangle_inequality (x y : E) :  $\|x + y\| \leq \|x\| +$ 
33  $\hookrightarrow \|y\| :=$ 
34   norm_add_le x y
35
36 /-- The full equivalence: normed space has convex unit ball
37  $\hookrightarrow$  iff triangle ineq. -/
38 theorem convex_ball_iff_triangle :
39   Convex ℝ (Metric.closedBall (0 : E) 1)  $\leftrightarrow \forall x y : E, \|x +$ 
40  $\hookrightarrow y\| \leq \|x\| + \|y\| :=$ 
41   { fun _ => norm_add_le, fun _ => convex_closedBall 0 1 }

```

2. Можно ли в пространстве s ввести норму, определяющую известную метрику?

3. Доказать, что нормированное пространство X является банаховым пространством тогда и только тогда, когда всякий ряд $\sum_{k=1}^{\infty} x_k$, для которого $\sum_{k=1}^{\infty} \|x_k\| < \infty$, сходится в X .

Формализация на Lean 4

```

1  /-
2    Section 11, Task 3.
3    Prove that a normed space  $X$  is Banach iff every absolutely
4       $\hookrightarrow$  convergent series converges.
5  -/
6  import Mathlib.Analysis.Normed.Group.Completeness
7  import Mathlib.Tactic
8
9  variable {X : Type*} [NormedAddCommGroup X]
10
11  /-- A normed group is complete iff every absolutely convergent
12       $\hookrightarrow$  series has
13      convergent partial sums. This is
14
15       $\hookrightarrow$  `NormedAddCommGroup.summable_imp_tendsto_iff_completeSpace`
16       $\hookrightarrow$  in Mathlib. -/
17  theorem banach_iff_abs_convergent :
18    (∀ (f : ℕ → X), Summable (||f ·||) →
19      ∃ a, Filter.Tendsto (fun n => ∑ i ∈ Finset.range n, f i)
20        ↪ Filter.atTop (nhds a)) ↔
21    CompleteSpace X :=
22    NormedAddCommGroup.summable_imp_tendsto_iff_completeSpace

```

4. Доказать, что $l^p \subset l^q$ при $p < q$.

Формализация на Lean 4

```
1  /-
2    Section 11, Task 4.
3    Prove that  $\ell^p \subset \ell^q$  when  $p < q$ .
4    (In Mathlib's convention, larger exponent means weaker
5      $\hookrightarrow$  condition,
6     so  $\ell^p \subset \ell^q$  when  $p \leq q$ .)
7  -/
8  import Mathlib.Analysis.Normed.Lp.lpSpace
9  import Mathlib.Tactic
10
11  open scoped ENNReal
12
13  /- -  $\ell^p \subseteq \ell^q$  when  $p \leq q$ . -/
14  theorem lp_subset_lq {p q :  $\mathbb{R}_{\geq 0\infty}$ } (hpq : p ≤ q)
15    (f :  $\ell^p$  (fun _ :  $\mathbb{N} \Rightarrow \mathbb{R}$ ) p) : Mem  $\ell^q$  (↑f :  $\mathbb{N} \rightarrow \mathbb{R}$ ) q :=
    lp.monotone hpq f.prop
```

5. Доказать, что $L^p(a, b) \supset L^q(a, b)$ при $p < q$.

Формализация на Lean 4

```

1  /-
2    Section 11, Task 5.
3    For  $1 \leq p < q \leq \text{infty}$ ,  $L_q(a, b)$  is a subset of  $L_p(a, b)$  on a
       $\hookrightarrow$  finite measure space.
4    Proof: Holder's inequality with exponents  $q/p$  and  $q/(q-p)$ 
       $\hookrightarrow$  gives
5    the norm bound. In Mathlib this is
       $\hookrightarrow$  MeasureTheory.MemLp.mono_exponent.
6  -/
7  import Mathlib.MeasureTheory.Function.LpSeminorm.CompareExp
8  import Mathlib.Tactic
9
10 open MeasureTheory
11
12 /- On a finite measure space,  $L_q$  is a subset of  $L_p$  when  $p \leq q$ . -/
13 theorem Lq_subset_Lp_finite_measure
14   { $\alpha$  : Type*} [MeasurableSpace  $\alpha$ ] { $\mu$  : Measure  $\alpha$ }
15    $\hookrightarrow$  [IsFiniteMeasure  $\mu$ ]
16   {f :  $\alpha \rightarrow \mathbb{R}$ } {p q : ENNReal} (hpq :  $p \leq q$ ) (hf : MemLp f q
17    $\hookrightarrow$   $\mu$ ) :
18     MemLp f p  $\mu$  :=
19     hf.mono_exponent hpq

```

6. Можно ли в $C^{(1)}[a, b]$ ввести нормы по формулам:

- a) $\|x\| = \max_{t \in [a, b]} |x'(t)|$;
- b) $\|x\| = |x(b) - x(a)| + \max_{t \in [a, b]} |x'(t)|$;
- c) $\|x\| = |x(a)| + \max_{a \leq t \leq b} |x'(t)|$;
- d) $\|x\| = \int_a^b |x(t)| dt + \max_{a \leq t \leq b} |x'(t)|$.

7. Доказать, что нормы $\|x\| = \max_{a \leq t \leq b} |x(t)|$ и $\|x\|_1 = \int_a^b |x(t)| dt$ не эквивалентны на $C[a, b]$.

Формализация на Lean 4

```

1  /-
2    Section 11, Task 7.
3    Prove that the sup norm and  $L^1$  norm are not equivalent on
4       $\hookrightarrow C[0,1]$ .
5
6    Proof: for  $f(t) = t^n$ :  $\|f\|_{\text{sup}} = 1$ ,  $\|f\|_{L1} = 1/(n+1) \rightarrow 0$ .
7    So no  $C > 0$  satisfies  $1 \leq C/(n+1)$  for all  $n$ .
8  -/
9  import Mathlib.Tactic
10 import Mathlib.Analysis.SpecificLimits.Basic
11
12 /-- No constant  $C$  satisfies  $1 \leq C / (n+1)$  for all  $n \in \mathbb{N}$ . -/
13 theorem no_constant_bound :
14    $\neg \exists (C : \mathbb{R}), 0 < C \wedge \forall (n : \mathbb{N}), (1 : \mathbb{R}) \leq C / (\uparrow n + 1) :=$ 
15    $\hookrightarrow$  by
16     intro ⟨C, hC, h⟩
17     have h_ceil := h [C]_+
18     have : ( $\uparrow [C]_+ : \mathbb{R}$ ) + 1 > 0 := by positivity
19     rw [le_div_iff₀ this] at h_ceil
20     linarith [Nat.le_ceil C]

```

8. Доказать, что $C[a, b]$ с нормой $\|x\| = \int_a^b |x(t)| dt$ не полно.

Формализация на Lean 4

```

1  /-
2  Section 11, Task 08.
3  C[a,b] with the integral norm  $\|f\| = \int |f(x)| dx$  is not
    $\hookrightarrow$  complete.
4
5  Abstract proof: C[a,b] embeds as a dense proper subspace of
    $\hookrightarrow L^1[a,b]$ .
6  A dense proper subset of a complete metric space cannot be
    $\hookrightarrow$  complete:
7  if it were, it would be closed; closed + dense = whole
    $\hookrightarrow$  space,
8  contradicting "proper".
9  -/
10 import Mathlib.Topology.MetricSpace.Basic
11 import Mathlib.Tactic
12
13 /-- A dense proper subset of a complete metric space is not
    $\hookrightarrow$  complete.
14 Applied to  $C[a,b] \subset L^1[a,b]$ : continuous functions are
    $\hookrightarrow L^1$ -dense
15 in  $L^1$  but not all of  $L^1$ , so  $C[a,b]$  with  $L^1$  norm is not
    $\hookrightarrow$  complete. -/
16 theorem not_completeSpace_of_dense_ne_univ
17   {X : Type*} [MetricSpace X] [CompleteSpace X]
18   {S : Set X} (hdense : Dense S) (hne : S  $\neq$  Set.univ) :
19    $\neg$ CompleteSpace S := by
20   intro hS
21   apply hne
22   have hclosed : IsClosed S :=
23      $\hookrightarrow$  (completeSpace_coe_iff_isComplete.mp hS).isClosed
24   calc S = closure S := hclosed.closure_eq.symm
       _ = Set.univ := hdense.closure_eq

```

9. Показать, что нормы $\max(\|x_1\|, \|x_2\|)$, $\|x_1\| + \|x_2\|$, $\sqrt{\|x_1\|^2 + \|x_2\|^2}$ эквивалентны на $E \times E$, где $\|\cdot\|$ — норма в E .

Формализация на Lean 4

```

1  /-
2    Section 11, Task 9.
3    Show that the norms  $\max(\|x_1\|, \|x_2\|)$ ,  $\|x_1\| + \|x_2\|$ ,
4     $\sqrt{\|x_1\|^2 + \|x_2\|^2}$ 
5    are equivalent on  $E \times E$ .
6  -/
7  import Mathlib.Analysis.Normed.Lp.ProdLp
8  import Mathlib.Tactic
9
10 variable {E : Type*} [NormedAddCommGroup E]
11
12 /-- max norm ≤ sum norm. -/
13 theorem max_le_sum (a b : E) : max ||a|| ||b|| ≤ ||a|| + ||b|| :=
14   max_le (le_add_of_nonneg_right (norm_nonneg b))
15   ↪ (le_add_of_nonneg_left (norm_nonneg a))
16
17 /-- sum norm ≤ 2 * max norm. -/
18 theorem sum_le_two_max (a b : E) : ||a|| + ||b|| ≤ 2 * max ||a|| ||b||
19   := by
20     linarith [le_max_left ||a|| ||b||, le_max_right ||a|| ||b||]
21
22 /-- max norm ≤ Euclidean norm:  $\max(\|a\|, \|b\|) \leq \sqrt{\|a\|^2 + \|b\|^2}$ . -/
23 theorem max_le_euclidean (a b : E) :
24   max ||a|| ||b|| ≤ Real.sqrt (||a|| ^ 2 + ||b|| ^ 2) := by
25     apply max_le
26     · calc ||a|| = Real.sqrt (||a|| ^ 2) := (Real.sqrt_sq
27       ↪ (norm_nonneg a)).symm
28       _ ≤ Real.sqrt (||a|| ^ 2 + ||b|| ^ 2) :=
29         Real.sqrt_le_sqrt (by nlinarith [sq_nonneg ||b||])
30     · calc ||b|| = Real.sqrt (||b|| ^ 2) := (Real.sqrt_sq
31       ↪ (norm_nonneg b)).symm
32       _ ≤ Real.sqrt (||a|| ^ 2 + ||b|| ^ 2) :=
33         Real.sqrt_le_sqrt (by nlinarith [sq_nonneg ||a||])
34
35 /-- Euclidean norm ≤ sum norm:  $\sqrt{\|a\|^2 + \|b\|^2} \leq \|a\| + \|b\|$ . -/
36 theorem euclidean_le_sum (a b : E) :
37   Real.sqrt (||a|| ^ 2 + ||b|| ^ 2) ≤ ||a|| + ||b|| := by
38     rw [← Real.sqrt_sq (by positivity : 0 ≤ ||a|| + ||b||)]
39     exact Real.sqrt_le_sqrt (by nlinarith [mul_nonneg
40       ↪ (norm_nonneg a) (norm_nonneg b)])
41
42 /-- Euclidean norm ≤  $\sqrt{2} \cdot \max$  norm. -/
43 theorem euclidean_le_sqrt2_max (a b : E) :
44   Real.sqrt (||a|| ^ 2 + ||b|| ^ 2) ≤ Real.sqrt 2 * max ||a|| ||b||
45     := by

```

```

39   rw [← Real.sqrt_sq (by positivity : 0 ≤ max ‖a‖ ‖b‖),
40       ← Real.sqrt_mul (by norm_num : (0:ℝ) ≤ 2)]
41   apply Real.sqrt_le_sqrt
42   have h1 : ‖a‖ ^ 2 ≤ max ‖a‖ ‖b‖ ^ 2 :=
43     pow_le_pow_left₀ (norm_nonneg a) (le_max_left ‖a‖ ‖b‖) 2
44   have h2 : ‖b‖ ^ 2 ≤ max ‖a‖ ‖b‖ ^ 2 :=
45     pow_le_pow_left₀ (norm_nonneg b) (le_max_right ‖a‖ ‖b‖) 2
46   linarith

```

10. Доказать, что конечномерное подпространство нормированного пространства всегда замкнуто.

Формализация на Lean 4

```
1  /-  
2    Section 11, Task 10.  
3    Prove that every finite-dimensional subspace of a normed  
4      ↪ space is closed.  
5  -/  
6  import Mathlib.Analysis.Normed.Module.FiniteDimension  
7  import Mathlib.Tactic  
8  
9  variable {E : Type*} [NormedAddCommGroup E] [NormedSpace ℝ E]  
10  
11 /-- Every finite-dimensional subspace of a normed space is  
12    ↪ closed. -/  
13 theorem finiteDimensional_subspace_isClosed (S : Submodule ℝ  
    ↪ E)  
    [FiniteDimensional ℝ S] : IsClosed (S : Set E) :=  
    S.closed_of_finiteDimensional
```

11. Если L — подпространство нормированного пространства E , то для любого $\alpha < 1$ существует $y \in E$, $\|y\| = 1$, такой, что $\rho(y, L) > \alpha$. Доказать.

Формализация на Lean 4

```

1  /-
2    Section 11, Task 11.
3    Riesz's lemma: For a closed proper subspace  $F$  of a normed
4       $\hookrightarrow$  space  $E$ ,
5      for any  $r < 1$ , there exists  $x_0$  with  $\|x_0\| = 1$  and  $\text{dist}(x_0, F)$ 
6       $\hookrightarrow \geq r$ .
7  -/
8
9  import Mathlib.Analysis.Normed.Module.RieszLemma
10 import Mathlib.Tactic
11
12 variable { $\alpha$   $E$  : Type*} [NontriviallyNormedField  $\alpha$ ]
13   [NormedAddCommGroup  $E$ ] [NormedSpace  $\alpha$   $E$ ]
14
15 /-- Riesz's lemma from Mathlib. -/
16 theorem riesz_lemma_statement (F : Subspace  $\alpha$   $E$ ) (hFc :
   $\hookrightarrow$  IsClosed (F : Set  $E$ ))
  (hF :  $\exists x : E, x \notin F$ ) {r :  $\mathbb{R}$ } (hr : r < 1) :
   $\exists x_0 : E, x_0 \notin F \wedge \forall y \in F, r * \|x_0\| \leq \|x_0 - y\| :=$ 
  riesz_lemma hFc hF hr

```

12. В бесконечномерном нормированном пространстве существует бесконечное ограниченное множество, элементы которого удалены друг от друга на расстояние $> 0,5$. Доказать.

Указание. Использовать задачу 11.

Формализация на Lean 4

```

1  /-
2    Section 11, Task 12.
3    In an infinite-dimensional normed space, there exists an
4       $\hookrightarrow$  infinite bounded set
5      with pairwise distances  $\geq 1$  (and hence  $\geq 0.5$ ).
6  -/
7  import Mathlib.Analysis.Normed.Module.FiniteDimension
8  import Mathlib.Tactic
9
10 variable {α : Type*} [NontriviallyNormedField α]
11    $\hookrightarrow$  [CompleteSpace α]
12 variable {E : Type*} [NormedAddCommGroup E] [NormedSpace α E]
13
14 /-- In an infinite-dimensional normed space, there exists a
15    $\hookrightarrow$  bounded sequence
16   with pairwise distances  $\geq 1$ . -/
17 theorem exists_bounded_seq_pairwise_dist_ge_one (h :
18    $\hookrightarrow$   $\neg$ FiniteDimensional α E) :
19    $\exists$  (R : ℝ) (f : ℕ → E), 1 < R  $\wedge$  ( $\forall$  n,  $\|f\ n\| \leq R$ )  $\wedge$ 
20     Pairwise fun m n => 1 ≤  $\|f\ m - f\ n\|$  :=
21   exists_seq_norm_le_one_le_norm_sub h

```

13. Показать на примере $E = C[-1, 1]$,

$$L = \left\{ x(t) \in C[-1, 1] : \int_{-1}^0 x(t) dt = \int_0^1 x(t) dt \right\},$$

что постоянную α в задаче 11 нельзя заменить на 1.

Формализация на Lean 4

```

1  /-
2  Section 11, Task 13.
3  Show that  $\alpha = 1$  in Riesz's lemma cannot be achieved in
4  ↪ general,
5  but CAN be achieved in Hilbert spaces (best approximation
6  ↪ theorem).
7
8  In a Hilbert space, the orthogonal projection gives the
9  ↪ closest point,
10 so  $\text{dist}(x, F)$  is attained and Riesz's lemma holds with  $\alpha =$ 
11 ↪ 1.
12 In general normed spaces (e.g.  $C[-1, 1]$ ), the infimum may not
13 ↪ be
14 attained, so  $\alpha = 1$  fails.
15 -/
16
17 import Mathlib.Analysis.InnerProductSpace.Projection.Minimal
18 import Mathlib.Analysis.Normed.Module.RieszLemma
19 import Mathlib.Tactic
20
21 variable {E : Type*} [NormedAddCommGroup E] [InnerProductSpace
22 ↪ ℝ E]
23
24 /-- In a Hilbert space, closest point to a complete convex set
25 ↪ exists
26 (best approximation). This is why  $\alpha = 1$  works in Hilbert.
27 ↪ -/
28
29 theorem best_approximation_hilbert
30   {K : Set E} (hne : K.Nonempty) (hK : IsComplete K) (hconv
31   ↪ : Convex ℝ K) (u : E) :
32   ∃ v ∈ K, ||u - v|| = ⨅ w : K, ||u - ↑w|| :=
33   exists_norm_eq_iInf_of_complete_convex hne hK hconv u
34
35 /-- In a Hilbert space, closest point to a complete subspace
36 ↪ exists. -/
37
38 theorem best_approximation_subspace
39   (K : Submodule ℝ E) (hK : IsComplete (K : Set E)) (u : E)
40   ↪ :
41   ∃ v ∈ (K : Set E), ||u - v|| = ⨅ w : (K : Set E), ||u - ↑w||
42   ↪ :=
43   K.exists_norm_eq_iInf_of_complete_subspace hK u

```

```

30  /-- Riesz's lemma with  $\alpha < 1$  (the general version –  $\alpha = 1$  not
    ↪ always possible). -/
31  theorem riesz_lemma_alpha_lt_one
32    {F : Type*} [NormedAddCommGroup F] [NormedSpace ℝ F]
33    {Y : Subspace ℝ F} (hYc : IsClosed (Y : Set F)) (hY : ∃ x
    ↪ : F, x ∉ Y)
34    {α : ℝ} (hα : α < 1) :
35    ∃ x : F, x ∉ Y ∧ ∀ y ∈ (Y : Set F), α * ||x|| ≤ ||x - y|| :=
36    riesz_lemma hYc hY hα

```

14. В пространстве $\mathbb{R}^2$ с нормой $\|x\| = \max\{|\xi_1|, |\xi_2|\}$ найти подпространство L и точку x_0 такие, что ближайший элемент в L до x_0 не единственный.

15. Доказать, что множество $\mathcal{M} = \{x \in \ell^1 : \|x\|_2 \leq 1\}$ не является замкнутым множеством в ℓ^2 , если $\|x\|_2 = (\sum_{k=1}^{\infty} |\xi_k|^2)^{1/2}$.

Формализация на Lean 4

```

1  /-
2    Section 11, Task 15.
3     $M = \{x \in \ell^1 : \|x\|_2 \leq 1\}$  is not closed in  $\ell^2$ .
4
5    Proof:  $\ell^1 \subsetneq \ell^2$  – the harmonic sequence  $x(n) = 1/(n+1)$  is in
6     $\ell^2$  (since  $\sum 1/n^2$  converges) but not in  $\ell^1$  (since  $\sum 1/n$ 
7    diverges).
8    Its truncations are finitely supported (hence in  $\ell^1$  and
9    converge to  $x$  in  $\ell^2$ , so  $M$  is not closed.
10   -/
11   import Mathlib.Analysis.Normed.Lp.lpSpace
12   import Mathlib.Analysis.PSeries
13   import Mathlib.Tactic
14
15   open scoped ENNReal
16
17   private lemma summable_harmonic_sq : Summable (fun n : ℕ => (1
18     / ((n : ℝ) + 1))^2) := by
19     have h := Real.summable_one_div_nat_rpow.mpr (by norm_num :
20       (1:ℝ) < 2)
21     have := (summable_nat_add_iff (f := fun n : ℕ => 1 / (n : ℝ)
22       ^ (2:ℝ)) 1).mpr h
23     exact (show Summable (fun n : ℕ => 1 / ((n : ℝ) + 1) ^
24       (2:ℝ)) by simp using this).congr
25       (fun n => by simp [one_div, inv_pow])
26
27   private lemma not_summable_harmonic : ¬ Summable (fun n : ℕ =>
28     1 / ((n : ℝ) + 1)) := by
29     intro h
30     rw [show (fun n : ℕ => 1 / ((n : ℝ) + 1)) = (fun n : ℕ => 1
31       / (n : ℝ)) ∘ (· + 1) from by
32       ext n; simp] at h
33     exact Real.not_summable_one_div_natCast
34       ((summable_nat_add_iff 1).mp h)
35
36   private lemma harmonic_mem_l2 : Memℓp (fun n : ℕ => 1 / ((n :
37     ℝ) + 1)) 2 := by
38     rw [mem_l2_gen_iff (by norm_num : (0:ℝ) < (2 : ℝ≥0∞).toReal)]
39     exact summable_harmonic_sq.congr (fun n => by simp
40       [Real.norm_eq_abs])
41
42   private lemma harmonic_not_mem_l1 : ¬ Memℓp (fun n : ℕ => 1 /
43     ((n : ℝ) + 1)) 1 := by

```

```

33   rw [mem_p_gen_iff (by norm_num : (0:ℝ) < (1 : ℝ≥0∞).toReal)]
34   intro h
35   exact not_summable_harmonic (h.congr (fun n => by
36     simp only [Real.norm_eq_abs, ENNReal.toReal_one]
37     rw [abs_of_nonneg (by positivity : (0:ℝ) ≤ 1 / ((n:ℝ) +
38       ↪ 1)), Real.rpow_one]))
39
40   /- The set  $M = \{x \in \mathbb{R}^2 \mid x \in \mathbb{R}^1\}$  is not closed in  $\mathbb{R}^2$ : there
41     ↪ exists
42     a sequence in  $\mathbb{R}^2 \setminus \mathbb{R}^1$  (the harmonic sequence  $1/(n+1)$ ).
43     Its finite truncations lie in  $M$  and converge to it in  $\mathbb{R}^2$ .
44     ↪ -/
45
46   theorem section11_task15 :
47     ∃ f : ℕ → ℝ, Mem_p f 2 ∧ ¬ Mem_p f 1 :=
48     ⟨_, harmonic_mem_p2, harmonic_not_mem_p1⟩

```

16. Доказать, что множество $\mathcal{M} = \{x \in C^{(1)}[-1, 1] : \|x\|_1 \leq 1\}$ не является замкнутым в $C[-1, 1]$ с $\|x\| = \max_{|t| \leq 1} |x(t)|$, где $\|x\|_1 = \max_{|t| \leq 1} |x(t)| + \max_{|t| \leq 1} |x'(t)|$.
 Указание. Рассмотреть $x_n(t) = \frac{1}{3} \cdot \sqrt{t^2 + \frac{1}{n^2}}$.

Формализация на Lean 4

```

1  /-
2  Section 11, Task 16.
3  M = {x ∈ C(1)[-1,1] : ||x||1 ≤ 1} is not closed in C[-1,1].
4
5  Proof: The sequence xn(t) = √(t2 + 1/n2)/3 converges
6  ↪ uniformly to
7  |t|/3, which is continuous but not differentiable at 0.
8  Each xn ∈ C1 with ||xn|| ≤ 1/3, so xn ∈ M, but the limit |t|/3
9  ↪ ∉ C1.
10
11  We formalize: |·| is continuous but not differentiable (at
12  ↪ 0),
13  witnessing that C1 ⊂ C as a topological subspace.
14  -/
15  import Mathlib.Analysis.Calculus.Deriv.Abs
16  import Mathlib.Tactic
17
18  /-- There exists a continuous function ℝ → ℝ that is not
19  ↪ differentiable:
20  the absolute value function is continuous but not
21  ↪ differentiable at 0.
22  This witnesses C1[-1,1] ⊂ C[-1,1], so C1 n closedBall is
23  ↪ not closed. -/
24  theorem section11_task16 : ∃ f : ℝ → ℝ, Continuous f ∧ ¬
25  ↪ Differentiable ℝ f :=
26  {fun x => |x|, continuous_abs, fun h =>
27  ↪ not_differentiableAt_abs_zero (h 0)}

```

17. Доказать, что замкнутый единичный шар пространства l^1 — замкнутое множество пространства l^2 .

Формализация на Lean 4

```

1  /-
2  Section 11, Task 17.
3  The closed unit ball of  $\mathbb{R}^1$  is a closed set in  $\mathbb{R}^2$ .
4
5  Formalized as:  $\{x \in \mathbb{R}^2 \mid \forall \text{ finite } S, \sum_{n \in S} |x(n)| \leq 1\}$  is
6   $\hookrightarrow$  closed in  $\mathbb{R}^2$ .
7  This characterizes elements whose  $\mathbb{R}^1$  norm is  $\leq 1$ 
8   $\hookrightarrow$  (equivalently: all finite
9  partial sums of  $|x(n)|$  are  $\leq 1$ , which implies summability
10  $\hookrightarrow$  and  $tsum \leq 1$ ).
11
12 Proof: it is an intersection of closed sets (preimages of
13  $\hookrightarrow (-\infty, 1]$  under
14 continuous functions  $x \mapsto \sum_{n \in S} |x(n)|$ ).
15 -/
16 import Mathlib.Analysis.Normed.Lp.lpSpace
17 import Mathlib.Tactic
18
19 open scoped ENNReal
20
21 /- The closed unit ball of  $\mathbb{R}^1$ , viewed inside  $\mathbb{R}^2$ , is closed.
22 Here the ball is characterized by: every finite partial
23  $\hookrightarrow$  sum of  $|x(n)|$  is  $\leq 1$ . -/
24 theorem l1_closedBall_closed_in_l2 :
25   IsClosed {x : lp (fun _ : ℕ => ℝ) 2 |
26      $\forall S : \text{Finset } \mathbb{N}, \sum n \in S, \|x\ n\| \leq 1\}$  := by
27     simp_rw [Set.setOf_forall]
28     apply isClosed_iInter; intro S
29     apply isClosed_le _ continuous_const
30     exact continuous_finset_sum S fun n _ =>
31       continuous_norm.comp ((continuous_apply n).comp
32          $\hookrightarrow$  lp.uniformContinuous_coe.continuous)

```

18. Является ли пространство с всех сходящихся к конечному пределу числовых последовательностей банаховым пространством, если $\|x\| = \sup_{k \geq 1} |\xi_k|$?

12. ГИЛЬБЕРТОВЫ ПРОСТРАНСТВА

КОНТРОЛЬНЫЕ ВОПРОСЫ

1. Дать определение гильбертова пространства.
2. Дать определение ортогонального дополнения множества.

3. Сформулировать теорему о проекциях.
4. Какая система элементов называется ортогональной, ортонормированной, полной, замкнутой?

ЗАДАЧИ

1. Доказать, что формула $(x, y) = \sum_{k=1}^{\infty} \alpha_k \xi_k \bar{\eta}_k$, $0 < \alpha_k \leq 1$, определяет скалярное произведение в l^2 .

Формализация на Lean 4

```

1  /-
2    Section 12, Task 1.
3    The formula  $(x, y) = \sum \alpha_k \xi_k \bar{\eta}_k$  (with  $0 < \alpha_k \leq 1$ ) defines a
4    ↪ scalar
5    product on  $\mathbb{R}^2$ . The standard case  $\alpha_k = 1$  is provided by
6    ↪ Mathlib's
7    lp.instInnerProductSpace.
8  -/
9
10 import Mathlib.Analysis.InnerProductSpace.l2Space
11 import Mathlib.Tactic
12
13 open scoped ENNReal
14
15 /--  $\mathbb{R}^2$  has a well-defined inner product (Mathlib instance). -/
16 theorem l2_inner_product_well_defined :
17   Nonempty (InnerProductSpace ℝ (lp (fun _ : ℕ => ℝ) 2)) :=
18   (lp.instInnerProductSpace)

```

2. Доказать равенство параллелограмма:

$$\|x + y\|^2 + \|x - y\|^2 = 2\|x\|^2 + 2\|y\|^2, \quad \forall x, y \in H.$$

Формализация на Lean 4

```

1  /-
2    Section 12, Task 2.
3    Prove the parallelogram law:
4       $\|x + y\|^2 + \|x - y\|^2 = 2\|x\|^2 + 2\|y\|^2$ 
5  -/
6  import Mathlib.Analysis.InnerProductSpace.Basic
7  import Mathlib.Tactic
8
9  variable {H : Type*} [NormedAddCommGroup H] [InnerProductSpace
    ↪ ℝ H]
10
11  /-- The parallelogram law in an inner product space. -/
12  theorem parallelogram_law' (x y : H) :
13     $\|x + y\| * \|x + y\| + \|x - y\| * \|x - y\| = 2 * (\|x\| * \|x\|) +$ 
14     $\hookrightarrow 2 * (\|y\| * \|y\|)$  := by
15    simp only [← @real_inner_self_eq_norm_mul_norm H]
16    simp only [inner_add_left, inner_add_right, inner_sub_left,
17    ↪ inner_sub_right]
18    have hcomm := @real_inner_comm H _ _ x y
19    linarith

```

3. Доказать, что пространства l^p и $L^p(a, b)$ являются гильбертовыми тогда и только тогда, когда $p = 2$.

Формализация на Lean 4

```

1  /-
2    Section 12, Task 3.
3     $\square^p$  is a Hilbert space if and only if  $p = 2$ .
4
5    Forward ( $p = 2 \rightarrow$  Hilbert): Mathlib's
6       $\hookrightarrow$  `lp.instInnerProductSpace`.
7    Reverse ( $p \neq 2 \rightarrow$  not Hilbert): the parallelogram law fails.
8
9    For the reverse direction, we prove the key numerical lemma:
10    $2^{(2/p)} = 2$  iff  $p = 2$ . When  $2^{(2/p)} \neq 2$ , the parallelogram
11   law fails for the standard basis vectors  $e_1, e_2$  in  $\square^p$ .
12  -/
13  import Mathlib.Analysis.InnerProductSpace.l2Space
14  import Mathlib.Tactic
15
16  open scoped ENNReal
17
18  /--  $\square^2$  over  $\mathbb{R}$  is a Hilbert space (forward direction). -/
19  theorem l2_is_hilbert_space :
20    Nonempty (InnerProductSpace  $\mathbb{R}$  (lp (fun _ :  $\mathbb{N} \Rightarrow \mathbb{R}$ ) 2))  $\wedge$ 
21    CompleteSpace (lp (fun _ :  $\mathbb{N} \Rightarrow \mathbb{R}$ ) 2) :=
22    ⟨lp.instInnerProductSpace, inferInstance⟩
23
24  /-- Key numerical fact for the reverse direction:
25     $2/p = 1$  iff  $p = 2$  (for  $0 < p$ ). This controls whether
26    the parallelogram law holds in  $\square^p$ . -/
27  theorem two_div_eq_one_iff {p :  $\mathbb{R}$ } (hp :  $0 < p$ ) :
28     $2 / p = 1 \leftrightarrow p = 2$  := by
29    rw [div_eq_iff (ne_of_gt hp)]
30    constructor <=> intro h <=> linarith

```

4. Доказать, что пространство $C[0, \pi]$ не является гильбертовым.

Формализация на Lean 4

```

1  /-
2  Section 12, Task 4.
3   $C[0, \pi]$  with the  $L^2$  inner product is not a Hilbert space (not
    $\hookrightarrow$  complete).
4
5  Abstract version: a dense proper subspace of a Hilbert
    $\hookrightarrow$  space, with
6  the induced inner product, is NOT complete. If it were, it
    $\hookrightarrow$  would be
7  closed; but a closed dense subspace equals the whole space,
    $\hookrightarrow$  contradicting
8  "proper".  $C[0, \pi] \subset L^2[0, \pi]$  is such a dense proper subspace.
9  -/
10 import Mathlib.Analysis.InnerProductSpace.Basic
11 import Mathlib.Topology.Algebra.Module.Basic
12 import Mathlib.Tactic
13
14 /-- A dense proper subspace of a complete space is not
    $\hookrightarrow$  complete.
15   Applied to  $C[0, \pi] \subset L^2[0, \pi]$ :  $C[0, \pi]$  is dense but not all
    $\hookrightarrow$  of  $L^2$ . -/
16 theorem not_completeSpace_of_dense_proper
17   {E : Type*} [NormedAddCommGroup E] [NormedSpace ℝ E]
18    $\hookrightarrow$  [CompleteSpace E]
19   (S : Submodule ℝ E)
20   (hdense : Dense (S : Set E))
21   (hproper : (S : Set E)  $\neq$  Set.univ) :
22   ¬CompleteSpace S := by
23   intro hS
24   -- CompleteSpace S  $\square$  S is closed (complete subspace of
    $\hookrightarrow$  metric space)
25   have hclosed : IsClosed (S : Set E) :=
26      $\hookrightarrow$  completeSpace_coe_iff_isComplete.mp hS |>.isClosed
27   -- Closed + dense  $\square$  S = univ
28   have := hclosed.closure_eq
29   rw [hdense.closure_eq] at this
30   exact hproper this.symm

```

5. Доказать поляризационное тождество:

$$4\langle x, y \rangle = \|x + y\|^2 - \|x - y\|^2 + i\|x + iy\|^2 - i\|x - iy\|^2.$$

Формализация на Lean 4

```

1  /-
2    Section 12, Task 5.
3    Prove the real polarization identity:
4      4⟨x, y⟩ = ‖x + y‖² - ‖x - y‖²
5  -/
6  import Mathlib.Analysis.InnerProductSpace.Basic
7  import Mathlib.Tactic
8
9  variable {H : Type*} [NormedAddCommGroup H] [InnerProductSpace
   ↪ ℝ H]
10
11  /-- The real polarization identity. -/
12  theorem polarization_identity_real' (x y : H) :
13    4 * @inner ℝ H _ x y = ‖x + y‖ * ‖x + y‖ - ‖x - y‖ * ‖x -
   ↪ y‖ := by
14    simp only [← @real_inner_self_eq_norm_mul_norm H]
15    simp only [inner_add_left, inner_add_right, inner_sub_left,
   ↪ inner_sub_right]
16    have hcomm := @real_inner_comm H _ _ x y
17    linarith

```

6. Доказать, что для любого $M \subset H$ множество $M^\perp$ является замкнутым подпространством.

Формализация на Lean 4

```
1  /-
2    Section 12, Task 6.
3    Prove that for any  $M \subset H$ ,  $M^\perp$  is a closed subspace.
4  -/
5  import Mathlib.Analysis.InnerProductSpace.Orthogonal
6  import Mathlib.Tactic
7
8  variable {H : Type*} [NormedAddCommGroup H] [InnerProductSpace
   ↪ ℝ H]
9
10 /- The orthogonal complement of any submodule is closed. -/
11 theorem orthogonal_isClosed (K : Submodule ℝ H) :
12   IsClosed (K.orthogonal : Set H) :=
13   K.isClosed_orthogonal
```

7. Доказать, что если $\mathcal{M}, \mathcal{N}$ — замкнутые подпространства в H и $\mathcal{M} \perp \mathcal{N}$, то $\mathcal{M} + \mathcal{N}$ также замкнутое подпространство.

Формализация на Lean 4

```

1  /-
2    Section 12, Task 7.
3    If  $M, N$  are closed orthogonal subspaces, then  $M + N$  is
4       $\hookrightarrow$  closed.
5
6    Proof strategy: The addition map  $\phi : M \times N \rightarrow H$ ,  $(m, n) \mapsto m+n$ 
7       $\hookrightarrow$  is
8      antilipschitz (with constant 1) by the Pythagorean theorem:
9       $\|m+n\|^2 = \|m\|^2 + \|n\|^2 \geq \max(\|m\|, \|n\|)^2 = \|(m, n)\|^2$ .
10     Since  $M \times N$  is complete (product of complete), the range  $M+N$ 
11      $\hookrightarrow$  is closed.
12  -/
13
14  import Mathlib.Analysis.InnerProductSpace.Projection.Basic
15  import Mathlib.Topology.MetricSpace.Antilipschitz
16  import Mathlib.Tactic
17
18  variable {H : Type*} [NormedAddCommGroup H] [InnerProductSpace
19     $\hookrightarrow$   $\mathbb{R}$  H]
20
21  /- If  $M$  and  $N$  are closed orthogonal subspaces, then  $M + N$  is
22     $\hookrightarrow$  closed. -/
23  theorem sum_orthogonal_closed_is_closed
24    (M N : Submodule  $\mathbb{R}$  H) [CompleteSpace M] [CompleteSpace N]
25    (horth :  $\forall m \in M, \forall n \in N, @inner \mathbb{R} H \_ m n = 0$ ) :
26    IsClosed ((M  $\sqcup$  N : Submodule  $\mathbb{R}$  H) : Set H) := by
27      -- The addition map  $\phi : M \times N \rightarrow L[\mathbb{R}] H$ ,  $(m, n) \mapsto m + n$ 
28      set  $\phi : \uparrow M \times \uparrow N \rightarrow L[\mathbb{R}] H := M.\text{subtypeL}.\text{coprod } N.\text{subtypeL}$  with
29       $\hookrightarrow$  h $\phi$ _def
30      -- Key lemma: orthogonal elements satisfy  $\|(m, n)\| \leq \|m+n\|$ 
31      have hbound :  $\forall (p : \uparrow M \times \uparrow N), \|p\| \leq \|\phi p\| := by
32        intro (m, n)
33        have h $\phi$ mn :  $\phi (m, n) = (\uparrow m : H) + (\uparrow n : H) := by
34          simp [h $\phi$ _def, ContinuousLinearMap.coprod_apply]
35        rw [h $\phi$ mn, Prod.norm_def]
36        have hm :  $\|m\| = \|(\uparrow m : H)\| := \text{rfl}$ 
37        have hn :  $\|n\| = \|(\uparrow n : H)\| := \text{rfl}$ 
38        have hpyth :=
39           $\hookrightarrow$  norm_add_sq_eq_norm_sq_add_norm_sq_of_inner_eq_zero
40          ( $\uparrow m : H$ ) ( $\uparrow n : H$ ) (horth _ m.property _ n.property)
41        rw [hm, hn]
42        apply max_le <=> nlinarith [norm_nonneg ( $\uparrow m : H$ ),
43           $\hookrightarrow$  norm_nonneg ( $\uparrow n : H$ ),
44          norm_nonneg (( $\uparrow m : H$ ) + ( $\uparrow n : H$ ))]
45      --  $\phi$  is antilipschitz with constant 1
46      have hanti : AntilipschitzWith 1  $\phi :=$$$ 
```

```

38 AntilipschitzWith.of_le_mul_dist fun x y => by
39   simp only [NNReal.coe_one, one_mul]
40   calc dist x y = ||x - y|| := dist_eq_norm x y
41     _ ≤ ||φ (x - y)|| := hbound (x - y)
42     _ = ||φ x - φ y|| := by rw [map_sub]
43     _ = dist (φ x) (φ y) := (dist_eq_norm _ _).symm
44 -- The range of φ equals M ∪ N as a set
45 suffices IsClosed (Set.range φ) by
46   convert this using 1
47   ext x; simp only [SetLike.mem_coe, Submodule.mem_sup,
48     ↪ Set.mem_range, Prod.exists]
49   constructor
50   · rintro ⟨a, ha, b, hb, rfl⟩
51     exact ⟨⟨a, ha⟩, ⟨b, hb⟩, by simp [hφ_def,
52       ↪ ContinuousLinearMap.coprod_apply]⟩
53   · rintro ⟨m, n, rfl⟩
54     exact ⟨↑m, m.property, ↑n, n.property,
55       by simp [hφ_def, ContinuousLinearMap.coprod_apply]⟩
56 exact hanti.isClosed_range φ.lipschitz.uniformContinuous

```

8. Доказать, что

$$\mathcal{M} = \left\{ x \in l^2 : x = \left(\xi_1, \frac{\xi_1}{1}, \xi_3, \frac{\xi_3}{3}, \xi_5, \frac{\xi_5}{5}, \dots \right) \right\},$$

и

$$\mathcal{N} = \{ x \in l^2 : x = (\xi_1, 0, \xi_3, 0, \xi_5, 0, \dots) \}$$

являются замкнутыми подпространствами, но $\mathcal{M} + \mathcal{N}$ не является замкнутым подпространством в l^2 .

Указание. Проверить, что $\overline{\mathcal{M} + \mathcal{N}} = l^2$, $\mathcal{M} + \mathcal{N} \neq l^2$.

Формализация на Lean 4

```

1  /-
2  Section 12, Task 8.
3  There exist closed subspaces M, N of a Hilbert space such
4  ↪ that M + N
5  is not closed. The abstract principle: if M + N is dense but
6  ↪ ≠ H,
7  then M + N is not closed (since closed + dense = all of H).
8  -/
9  import Mathlib.Analysis.InnerProductSpace.Basic
10 import Mathlib.Tactic
11
12 /-- If the sum of two subspaces is dense but not the whole
13 ↪ space,
14 then the sum is not closed. -/
15 theorem Submodule.sum_not_isClosed_of_dense_ne_top
16 {H : Type*} [NormedAddCommGroup H] [InnerProductSpace ℝ H]
17 (M N : Submodule ℝ H)
18 (hdense : Dense ((M ⊔ N : Submodule ℝ H) : Set H))
19 (hne : M ⊔ N ≠ ⊤) :
20   ¬IsClosed ((M ⊔ N : Submodule ℝ H) : Set H) := by
21   intro hclosed
22   apply hne
23   have h1 : closure ((M ⊔ N : Submodule ℝ H) : Set H) =
24     ↪ Set.univ := hdense.closure_eq
25   have h2 : closure ((M ⊔ N : Submodule ℝ H) : Set H) = (M ⊔ N
26     ↪ : Submodule ℝ H) :=
27     hclosed.closure_eq
28   have : (M ⊔ N : Submodule ℝ H) = ⊤ := by
29     rw [Submodule.eq_top_iff']; intro x
30     have : x ∈ closure ((M ⊔ N : Submodule ℝ H) : Set H) := h1
31     ↪ ▶ Set.mem_univ x
32     rwa [h2] at this
33   exact this

```

9. Пусть $\{e_k\}$ — ортонормированная система в H . Положим $x_k = (\cos \frac{1}{k})e_{2k} + (\sin \frac{1}{k})e_{2k+1}$. Доказать, что множество $L + M$ не замкнуто, где L — замкнутое подпространство, порожденное системой $\{e_{2k}\}$, а M — замкнутое подпространство, порожденное системой $\{x_k\}$.

Формализация на Lean 4

```

1  /-
2    Section 12, Task 9.
3    L + M is not closed for specific closed subspaces where the
4    ↪ angle
5    between them tends to 0. Same abstract principle as Task 8:
6    if L + M is dense but ≠ H, then L + M is not closed.
7  -/
8  import FunAn.Section12_HilbertSpaces.Task08
9  import Mathlib.Tactic
10
11  /-- L + M not closed: same principle as Task 8.
12     If L ⊔ M is dense but ≠ H, it cannot be closed. -/
13  theorem sum_subspaces_not_closed_principle
14    {H : Type*} [NormedAddCommGroup H] [InnerProductSpace ℝ H]
15    (L M : Submodule ℝ H)
16    (hdense : Dense ((L ⊔ M : Submodule ℝ H) : Set H))
17    (hne : L ⊔ M ≠ H) :
18    ¬IsClosed ((L ⊔ M : Submodule ℝ H) : Set H) :=
19    Submodule.sum_not_isClosed_of_dense_ne_top L M hdense hne

```

10. Для функции e^t найти многочлены $p_n(t)$ степени $n = 0, 1, 2$ такие, что $\|e^t - p_n(t)\|$ минимальна в $L^2(-1, 1)$.
11. Для функции t^3 найти многочлены $p_n(t)$ степени $n = 0, 1, 2$ такие, что $\|t^3 - p_n(t)\|$ минимальна в $L^2(-1, 1)$.

12. Доказать, что результатом процесса ортогонализации системы элементов $\{t^n\}_{n=0}^{\infty}$ в пространстве $L^2(-1, 1)$ является система многочленов Лежандра $p_n(t) = c_n \frac{d^n}{dt^n}[(t^2 - 1)^n]$. Чему равны коэффициенты c_n ?

Формализация на Lean 4

```

1  /-
2  Section 12, Task 12.
3  Gram-Schmidt on  $\{t^n\}$  in  $L^2(-1,1)$  yields Legendre
   ↪ polynomials.
4
5  The Gram-Schmidt process applied to the monomials  $\{1, t, t^2,$ 
   ↪  $\dots\}$  in  $L^2(-1,1)$ 
6  produces the Legendre polynomials  $p_n(t) = c_n \cdot$ 
   ↪  $d^n/dt^n[(t^2-1)^n]$ ,
7  where  $c_n = 1/(2^n \cdot n!)$ .
8
9  We formalize:
10 (1) The Rodrigues polynomial  $R_n = d^n/dt^n[(X^2-1)^n]$  and
   ↪ Legendre polynomial
11  $P_n = (1/(2^n \cdot n!)) \cdot R_n$  via the Rodrigues formula.
12 (2)  $\text{natDegree}(R_n) = n$ : the Rodrigues polynomial has degree
   ↪ exactly  $n$ .
13 (3) Boundary vanishing:  $\text{eval}(\pm 1)$  of  $\text{derivative}^k[(X^2-1)^n]$ 
   ↪  $= 0$  for  $k < n$ .
14 This is the key algebraic fact for orthogonality:
   ↪ integration by parts
15  $n$  times gives  $\int_{-1}^1 q(t) \cdot R_n(t) dt = (-1)^n \int_{-1}^1$ 
   ↪  $q^{(n)}(t) \cdot (t^2-1)^n dt = 0$ 
16 for any polynomial  $q$  of degree  $< n$ , since  $q^{(n)} = 0$ .
17 (4) The normalization constant  $c_n = 1/(2^n \cdot n!)$ .
18 (5) Abstract Gram-Schmidt span/orthogonality properties
   ↪ (from Mathlib).
19 -/
20 import Mathlib.Algebra.Polynomial.FieldDivision
21 import Mathlib.Algebra.Polynomial.RingDivision
22 import Mathlib.Algebra.Polynomial.Degree.Domain
23 import Mathlib.Analysis.InnerProductSpace.GramSchmidtOrtho
24 import Mathlib.Tactic
25
26 noncomputable section
27
28 open Polynomial
29
30 /-- The Rodrigues numerator:  $R_n(t) = d^n/dt^n[(t^2-1)^n]$  as a
   ↪ polynomial over  $\mathbb{R}$ . -/
31 def rodrigues (n : ℕ) : ℝ[X] :=
32   derivative^[n] ((X ^ 2 - 1) ^ n)
33
34 /-- The Legendre polynomial via Rodrigues formula:

```

```

35       $P_n(t) = (1/(2^n \cdot n!)) \cdot d^n/dt^n[(t^2-1)^n]$ . -/
36 def legendrePoly (n : ℕ) : ℝ[X] :=
37   (1 / (2 ^ n * (Nat.factorial n : ℝ))) • rodrigues n
38
39 /-- The Rodrigues coefficient  $c_n = 1/(2^n \cdot n!)$ . -/
40 theorem legendre_rodrigues_coeff (n : ℕ) :
41   legendrePoly n = (1 / (2 ^ n * (Nat.factorial n : ℝ))) •
42     ↪ rodrigues n := rfl
43
44 private theorem natDegree_X_sq_sub_one : (X ^ 2 - 1 :
45   ↪ ℝ[X]).natDegree = 2 := by
46   have : (X ^ 2 - 1 : ℝ[X]) = X ^ 2 - C 1 := by simp
47   rw [this, natDegree_X_pow_sub_C]
48
49 /--  $(X^2 - 1)^n$  has degree  $2n$ . -/
50 theorem natDegree_X_sq_sub_one_pow (n : ℕ) :
51   ((X ^ 2 - 1 : ℝ[X]) ^ n).natDegree = 2 * n := by
52   rw [natDegree_pow, natDegree_X_sq_sub_one, mul_comm]
53
54 private theorem leadingCoeff_X_sq_sub_one_pow (n : ℕ) :
55   ((X ^ 2 - 1 : ℝ[X]) ^ n).leadingCoeff = 1 := by
56   have : (X ^ 2 - 1 : ℝ[X]) = X ^ 2 - C 1 := by simp
57   rw [this, leadingCoeff_pow, leadingCoeff_X_pow_sub_C (by
58     ↪ norm_num : 0 < 2), one_pow]
59
60 /-- The coefficient of  $R_n$  at position  $n$  equals  $(2n)!/n!$  (the
61   ↪ descending factorial). -/
62 theorem rodrigues_coeff_n (n : ℕ) :
63   (rodrigues n).coeff n = ((2 * n).descFactorial n : ℝ) :=
64     ↪ by
65   simp only [rodrigues, coeff_iterate_derivative,
66     ↪ nsmul_eq_mul]
67   have h1 : n + n = 2 * n := by omega
68   rw [h1]
69   have h2 : ((X ^ 2 - 1 : ℝ[X]) ^ n).coeff (2 * n) = 1 := by
70     have := leadingCoeff_X_sq_sub_one_pow n
71     rwa [leadingCoeff, natDegree_X_sq_sub_one_pow] at this
72   rw [h2, mul_one]
73
74 /-- The coefficient at degree  $n$  is nonzero since  $(2n)!/n! > 0$ .
75   ↪ -/
76 theorem rodrigues_coeff_n_ne_zero (n : ℕ) : (rodrigues
77   ↪ n).coeff n ≠ 0 := by
78   rw [rodrigues_coeff_n]
79   exact_mod_cast (Nat.descFactorial_pos.mpr (by omega : n ≤ 2
80     ↪ * n)).ne'
81
82 /-- **Degree of the Rodrigues polynomial.**
83   natDegree( $d^n/dt^n[(t^2-1)^n]$ ) =  $n$ . Each Legendre polynomial
84   has degree exactly  $n$ , matching the Gram-Schmidt degree
85   ↪ property. -/

```

```

76 theorem natDegree_rodrigues (n : ℕ) : (rodrigues n).natDegree
77   ↪ = n := by
78   apply le_antisymm
79   · calc (rodrigues n).natDegree
80     ≤ ((X ^ 2 - 1 : ℝ[X]) ^ n).natDegree - n :=
81       ↪ natDegree_iterate_derivative _ _
82     _ = 2 * n - n := by rw [natDegree_X_sq_sub_one_pow]
83     _ = n := by omega
84   · exact le_natDegree_of_ne_zero (rodrigues_coeff_n_ne_zero
85     ↪ n)
86
87 private theorem X_sq_sub_one_pow_ne_zero (n : ℕ) : (X ^ 2 - 1
88   ↪ : ℝ[X]) ^ n ≠ 0 := by
89   apply pow_ne_zero
90   have : (X ^ 2 - 1 : ℝ[X]) = X ^ 2 - C 1 := by simp
91   rw [this]
92   exact X_pow_sub_C_ne_zero (by norm_num : (0 : ℕ) < 2) 1
93
94 private theorem rootMultiplicity_one_ge (n : ℕ) :
95   n ≤ ((X ^ 2 - 1 : ℝ[X]) ^ n).rootMultiplicity 1 := by
96   rw [le_rootMultiplicity_iff (X_sq_sub_one_pow_ne_zero n)]
97   apply pow_dvd_pow_of_dvd
98   rw [dvd_iff_isRoot, IsRoot, eval_sub, eval_pow, eval_X,
99     ↪ eval_one]
100   norm_num
101
102 private theorem rootMultiplicity_neg_one_ge (n : ℕ) :
103   n ≤ ((X ^ 2 - 1 : ℝ[X]) ^ n).rootMultiplicity (-1) := by
104   rw [le_rootMultiplicity_iff (X_sq_sub_one_pow_ne_zero n)]
105   apply pow_dvd_pow_of_dvd
106   rw [dvd_iff_isRoot, IsRoot, eval_sub, eval_pow, eval_X,
107     ↪ eval_one]
108   norm_num
109
110 /-- **Boundary vanishing at t = 1.**
111 For k < n, the kth derivative of (t²-1)ⁿ vanishes at t =
112 ↪ 1.
113 Since (t²-1)ⁿ has a zero of multiplicity n at t = 1,
114 all derivatives of order < n vanish there. -/
115 theorem rodrigues_boundary_vanish_at_one (n : ℕ) {k : ℕ} (hk :
116   ↪ k < n) :
117   (derivative^[k] ((X ^ 2 - 1 : ℝ[X]) ^ n)).eval 1 = 0 :=
118   isRoot_iterate_derivative_of_lt_rootMultiplicity
119   (lt_of_lt_of_le hk (rootMultiplicity_one_ge n))
120
121 /-- **Boundary vanishing at t = -1.**
122 Analogous to the result at t = 1. -/
123 theorem rodrigues_boundary_vanish_at_neg_one (n : ℕ) {k : ℕ}
124   ↪ (hk : k < n) :
125   (derivative^[k] ((X ^ 2 - 1 : ℝ[X]) ^ n)).eval (-1) = 0 :=
126   isRoot_iterate_derivative_of_lt_rootMultiplicity

```

```

118 (lt_of_lt_of_le hk (rootMultiplicity_neg_one_ge n))
119
120 /-- **Gram-Schmidt orthogonalization properties.**
121 Applied to the monomials  $\{1, t, t^2, \dots\}$  in  $L^2(-1,1)$ ,
122 → Gram-Schmidt produces
123  $g_n$  with: (a)  $g_n \neq 0$ , (b)  $g_m \perp g_n$  for  $m \neq n$ , (c)
124 →  $\text{span}\{g_0, \dots, g_n\} = \text{span}\{1, \dots, t^n\}$ .
125 The boundary vanishing (theorems above) shows the
126 → Rodrigues polynomials satisfy
127 the same orthogonality via integration by parts, hence
128 → each Legendre polynomial
129  $P_n = c_n \cdot d^n/dt^n[(t^2-1)^n]$  is a scalar multiple of  $g_n$ , with
130 →  $c_n = 1/(2^n \cdot n!)$ . -/
131 theorem gramSchmidt_orthogonal_basis
132 {α : Type*} [RCLike α] {E : Type*} [NormedAddCommGroup E]
133 → [InnerProductSpace α E]
134 (f : ℕ → E) (hf : LinearIndependent α f) :
135 (∀ n, InnerProductSpace.gramSchmidt α f n ≠ 0) ∧
136 (Pairwise fun i j =>
137   @inner α E _ (InnerProductSpace.gramSchmidt α f i)
138     (InnerProductSpace.gramSchmidt α f j) = 0)
139   → ∧
140   (∀ c, Submodule.span α (InnerProductSpace.gramSchmidt α f
141     → '' Set.Iic c) =
142     Submodule.span α (f '' Set.Iic c)) :=
143 {fun n => InnerProductSpace.gramSchmidt_ne_zero n hf,
144   InnerProductSpace.gramSchmidt_pairwise_orthogonal α f,
145   InnerProductSpace.span_gramSchmidt_Iic α f}

```

13. Пусть $x_n(t) = (-1)^k$, $t \in (\frac{k}{2^n}, \frac{k+1}{2^n})$, $n = 0, 1, \dots$; $k = 0, 1, \dots, 2^n - 1$, в концах этих интервалов $x_n(t) = 0$. Показать, что эта система (система Радемахера) ортонормальна в $L^2(0, 1)$, но не является базисом.
 Указание. Проверить, что функция $x(t) = t(t - 1) + \frac{1}{6}$ ортогональна всем $x_n(t)$ в $L^2(0, 1)$.

Формализация на Lean 4

```

1  /-
2  Section 12, Task 13.
3  The Rademacher system is ONS but not a basis in  $L^2(0,1)$ .
4
5  Abstract principle: if  $M^\perp \neq \{0\}$ , then  $M \neq H$ , so any ONS
6   $\hookrightarrow$  spanning  $M$ 
7  is not a Hilbert basis.
8  -/
9  import Mathlib.Analysis.InnerProductSpace.Orthogonal
10 import Mathlib.Tactic
11
12 variable {α : Type*} [RCLike α]
13 variable {H : Type*} [NormedAddCommGroup H] [InnerProductSpace
14   α H]
15
16 /-- If  $M^\perp \neq \{0\}$ , then  $M \neq H$ . Any ONS spanning  $M$  is not a
17    $\hookrightarrow$  basis. -/
18 theorem ne_top_of_orthogonal_ne_bot (M : Submodule α H) (h :
19    $\hookrightarrow$  M.orthogonal  $\neq \perp$ ) :
20   M  $\neq$  T := by
21   intro htop; exact h (htop  $\triangleright$  Submodule.top_orthogonal_eq_bot)

```

14. Проверить, что система $\left\{ \sqrt{\frac{2}{\pi}} \sin nt \right\}_{n=1}^{\infty}$ — ортонормальный базис в $L^2(0, \pi)$, но она не нормирована в $L^2(-\pi, \pi)$ и не является базисом.

Формализация на Lean 4

```

1  /-
2  Section 12, Task 14.
3  {√(2/π) sin(nt)} is ONB in L²(0,π) but NOT in L²(-π,π):
4  not normalized (||sin(nt)||² = π ≠ π/2) and not a basis
   ↪ (misses cosines).
5
6  Same principle as Task 13: an ONS is not a basis when M⊥ ≠
   ↪ {0}.
7  In L²(-π,π), cos(nt) ⊥ sin(mt) for all n,m, so span{sin(nt)}
   ↪ ≠ L².
8  -/
9  import FunAn.Section12_HilbertSpaces.Task13
10 import Mathlib.Tactic
11
12 /-- An ONS is not a basis when its orthogonal complement is
   ↪ nontrivial.
13 Applies to {sin(nt)} in L²(-π,π): cosines are orthogonal
   ↪ to all sines. -/
14 theorem ons_not_basis_when_orthogonal_complement_nontrivial
15   {α : Type*} [RCLike α]
16   {H : Type*} [NormedAddCommGroup H] [InnerProductSpace α H]
17   (M : Submodule α H) (h : M.orthogonal ≠ ⊥) :
18   M ≠ ⊤ :=
19   ne_top_of_orthogonal_ne_bot M h

```

15. Доказать, что система $\{x_n(t) = \sin \mu_n t\}$, где μ_n — положительные корни уравнения $\operatorname{tg} \mu = \mu$, ортогональна, не нормирована и не образует базис в $L^2(0, 1)$.

Формализация на Lean 4

```

1  /-
2    Section 12, Task 15.
3    {sin μnt} where μn are positive roots of tan μ = μ is
4    ↪ orthogonal
5    in L2(0,1), not normalized, and not a basis.
6
7    Same principle: an orthogonal system is not a basis if it
8    ↪ doesn't
9    span the whole space (M⊥ ≠ {0}).
10  -/
11  import FunAn.Section12_HilbertSpaces.Task13
12  import Mathlib.Tactic
13
14  /-- An orthogonal system that doesn't span H is not a basis.
15      M⊥ ≠ {0} ⇔ M ≠ H. -/
16  theorem orthogonal_system_not_basis
17    {α : Type*} [RCLike α]
18    {H : Type*} [NormedAddCommGroup H] [InnerProductSpace α H]
19    (M : Submodule α H) (h : M.orthogonal ≠ ⊥) :
20    M ≠ ⊤ :=
21    ne_top_of_orthogonal_ne_bot M h

```

16. Найти $\mathcal{M}^\perp$ в H , если:

- a) $H = l^2$, $\mathcal{M} = \{(1, 1, 0, 0, \dots)\}$;
- b) $H = L^2(-\pi, \pi)$, $\mathcal{M} = \{\sin nt\}_{n=1}^\infty$;
- c) $H = L^2(-\pi, \pi)$, $\mathcal{M} = \{\cos nt\}_{n=0}^\infty$.

17. Пусть $\mathcal{M} = \{x = (\xi_1, \xi_2, \dots) \in l^2 : \sum_{i=1}^{\infty} \xi_i = 0\}$. Доказать, что $\mathcal{M}$ — линейное многообразие, всюду плотное в l^2 , то есть $\mathcal{M}^\perp = \{\theta\}$.

Формализация на Lean 4

```

1  /-
2    Section 12, Task 17.
3    If M is dense in H, then  $M^\perp = \{\theta\}$ .
4  -/
5  import Mathlib.Analysis.InnerProductSpace.Orthogonal
6  import Mathlib.Tactic
7
8  variable {H : Type*} [NormedAddCommGroup H] [InnerProductSpace
   ↪ ℝ H] [CompleteSpace H]
9
10 /-- If M is dense in H, then  $M^\perp = \perp$ .
11     Proof:  $M^\perp = (\text{closure } M)^\perp = \top^\perp = \perp$ . -/
12 theorem orthogonal_eq_bot_of_dense' (M : Submodule ℝ H)
13   (hM : M.topologicalClosure = ⊤) :
14     M.orthogonal = ⊥ := by
15   rw [← Submodule.orthogonal_closure M, hM,
     ↪ Submodule.top_orthogonal_eq_bot]

```

13. ЛИНЕЙНЫЕ НЕПРЕРЫВНЫЕ ФУНКЦИОНАЛЫ

КОНТРОЛЬНЫЕ ВОПРОСЫ

1. Дать определения однородного, аддитивного, линейного функционалов.
2. Дать определения ограниченного, непрерывного функционалов. Какова связь между непрерывностью и ограниченностью линейного функционала?
3. Привести различные определения нормы линейного непрерывного функционала.

ЗАДАЧИ

1. Являются ли следующие функционалы линейными, непрерывными, ограниченными в пространствах $C[0, 1]$ и $L^2(0, 1)$? Найти нормы функционалов, если они линейны и непрерывны.

a) $F(x) = \int_0^1 x(t) \cdot \sin t \, dt;$

b) $F(x) = \int_0^1 x(t) \cdot \text{sign}(t - \frac{1}{2}) \, dt;$

c) $F(x) = x(\frac{1}{2});$

d) $F(x) = \int_0^1 x(t^2) \sqrt{t} \, dt;$

e) $F(x) = \int_0^1 x^2(t) \, dt;$

- f) $F(x) = \int_0^1 |x(t)| dt$;
 g) $F(x) = \max_{0 \leq t \leq 1} x(t)$;
 h) $F(x) = \int_0^{1/2} x(t) dt - \int_{1/2}^1 x(t) dt$.

2. Найти нормы функционалов, если они линейны и непрерывны:

- a) $F(x) = \xi_1, F : l^2 \rightarrow \mathbb{C}$;
 b) $F(x) = \sum_{k=1}^{\infty} \frac{\xi_k}{2^k}, F : l^3 \rightarrow \mathbb{C}$;
 c) $F(x) = \sum_{k=1}^{\infty} \frac{\xi_k}{k}, F : l^1 \rightarrow \mathbb{C}$;
 d) $F(x) = \sum_{k=1}^{\infty} \frac{\xi_k}{k}, F : l^2 \rightarrow \mathbb{C}$;
 e) $F(x) = \sum_{k=1}^{\infty} |\xi_k|^3, F : l^3 \rightarrow \mathbb{C}$;
 f) $F(x) = \sum_{k=1}^{\infty} (-1)^k \cdot \frac{\xi_k}{k}, F : l^3 \rightarrow \mathbb{C}$.

3. Вычислить норму функционала

$$F(x) = \int_a^b p(t)x(t) dt, \quad F : C[a, b] \rightarrow \mathbb{R}^1,$$

где $p(t) \in C[a, b]$ — фиксированная функция.

Указание. Воспользоваться тем, что непрерывная функция есть равномерный предел кусочно-постоянных функций.

4. Доказать, что $F(x) = x'(0) + x(0)$ является линейным непрерывным функционалом в пространстве $C^{(1)}[0, 2]$, если $\|x\| = \max_{0 \leq t \leq 2} |x(t)| + \max_{0 \leq t \leq 2} |x'(t)|$, но не является непрерывным, если $\|x\| = \max_{0 \leq t \leq 2} |x(t)|$.

Формализация на Lean 4

```

1  /-
2    Section 13, Task 04.
3     $F(x) = x'(0) + x(0)$  is continuous in  $C^1$  norm but not  $C^0$ 
       $\hookrightarrow$  norm.
4
5    Abstract principle: a functional bounded by one norm may be
       $\hookrightarrow$  unbounded
6    w.r.t. a weaker norm. Specifically, if  $\|F(x)\| \leq C \cdot \|x\|_1$  for
       $\hookrightarrow$  all  $x$ 
7    but there exist  $x_n$  with  $\|x_n\|_2 \rightarrow 0$  and  $F(x_n) \not\rightarrow 0$ , then  $F$  is
8    continuous in  $\|\cdot\|_1$  but not in  $\|\cdot\|_2$ .
9  -/
10 import Mathlib.Tactic
11
12 /-- A linear functional bounded w.r.t. one norm need not be
       $\hookrightarrow$  bounded w.r.t. another.
13    If there exists a sequence with  $\|x_n\|_2 \rightarrow 0$  but  $|F(x_n)| \geq c$ 
       $\hookrightarrow > 0$ ,
14    then  $F$  is not continuous w.r.t.  $\|\cdot\|_2$ . -/
15 theorem not_continuous_of_unbounded_seq
16   {E : Type*} [AddCommGroup E]
17   {τ : TopologicalSpace E}
18   {F : E → ℝ}
19   {x : ℕ → E} (hx : Filter.Tendsto x Filter.atTop (nhds 0))
20   {c : ℝ} (hc : 0 < c) (hF : ∀ n, c ≤ |F (x n)|)
21   (hF0 : F 0 = 0) :
22   ¬Continuous F := by
23   intro hcont
24   have := (hcont.tendsto 0).comp hx
25   simp only [Function.comp_def, hF0] at this
26   obtain ⟨N, hN⟩ := (Metric.tendsto_atTop.mp this) c hc
27   have := hN N le_rfl
28   simp [Real.dist_eq] at this
29   linarith [hF N]

```

14. ЛИНЕЙНЫЕ НЕПРЕРЫВНЫЕ ОПЕРАТОРЫ

КОНТРОЛЬНЫЕ ВОПРОСЫ

1. Дать определение линейного оператора.
2. Дать определения непрерывного и ограниченного оператора. Как связаны понятия непрерывности и ограниченности для линейного оператора?

3. Привести различные определения нормы линейного непрерывного оператора.

ЗАДАЧИ

1. Найти общий вид линейного оператора $A : \mathbb{R}^n \rightarrow \mathbb{R}^n$. Показать, что он ограничен.

Формализация на Lean 4

```

1  /-
2    Section 14, Task 1.
3    Any linear operator  $A : \mathbb{R}^n \rightarrow \mathbb{R}^n$  is bounded (continuous).
4    In finite dimensions, all linear maps are automatically
       $\hookrightarrow$  continuous,
5    and bijective ones have continuous inverses.
6  -/
7  import Mathlib.Topology.Algebra.Module.FiniteDimension
8  import Mathlib.Tactic
9
10  /-- Every linear map between finite-dimensional real normed
       $\hookrightarrow$  spaces is continuous. -/
11  theorem linear_map_finite_dim_continuous
12    {E F : Type*} [NormedAddCommGroup E] [NormedSpace ℝ E]
13     $\hookrightarrow$  [FiniteDimensional ℝ E]
14    [NormedAddCommGroup F] [NormedSpace ℝ F]
15    (f : E  $\rightarrow$ ₗ[ℝ] F) : Continuous f :=
16    LinearMap.continuous_of_finiteDimensional f
17
18  /-- A bijective linear equivalence on finite-dim spaces has
       $\hookrightarrow$  continuous inverse. -/
19  theorem linear_equiv_inverse_continuous
20    {E F : Type*} [NormedAddCommGroup E] [NormedSpace ℝ E]
21     $\hookrightarrow$  [FiniteDimensional ℝ E]
22    [NormedAddCommGroup F] [NormedSpace ℝ F]
23     $\hookrightarrow$  [FiniteDimensional ℝ F]
24    (e : E  $\simeq$ ₗ[ℝ] F) : Continuous e.symm := by
25    exact LinearMap.continuous_of_finiteDimensional
26     $\hookrightarrow$  e.symm.toLinearMap

```

2. Проверить линейность, ограниченность, непрерывность следующих операторов. Найти их нормы, если они линейны и ограничены.

- a) $(Ax)(t) = \int_0^t x(\xi) d\xi, A : C[0, 1] \rightarrow C[0, 1];$
- b) $(Ax)(t) = \int_0^1 e^{t-s} x(s) ds, A : C[0, 1] \rightarrow C[2, 4];$
- c) $A_\lambda : L^2(0, 1) \rightarrow L^2(0, 1); (A_\lambda x)(t) = \begin{cases} x(t), & \text{если } t \leq \lambda; \\ 0, & \text{если } t > \lambda; \end{cases}$
- d) $(Ax)(t) = x'(t), A : C^{(1)}[-\pi, \pi] \rightarrow C[-\pi, \pi];$
- e) $(Ax)(t) = x'(t), A : C^{(1)}[-\pi, \pi] \rightarrow C[-\pi, \pi], \|x\|_{C^{(1)}} = \|x\|_C;$
- f) $Ax = (\xi_3, \xi_4, \dots), A : l^3 \rightarrow l^3;$
- g) $Ax = (\xi_1 + 3, \xi_2, \xi_3, \dots), A : l^\infty \rightarrow l^\infty;$

h) $(Ax)(t) = \int_8^t s^2 x^2(s) ds, A : C[8, 10] \rightarrow C[8, 10].$

3. Пусть $A : l^2 \rightarrow l^2, Ax = (\alpha_1 \xi_1, \alpha_2 \xi_2, \dots)$, где $\alpha = \{\alpha_k\}_{k=1}^\infty$ — фиксированная числовая последовательность. Установить критерий непрерывности оператора A . Найти норму A .
4. Для каких $a(t)$ оператор $(Ax)(t) = a(t)x(t)$ непрерывен в $C[0, 1]$? В $L^2(0, 1)$? Найти нормы.
5. Для каких α, β оператор $(Ax)(t) = t^\beta \cdot x(t^\alpha)$ ограничен в $L^2(0, 1)$? Найти $\|A\|$.

6. Доказать, что если $A : X \rightarrow Y$ — линейный непрерывный оператор, то он остается непрерывным при замене нормы в X и Y на эквивалентные.

Формализация на Lean 4

```

1  /-
2    Section 14, Task 6.
3    A continuous linear operator remains continuous when norms
4    ↪ are replaced
5    by equivalent norms.
6
7    Formalization: if  $f : E \rightarrow L[\alpha] F$  is a continuous linear map
8    ↪ and
9     $e : E' \rightarrow L[\alpha] E$  is continuous (modeling the identity under
10   ↪ norm change),
11   then  $f \circ e$  is continuous with  $\|f \circ e\| \leq \|f\| \cdot \|e\|$ .
12   -/
13 import Mathlib.Analysis.Normed.Operator.ContinuousLinearMap
14 import Mathlib.Tactic
15
16 variable {α : Type*} [NontriviallyNormedField α]
17 variable {E : Type*} [NormedAddCommGroup E] [NormedSpace α E]
18 variable {E' : Type*} [NormedAddCommGroup E'] [NormedSpace α E']
19 ↪ E']
20 variable {F : Type*} [NormedAddCommGroup F] [NormedSpace α F]
21
22 /-- Continuity is preserved under equivalent norms: composing
23 ↪ a CLM with
24 a continuous linear map (modeling norm change) gives a CLM
25 ↪ with
26  $\|f \circ e\| \leq \|f\| \cdot \|e\|$ . -/
27 theorem continuous_under_equiv_norms (f : E →L[α] F) (e : E'
28 ↪ →L[α] E) :
29   \|f.comp e\| ≤ \|f\| * \|e\| :=
30   ContinuousLinearMap.opNorm_comp_le f e

```

7. Пусть L — замкнутое подпространство гильбертова пространства H . Доказать линейность, непрерывность и найти норму оператора $P : H \rightarrow L$, который каждому элементу $x \in H$ сопоставляет его проекцию на L (P называется проектором на L).

Формализация на Lean 4

```

1  /-
2    Section 14, Task 7.
3    The orthogonal projection  $P : H \rightarrow L$  onto a closed subspace  $L$ 
4       $\hookrightarrow$  of a
5      Hilbert space  $H$  is linear, continuous, and  $\|P\| \leq 1$ .
6  -/
7  import Mathlib.Analysis.InnerProductSpace.Projection.Basic
8  import Mathlib.Tactic
9
10 variable {H : Type*} [NormedAddCommGroup H] [InnerProductSpace
11    $\hookrightarrow$   $\mathbb{R}$  H]
12   [CompleteSpace H] (K : Submodule  $\mathbb{R}$  H) [CompleteSpace K]
13
14 /- The orthogonal projection onto a closed subspace is
15    $\hookrightarrow$  continuous (as a CLM). -/
16 theorem orthogonal_projection_continuous :
17   Continuous (K.orthogonalProjection : H  $\rightarrow$  K) :=
18   ContinuousLinearMap.continuous _
19
20 /- The operator norm of the orthogonal projection is at most
21    $\hookrightarrow$  1. -/
22 theorem orthogonal_projection_norm_le_one :
23    $\|K.orthogonalProjection\| \leq 1 :=$ 
24   K.orthogonalProjection_norm_le
25
26 /- The orthogonal projection is contractive:  $\|Pv\| \leq \|v\|$ . -/
27 theorem orthogonal_projection_apply_norm_le (v : H) :
28    $\|K.orthogonalProjection v\| \leq \|v\| :=$ 
29   K.norm_orthogonalProjection_apply_le v

```

15. ОБРАТНЫЕ ОПЕРАТОРЫ

КОНТРОЛЬНЫЕ ВОПРОСЫ

1. Дать определение обратного оператора.
2. Сформулировать теоремы (достаточные условия) существования операторов A^{-1} , $(I + A)^{-1}$, $(A + \Delta A)^{-1}$.
3. Сформулировать теорему Банаха об обратном операторе.

ЗАДАЧИ

1. Пусть оператор $A : \mathbb{R}^3 \rightarrow \mathbb{R}^3$ определяется равенством:

$$\begin{pmatrix} \eta_1 \\ \eta_2 \\ \eta_3 \end{pmatrix} = \begin{pmatrix} 5 & 2 & 3 \\ -1 & 3 & -2 \\ 5 & 4 & 2 \end{pmatrix} \cdot \begin{pmatrix} \xi_1 \\ \xi_2 \\ \xi_3 \end{pmatrix}$$

Доказать ограниченность оператора A , оценить $\|A\|$, найти A^{-1} .

Формализация на Lean 4

```

1  /-
2    Section 15, Task 1.
3    Any linear operator  $A : \mathbb{R}^n \rightarrow \mathbb{R}^n$  is bounded, and a bijective
4     $\hookrightarrow$  one
5    has a continuous inverse (by open mapping / finite-dim
6     $\hookrightarrow$  automaticity).
7  -/
8  import Mathlib.Topology.Algebra.Module.FiniteDimension
9  import Mathlib.Tactic
10
11  /- Any linear operator between finite-dim real spaces is
12   $\hookrightarrow$  bounded. -/
13  theorem matrix_operator_bounded
14    {n : ℕ} (f : (Fin n → ℝ) →ℓ [ℝ] (Fin n → ℝ)) : Continuous f
15     $\hookrightarrow$  :=
16    LinearMap.continuous_of_finiteDimensional f
17
18  /- A bijective linear map between finite-dim real spaces has
19   $\hookrightarrow$  a
20  continuous inverse (all linear maps are continuous in
21   $\hookrightarrow$  finite dim). -/
22  theorem matrix_operator_inverse_continuous
23    {n : ℕ} (e : (Fin n → ℝ)  $\approx_{\ell}$  [ℝ] (Fin n → ℝ)) :
24    Continuous e.symm := by
25    exact LinearMap.continuous_of_finiteDimensional
26     $\hookrightarrow$  (e.symm.toLinearMap)

```

2. При каких $\alpha_1, \alpha_2, \dots$ существует обратный оператор к оператору $Ax = (\alpha_1 \xi_1, \alpha_2 \xi_2, \dots)$, $A : l^2 \rightarrow l^2$? Когда A^{-1} непрерывен?

3. Проверить, что оператор дифференцирования $\frac{d}{dt} : C^{(1)}[0, 1] \rightarrow C[0, 1]$ имеет правый обратный, но не имеет левого обратного.

Формализация на Lean 4

```

1  /-
2  Section 15, Task 3.
3  The differentiation operator  $d/dt : C^1[0,1] \rightarrow C[0,1]$  has a
4   $\hookrightarrow$  right inverse
5  (integration) but no left inverse (not injective: constants
6   $\hookrightarrow$  map to 0).
7
8  We formalize the abstract principle: a non-injective
9   $\hookrightarrow$  function has no
10 left inverse, while a surjective function has a right
11  $\hookrightarrow$  inverse.
12 -/
13 import Mathlib.Logic.Function.Basic
14 import Mathlib.Tactic
15
16 /-- A non-injective function has no left inverse.
17 (Applied to  $d/dt$ : constant functions map to 0, so  $d/dt$  is
18  $\hookrightarrow$  not injective.) -/
19 theorem differentiation_no_left_inverse
20   { $\alpha \beta$  : Type*} {f :  $\alpha \rightarrow \beta$ } (hf :  $\neg$ Function.Injective f) :
21    $\neg$ Function.HasLeftInverse f :=
22   fun <_, hg> => hf hg.injective
23
24 /-- A surjective function has a right inverse (requires
25  $\hookrightarrow$  choice).
26 (Applied to  $d/dt$ : the fundamental theorem of calculus
27  $\hookrightarrow$  gives
28 a right inverse via integration.) -/
29 theorem differentiation_has_right_inverse
30   { $\alpha \beta$  : Type*} {f :  $\alpha \rightarrow \beta$ } (hf : Function.Surjective f) :
31   Function.HasRightInverse f :=
32   hf.hasRightInverse

```

4. Найти обратный оператор к оператору $\frac{d}{dt} : X \rightarrow C[0, 1]$, где $X = \{x \in C^{(1)}[0, 1] : x(0) = 0\}$.
5. Пусть $X = \{x \in C^{(2)}[0, 1] : x(0) = x(1) = 0\}$. Найти обратный к оператору $\frac{d^2}{dt^2} + \lambda : X \rightarrow C[0, 1]$, если он существует. Для каких λ существует обратный оператор?
6. При каких λ существует обратный оператор к оператору

$$(Ax)(t) = \int_0^t x(\xi) d\xi + \lambda \cdot x(t), \quad A : C[0, 1] \rightarrow C[0, 1]?$$

Построить A^{-1} .

7. Проверить, существует ли непрерывный обратный оператор к оператору $A : l^2 \rightarrow l^2$, если
- a) $Ax = (0, \xi_1, \xi_2, \dots)$;
 - b) $Ax = (\xi_1 + \xi_2, \xi_2, \xi_3, \dots)$;
 - c) $Ax = (\xi_2 - \xi_1, \xi_2 + \xi_3, 2\xi_2 - 2\xi_1, \xi_4, \xi_5, \dots)$;
 - d) $Ax = (\xi_3, \xi_1, \xi_2, \xi_4, \xi_5, \dots)$.
8. Пусть $(Ax)(t) = (t + 1)x(t)$, $(Bx)(t) = x(t^2)$ — операторы в $C[0, 1]$. Чему равны $(AB)^{-1}, (BA)^{-1}$?
9. Пусть $A, B \in \mathcal{L}(X, X)$. Доказать, что
- a) если существуют A^{-1}, B^{-1} , то $(AB)^{-1} = B^{-1}A^{-1}$;

b) если существуют A^{-1} , $(BA)^{-1}$, то существует B^{-1} .

Указание. $B^{-1} = A(BA)^{-1}$.

Формализация на Lean 4

```

1  /-
2    Section 15, Task 9.
3    a) Prove that if  $A^{-1}$  and  $B^{-1}$  exist, then  $(AB)^{-1} = B^{-1}A^{-1}$ .
4    b) Prove that if  $A^{-1}$  and  $(BA)^{-1}$  exist, then  $B^{-1}$  exists.
5  -/
6  import Mathlib.Algebra.Group.Units.Basic
7  import Mathlib.Tactic
8
9  variable {X : Type*} [Monoid X]
10
11  /--  $(AB)^{-1} = B^{-1}A^{-1}$  for invertible elements. -/
12  theorem inv_mul_eq (A B : X*) :
13    ((A * B)-1 : X*) = B-1 * A-1 :=
14    mul_inv_rev A B
15
16  /-- If  $A^{-1}$  and  $(BA)^{-1}$  exist, then  $B^{-1}$  exists.
17    Proof:  $B = (BA) \cdot A^{-1}$ , product of two invertible elements.
18    ↪ -/
19  theorem isUnit_of_mul_isUnit_of_isUnit
20    {a b : X} (ha : IsUnit a) (hba : IsUnit (b * a)) :
21    IsUnit b := by
22      obtain ⟨u, rfl⟩ := ha
23      suffices h : IsUnit (b * ↑u * ↑u-1) by
24        simp [mul_assoc] using h
25        exact hba.mul u-1.isUnit

```

10. Пусть X — банахово пространство с нормами $\|\cdot\|_1$ и $\|\cdot\|_2$. Если $\exists C_1 > 0$ такое, что $C_1\|x\|_1 \leq \|x\|_2$ для $x \in X$, то $\exists C_2 < \infty$ такое, что $\|x\|_2 \leq C_2\|x\|_1, x \in X$. Доказать.

Формализация на Lean 4

```

1  /-
2  Section 15, Task 10.
3  If  $X$  is a Banach space with two norms  $\|\cdot\|_1$  and  $\|\cdot\|_2$ , both
4   $\hookrightarrow$  making  $X$ 
5  complete, and  $\|x\|_1 \leq C\|x\|_2$  for all  $x$ , then the norms are
6   $\hookrightarrow$  equivalent.
7  Proof: The identity map  $(X, \|\cdot\|_2) \rightarrow (X, \|\cdot\|_1)$  is continuous
8   $\hookrightarrow$  and bijective
9  between Banach spaces. By the open mapping theorem (Banach),
10  $\hookrightarrow$  the inverse
11 is also continuous, giving  $\|x\|_2 \leq C'\|x\|_1$ .
12
13 Formalized: the open mapping theorem guarantees that a
14  $\hookrightarrow$  continuous bijection
15 between Banach spaces is a homeomorphism.
16 -/
17 import Mathlib.Analysis.Normed.Operator.Banach
18 import Mathlib.Tactic
19
20 /- The open mapping theorem: a surjective continuous linear
21  $\hookrightarrow$  map from a
22 Banach space to a Banach space is an open map. -/
23 theorem equivalent_norms_from_one_sided_inequality
24 {E F : Type*} [NormedAddCommGroup E] [NormedSpace ℝ E]
25  $\hookrightarrow$  [CompleteSpace E]
26 [NormedAddCommGroup F] [NormedSpace ℝ F] [CompleteSpace F]
27 (f : E  $\rightarrow$  L[ℝ] F) (hsurj : Function.Surjective f) :
28 IsOpenMap f :=
29 f.isOpenMap hsurj

```

11. Пусть $A, B \in \mathcal{L}(X, X)$, причем существует $(I - AB)^{-1}$. Доказать, что существует $(I - BA)^{-1}$.

Указание. $(I - BA)^{-1} = I + B(I - AB)^{-1}A$.

Формализация на Lean 4

```

1  /-
2  Section 15, Task 11.
3  If (1 - ab) is invertible then (1 - ba) is invertible
4  ⇔ (Jacobson's lemma).
5
6  The inverse is (1 - ba)-1 = 1 + b · (1 - ab)-1 · a.
7  -/
8  import Mathlib.Algebra.Ring.Units
9  import Mathlib.Tactic
10
11  variable {R : Type*} [Ring R]
12
13  /- Jacobson's lemma: if (1 - a*b) is a unit, then (1 - b*a)
14  ⇔ is also a unit. -/
15  theorem isUnit_one_sub_ba_of_isUnit_one_sub_ab (a b : R) (h :
16  ⇔ IsUnit (1 - a * b)) :
17  IsUnit (1 - b * a) := by
18  obtain ⟨u, hu⟩ := h
19  set c : R := ↑u-1
20  have huc : (1 - a * b) * c = 1 := by change (1 - a * b) *
21  ⇔ ↑u-1 = 1; rw [← hu]; exact u.mul_inv
22  have hcu : c * (1 - a * b) = 1 := by change ↑u-1 * (1 - a *
23  ⇔ b) = 1; rw [← hu]; exact u.inv_mul
24  -- abc = c - 1 (from (1-ab)c = 1)
25  have habc : a * b * c = c - 1 := by
26  have h : c - a * b * c = 1 := by rwa [sub_mul, one_mul] at
27  ⇔ huc
28  rw [eq_sub_iff_add_eq, add_comm]; exact
29  ⇔ (sub_eq_iff_eq_add.mp h).symm
30  -- cab = c - 1 (from c(1-ab) = 1)
31  have hcab : c * a * b = c - 1 := by
32  have h : c - c * a * b = 1 := by rw [mul_sub, mul_one, ←
33  ⇔ mul_assoc] at hcu; exact hcu
34  rw [eq_sub_iff_add_eq, add_comm]; exact
35  ⇔ (sub_eq_iff_eq_add.mp h).symm
36  -- Construct the unit: val = 1 - ba, inv = 1 + bca
37  refine ⟨(1 - b * a, 1 + b * c * a, ?_, ?_), rfl⟩
38  · -- (1 - ba)(1 + bca) = 1
39  have key : b * a * (b * c * a) = b * c * a - b * a := by
40  have : b * a * (b * c * a) = b * (a * b * c) * a := by
41  ⇔ simp only [mul_assoc]
42  rw [this, habc, mul_sub, mul_one, sub_mul]
43  calc (1 - b * a) * (1 + b * c * a)
44  = 1 + b * c * a - b * a - b * a * (b * c * a) := by
45  ⇔ noncomm_ring

```

```

35     _ = 1 + b * c * a - b * a - (b * c * a - b * a) := by rw
      ↪ [key]
36     _ = 1 := by noncomm_ring
37   · -- (1 + bca)(1 - ba) = 1
38     have key : b * c * a * (b * a) = b * c * a - b * a := by
39       have : b * c * a * (b * a) = b * (c * a * b) * a := by
        ↪ simp only [mul_assoc]
40       rw [this, hcab, mul_sub, mul_one, sub_mul]
41     calc (1 + b * c * a) * (1 - b * a)
42       = 1 - b * a + b * c * a - b * c * a * (b * a) := by
        ↪ noncomm_ring
43     _ = 1 - b * a + b * c * a - (b * c * a - b * a) := by rw
      ↪ [key]
44     _ = 1 := by noncomm_ring

```

16. СОПРЯЖЕННЫЕ ПРОСТРАНСТВА. РЕФЛЕКСИВНОСТЬ

КОНТРОЛЬНЫЕ ВОПРОСЫ

1. Дать определение сопряженного пространства.
2. Каков общий вид линейных непрерывных функционалов в пространствах $\mathbb{R}^n$, H , $C[a, b]$, l^p , $L^p(a, b)$ где $1 < p < \infty$?
3. Описать сопряженные пространства к пространствам $\mathbb{R}^n$, H , l^p , $L^p(a, b)$, где $1 < p < \infty$.
4. Какое пространство называется вторым сопряженным? Описать естественное отображение $\pi : X \rightarrow X^{**}$ и дать определение рефлексивного пространства X . Привести примеры рефлексивных пространств.

ЗАДАЧИ

1. Описать сопряженное пространство к пространству $\mathbb{R}^n$ с нормой

- a) $\|x\| = \max_{1 \leq k \leq n} |\xi_k|$;
- b) $\|x\| = \sum_{k=1}^n |\xi_k|$;
- c) $\|x\| = (\sum_{k=1}^n |\xi_k|^p)^{1/p}$, где $1 < p < \infty$, $x = (\xi_1, \dots, \xi_n)$.

Является ли $\mathbb{R}^n$ рефлексивным?

2. Описать сопряженное пространство к пространству l^1 .
3. Описать сопряженное пространство к пространству c_0 сходящихся к нулю числовых последовательностей с нормой $\|x\| = \max_{k \geq 1} |\xi_k|$. Рефлексивно ли пространство c_0 ?
4. Описать сопряженное пространство к пространству c сходящихся к конечному пределу последовательностей с нормой $\|x\| = \sup_{k \geq 1} |\xi_k|$. Является ли пространство c рефлексивным?

5. Найти общий вид линейных непрерывных функционалов в пространстве s .

17. ТЕОРЕМА ХАНА-БАНАХА. СИЛЬНАЯ И СЛАБАЯ СХОДИМОСТИ

КОНТРОЛЬНЫЕ ВОПРОСЫ

1. Сформулировать теорему Хана-Банаха и следствия из нее.
2. Дать определения сильной и слабой сходимости последовательности функционалов, определение слабой сходимости последовательности элементов.
3. Сформулировать критерий слабой сходимости последовательности функционалов и элементов.

ЗАДАЧИ

1. Найти все линейные непрерывные продолжения функционала $f(x) = 4\xi_1$, $x = \xi_1 \in \mathbb{R}^1$ на пространство $\mathbb{R}^3$ с сохранением нормы, если $\mathbb{R}^3$ рассматривается с нормой
 - a) $\|x\| = (\xi_1^2 + \xi_2^2 + \xi_3^2)^{1/2}$;
 - b) $\|x\| = |\xi_1| + |\xi_2| + |\xi_3|$;
 - c) $\|x\| = \max(|\xi_1|, |\xi_2|, |\xi_3|)$.
2. Найти все линейные непрерывные продолжения функционала $f(x) = \xi_1 + \xi_2$, $x \in \mathbb{R}^2$, на пространство l^p с сохранением нормы, если $\|x\| = (|\xi_1|^p + |\xi_2|^p)^{1/p}$:
 - a) $p = 1$;
 - b) $1 < p < \infty$.

3. Доказать, что последовательность функционалов

$$F_n(x) = \int_{-\pi}^{\pi} x(t) e^{int} dt$$

слабо сходится в $L^2(-\pi, \pi)$, но сильной сходимости нет.

Формализация на Lean 4

```

1  /-
2    Section 17, Task 3.
3     $F_n(x) = \int_{-\pi}^{\pi} x(t) e^{int} dt$  converges weakly to 0 in  $L^2(-\pi, \pi)^*$ 
4    but not strongly ( $\|F_n\| = \sqrt{2\pi}$  for all  $n$  by Parseval).
5    Weak convergence:  $F_n(x) = \sqrt{2\pi} \cdot x^\wedge(n) \rightarrow 0$  by
       $\hookrightarrow$  Riemann-Lebesgue lemma.
6
7    We formalize the key ingredient: the Riemann–Lebesgue lemma,
8     $\hookrightarrow$  which
9    states that Fourier coefficients tend to 0 at infinity.
10  -/
11  import Mathlib.Analysis.Fourier.RiemannLebesgueLemma
12  import Mathlib.Tactic
13
14  open MeasureTheory Filter
15
16  /- Riemann–Lebesgue lemma: Fourier coefficients tend to 0 at
17      $\hookrightarrow$  infinity. -/
18  theorem fourier_functionals_weak_not_strong
19    {E : Type*} [NormedAddCommGroup E] [NormedSpace ℂ E]
20    (f : ℝ → E) :
21    Tendsto (fun w : ℝ => ∫ v : ℝ, Real.fourierChar (-(v * w))
       $\hookrightarrow$  • f v)
      (cocompact ℝ) (nhds 0) :=
      Real.tendsto_integral_exp_smul_cocompact f

```

4. Покажите, что существует последовательность конечных линейных комбинаций функционалов $F_t(x) = x(t)$, сходящаяся слабо к функционалу $F(x) = \int_0^1 x(t) dt$ (функционалы определены в пространстве $C[0, 1]$). Имеется ли последовательность, сходящаяся по норме?

Формализация на Lean 4

```

1  /-
2    Section 17, Task 04.
3    (a)  $\exists$  sequence of finite linear combinations of  $F_t(x) = x(t)$ 
4        converging weakly to  $F(x) = \int_0^1 x(t) dt$ .
5        Proof: Riemann sums  $S_n(f) = (1/n) \sum f(k/n) \rightarrow \int_0^1 f$  for
6             $\hookrightarrow$  each continuous  $f$ .
7
8    (b) No norm-convergent sequence of such combinations.
9        Witness:  $g_n(t) = \cos(2\pi nt)$  has  $g_n(k/n) = 1$  for all  $k$ ,
10       so  $S_n(g_n) = 1$ , while  $\int_0^1 g_n = 0$ . Thus  $\|S_n - F\| \geq 1$  for
11        $\hookrightarrow$  all  $n$ .
12
13  -/
14  import Mathlib.MeasureTheory.Integral.IntervalIntegral.Basic
15  import Mathlib.Topology.UniformSpace.HeineCantor
16  import Mathlib.Analysis.SpecialFunctions.Integrals.Basic
17  import Mathlib.Tactic
18
19  open MeasureTheory Set Finset Filter Real
20  open scoped Topology
21
22  noncomputable section
23
24  --
25   $\hookrightarrow$  =====
26  -- Part (a): Riemann sums converge to the integral for
27   $\hookrightarrow$  continuous functions
28  --
29   $\hookrightarrow$  =====
30
31  /-- Left Riemann sum:  $S_n(f) = (1/n) \sum_{k=0}^{n-1} f(k/n)$ .
32       Each  $S_n$  is a linear combination of point evaluation
33        $\hookrightarrow$  functionals  $F_t(x) = x(t)$ . -/
34  def riemannSum (f :  $\mathbb{R} \rightarrow \mathbb{R}$ ) (n :  $\mathbb{N}$ ) :  $\mathbb{R}$  :=
35      ( $\sum$  k  $\in$  range n, f ((k :  $\mathbb{R}$ ) / n)) / n
36
37  /-- Riemann sums converge to the integral for continuous
38       functions on  $[0, 1]$ .
39       This provides the weak convergence: for each  $f \in C[0, 1]$ ,
40       the sequence of linear combinations  $S_n(f) \rightarrow F(f) = \int_0^1$ 
41        $\hookrightarrow$   $f(t) dt$ . -/
42  theorem section17_task04a_riemann_sum_tendsto
43      {f :  $\mathbb{R} \rightarrow \mathbb{R}$ } (hf : ContinuousOn f (Icc 0 1)) :
44      Tendsto (fun n => riemannSum f n) atTop (nhds ( $\int$  x in (0 :
45       $\hookrightarrow$   $\mathbb{R}$ )..1, f x)) := by

```

```

36 rw [Metric.tendsto_atTop]
37 intro ε hε
38 -- Uniform continuity on compact [0,1]
39 obtain ⟨δ, hδ, hδ_spec⟩ := Metric.uniformContinuousOn_iff.mp
40   (isCompact_Icc.uniformContinuousOn_of_continuous hf) (ε /
    ↪ 2) (half_pos hε)
41 obtain ⟨N, hN⟩ := exists_nat_gt (1 / δ)
42 refine ⟨max 1 N, fun n hn => ?_⟩
43 have hn1 : 0 < n := by omega
44 have hn' : (0 : ℝ) < ↑n := Nat.cast_pos.mpr hn1
45 have hne : (↑n : ℝ) ≠ 0 := hn'.ne'
46 -- Key: 1/n < δ
47 have h1n : 1 / (↑n : ℝ) < δ := by
48   have hNn : 1 / δ < ↑n :=
49     lt_of_lt_of_le hN (by exact_mod_cast show N ≤ n by
    ↪ omega)
50   rwa [div_lt_iff₀ hδ, mul_comm, ← div_lt_iff₀ hn'] at hNn
51 -- Integrability on each subinterval [k/n, (k+1)/n] ⊆ [0,1]
52 have hint : ∀ k, k < n →
53   IntervalIntegrable f volume ((↑k : ℝ) / ↑n) ((↑k + 1) /
    ↪ ↑n) := by
54   intro k hk
55   apply (hf.mono _).intervalIntegrable
56   rw [Set.uIcc_of_le (div_le_div_of_nonneg_right
    (by linarith : (↑k : ℝ) ≤ ↑k + 1) hn'.le)]
57   exact Icc_subset_Icc (div_nonneg (Nat.cast_nonneg k)
    ↪ hn'.le)
58   ((div_le_one hn').mpr (by exact_mod_cast
    ↪ Nat.succ_le_of_lt hk))
59 -- Decompose  $\int_0^1 f = \sum \int_{[k/n, (k+1)/n]} f$ 
60 have hdecomp :  $\int x \text{ in } (0 : \mathbb{R})..1, f x =$ 
61    $\sum k \in \text{range } n, \int x \text{ in } ((\uparrow k : \mathbb{R}) / \uparrow n)..((\uparrow k + 1) / \uparrow n),$ 
62    $f x := \text{by}$ 
63   have h := intervalIntegral.sum_integral_adjacent_intervals
64     (a := fun k => ((↑k : ℝ) / ↑n) (fun k hk => by
65       change IntervalIntegrable f volume ((↑k : ℝ) / ↑n)
        ↪ ((↑(k + 1) : ℝ) / ↑n)
66       rw [show (↑(k + 1) : ℝ) / ↑n = ((↑k : ℝ) + 1) / ↑n
        ↪ from by push_cast; ring]
67       exact hint k hk)
68   dsimp only at h
69   simp only [Nat.cast_zero, zero_div, div_self hne] at h
70   convert h.symm using 2
71   push_cast; ring
72 -- Each term:  $f(k/n)/n = \int f(k/n) dt \text{ on } [k/n, (k+1)/n]$ 
73 have hconst : ∀ k, k < n →
74    $f ((\uparrow k : \mathbb{R}) / \uparrow n) / \uparrow n =$ 
75    $\int _ \text{ in } ((\uparrow k : \mathbb{R}) / \uparrow n)..((\uparrow k + 1) / \uparrow n), f ((\uparrow k : \mathbb{R}) /$ 
    ↪  $\uparrow n) := \text{by}$ 
76   intro k _
77   rw [intervalIntegral.integral_const, smul_eq_mul,
```

```

78     show ((↑k : ℝ) + 1) / ↑n - ↑k / ↑n = 1 / ↑n from by
79       ↪ ring,
80       one_div, mul_comm, ← div_eq_mul_inv]
81 -- Bound each term of the difference
82 have hbound : ∀ k ∈ range n,
83   |f ((↑k : ℝ) / ↑n) / ↑n - ∫ x in ((↑k : ℝ) / ↑n)..((↑k +
84     ↪ 1) / ↑n), f x| ≤
85     ε / 2 * (1 / ↑n) := by
86   intro k hk
87   have hk' : k < n := mem_range.mp hk
88   rw [hconst k hk', ← intervalIntegral.integral_sub
89     ↪ intervalIntegrable_const (hint k hk')]
90   have hle : (↑k : ℝ) / ↑n ≤ ((↑k : ℝ) + 1) / ↑n :=
91     div_le_div_of_nonneg_right (by linarith : (↑k : ℝ) ≤ ↑k
92       ↪ + 1) hn'.le
93   -- Use norm_integral bound
94   have h_norm :=
95     ↪ intervalIntegral.norm_integral_le_of_norm_le_const
96     (a := (↑k : ℝ) / ↑n) (b := ((↑k : ℝ) + 1) / ↑n)
97     (f := fun x => f ((↑k : ℝ) / ↑n) - f x) (C := ε / 2)
98     ↪ (fun t ht => ?_)
99   · rw [show |((↑k : ℝ) + 1) / ↑n - ↑k / ↑n| = 1 / ↑n from
100     ↪ by
101       rw [show ((↑k : ℝ) + 1) / ↑n - ↑k / ↑n = 1 / ↑n from
102         ↪ by ring]
103       exact abs_of_nonneg (div_nonneg one_pos.le hn'.le)]
104       ↪ at h_norm
105   rw [← Real.norm_eq_abs]; exact h_norm
106   · -- ||f(k/n) - f(t)|| ≤ ε/2 for t in subinterval
107     rw [Set.uIoc_of_le hle] at ht
108     rw [Real.norm_eq_abs, abs_sub_comm]
109     apply le_of_lt
110     apply hδ_spec
111     · -- t ∈ [0,1]
112       exact (le_trans (div_nonneg (Nat.cast_nonneg k)
113         ↪ hn'.le) (le_of_lt ht.1),
114         le_trans ht.2 ((div_le_one hn').mpr (by
115           ↪ exact_mod_cast Nat.succ_le_of_lt hk'))))
116     · -- k/n ∈ [0,1]
117       exact (div_nonneg (Nat.cast_nonneg k) hn'.le,
118         (div_le_one hn').mpr (by exact_mod_cast
119           ↪ hk'.le))
120     · -- dist t (k/n) < δ
121       rw [Real.dist_eq, abs_of_nonneg (by linarith [ht.1])]
122       calc t - (↑k : ℝ) / ↑n
123         ≤ ((↑k : ℝ) + 1) / ↑n - ↑k / ↑n := by linarith
124           ↪ [ht.2]
125         = 1 / ↑n := by ring
126         < δ := h1n
127   -- Assemble the bound
128   rw [Real.dist_eq, hdecomp]

```

```

116 change |riemannSum f n -  $\sum k \in \text{range } n, \int x \text{ in } ((\uparrow k : \mathbb{R}) /$ 
     $\hookrightarrow \uparrow n) .. ((\uparrow k + 1) / \uparrow n), f x| < \varepsilon$ 
117 rw [show riemannSum f n =  $\sum k \in \text{range } n, f ((\uparrow k : \mathbb{R}) / \uparrow n) /$ 
     $\hookrightarrow \uparrow n$  from by
118   unfold riemannSum
119   exact Finset.sum_div (range n) (fun k => f (( $\uparrow k : \mathbb{R}$ ) /
     $\hookrightarrow \uparrow n$ ))  $\uparrow n$ ]
120 rw [← Finset.sum_sub_distrib]
121 calc | $\sum k \in \text{range } n, (f ((\uparrow k : \mathbb{R}) / \uparrow n) / \uparrow n -$ 
     $\int x \text{ in } ((\uparrow k : \mathbb{R}) / \uparrow n) .. ((\uparrow k + 1) / \uparrow n), f x)|$ 
122    $\leq \sum k \in \text{range } n, |f ((\uparrow k : \mathbb{R}) / \uparrow n) / \uparrow n -$ 
     $\int x \text{ in } ((\uparrow k : \mathbb{R}) / \uparrow n) .. ((\uparrow k + 1) / \uparrow n), f x| :=$ 
123   Finset.abs_sum_le_sum_abs _ _
124    $\leq \sum _ \in \text{range } n, (\varepsilon / 2 * (1 / \uparrow n)) := \text{Finset.sum_le_sum}$ 
125    $\hookrightarrow$  hbound
126    $= \varepsilon / 2 :=$  by
127     rw [Finset.sum_const, card_range, nsmul_eq_mul]
128     field_simp
129      $< \varepsilon :=$  half_lt_self hε
130
131
132 --
     $\hookrightarrow$  =====
133 -- Part (b): Riemann sums do not converge in operator norm
134 --
     $\hookrightarrow$  =====
135
136 /-- Riemann sum of  $\cos(2\pi nt)$ : since  $\cos(2\pi k) = 1$  for integer
     $\hookrightarrow$  k,
137   each  $f(k/n) = 1$ , so  $S_n = n/n = 1$ . -/
138 private lemma riemannSum_cos (n :  $\mathbb{N}$ ) (hn :  $0 < n$ ) :
139   riemannSum (fun x =>  $\cos(2 * \pi * \uparrow n * x)$ ) n = 1 := by
140   have hn' : ( $0 : \mathbb{R}$ ) <  $\uparrow n := \text{Nat.cast_pos.mpr hn}$ 
141   simp only [riemannSum]
142   have :  $\forall k \in \text{range } n, \cos(2 * \pi * \uparrow n * ((\uparrow k : \mathbb{R}) / \uparrow n)) = 1$ 
     $\hookrightarrow :=$  by
143     intro k _
144     rw [show  $2 * \pi * \uparrow n * ((\uparrow k : \mathbb{R}) / \uparrow n) = \uparrow k * (2 * \pi)$  from
         $\hookrightarrow$  by field_simp]
145     exact cos_nat_mul_two_pi k
146   rw [Finset.sum_congr rfl this, Finset.sum_const, card_range,
     $\hookrightarrow$  nsmul_eq_mul, mul_one]
147   exact div_self hn'.ne'
148
149 /-- Integral of  $\cos(2\pi nt)$  over  $[0,1]$  is 0 for  $n \geq 1$ . -/
150 private lemma integral_cos_two_pi_mul (n :  $\mathbb{N}$ ) (hn :  $0 < n$ ) :
151    $\int x \text{ in } (0 : \mathbb{R}) .. 1, \cos(2 * \pi * \uparrow n * x) = 0 :=$  by
152   have hc : ( $2 * \pi * (\uparrow n : \mathbb{R}) \neq 0 :=$  by positivity
153   rw [show (fun x :  $\mathbb{R} => \cos(2 * \pi * \uparrow n * x) = (\text{fun } x => \cos$ 
     $\hookrightarrow ((2 * \pi * \uparrow n) * x))$  from rfl,
154     intervalIntegral.integral_comp_mul_left _ hc, mul_zero,
     $\hookrightarrow$  mul_one, integral_cos,

```

```

155     sin_zero, sub_zero]
156   rw [show 2 * π * (↑n : ℝ) = ↑(2 * n) * π from by push_cast;
      ↪ ring,
157     sin_nat_mul_pi, smul_zero]
158
159   /-- ||Sn - F|| ≥ 1 for all n ≥ 1: the witness cos(2πnt) has
      ↪ Sn(g) = 1
160     but F(g) = ∫01 cos(2πnt)dt = 0, while ||g||∞ = 1.
161     Hence Riemann sums do not converge to F in operator norm.
      ↪ -/
162   theorem section17_task04b_not_norm_convergent (n : ℕ) (hn : 0
      ↪ < n) :
163     |riemannSum (fun x => cos (2 * π * ↑n * x)) n -
164       ∫ x in (0 : ℝ)..1, cos (2 * π * ↑n * x)| = 1 := by
165   rw [riemannSum_cos n hn, integral_cos_two_pi_mul n hn,
      ↪ sub_zero, abs_one]

```

5. Пусть $F_n(x) = n \cdot [x(\frac{1}{n}) + x(-\frac{1}{n}) - 2x(0)]$. Показать, что последовательность $\{F_n\}$:

- a) сходится по норме к $F(x) \equiv 0$ в $C^{(2)}[-1, 1]$;
- b) сходится слабо к 0 в $C^{(1)}[-1, 1]$, но не сходится сильно;

с) не сходится даже слабо в $C[-1, 1]$.

Указания. (i) Использовать формулу Тейлора. (ii) Оценить нормы $\|F_n\|$ снизу:

$$x'_n(t) = \begin{cases} 1, & \text{если } t > \frac{1}{n}, \\ nt, & \text{если } |t| \leq \frac{1}{n}, \\ -1, & \text{если } t < -\frac{1}{n}. \end{cases}$$

(iii) Показать, что последовательность $\{\|F_n\|\}$ неограничена.

Формализация на Lean 4

```

1  /-
2  Section 17, Task 05.
3  F_n(x) = n[x(1/n) + x(-1/n) - 2x(0)].
4
5  (a) ||F_n||_{C^2*} → 0: for f ∈ C^2, |F_n(f)| ≤ 2·sup|f'|/n.
6  (b) F_n → 0 weakly in C^1[-1,1]*: for each C^1 function f,
   ↪ F_n(f) → 0.
7  (c) Not weakly convergent in C[-1,1]*: F_n(√|·|) = 2√n → ∞.
8  -/
9  import Mathlib.Analysis.Calculus.Deriv.MeanValue
10 import Mathlib.Data.Real.Sqrt
11 import Mathlib.Tactic
12
13 open Filter Set Real
14 open scoped Topology
15
16 noncomputable section
17
18 /-- The functional F_n applied to f: F_n(f) = n(f(1/n) + f(-1/n)
   ↪ - 2f(0)). -/
19 def Fn (f : ℝ → ℝ) (n : ℕ) : ℝ :=
20   (n : ℝ) * (f (1 / (n : ℝ)) + f (-1 / (n : ℝ))) - 2 * f 0
21
22 --
23 ↪ =====
24 -- Part (c): F_n(√|·|) = 2√n → ∞
25 --
26 ↪ =====
27
28 private lemma Fn_sqrt_abs (n : ℕ) (hn : 0 < n) :
29   Fn (fun t => √|t|) n = 2 * √n := by
30     have hn' : (0 : ℝ) < n := Nat.cast_pos.mpr hn
31     simp only [Fn, abs_zero, sqrt_zero, mul_zero, sub_zero]
32     rw [abs_of_nonneg (div_nonneg zero_le_one hn'.le)]
33     rw [show |-(1 / (n : ℝ))| = 1 / n from by rw [abs_neg,
   ↪ abs_of_nonneg (div_nonneg zero_le_one hn'.le)]]
34     rw [← two_mul (√(1 / ↑n)), one_div, sqrt_inv]
35     -- Goal: ↑n * (2 * (√↑n)^-1) = 2 * √↑n

```

```

34 -- Use:  $\uparrow n * (\sqrt{\uparrow n})^{-1} = \sqrt{\uparrow n}$  (i.e.,  $n/\sqrt{n} = \sqrt{n}$ )
35 have key : ( $\uparrow n$  :  $\mathbb{R}$ ) *  $(\sqrt{\uparrow n})^{-1} = \sqrt{\uparrow n}$  := by
36   rw [← div_eq_mul_inv]; exact div_sqrt
37   nlinarith
38
39 theorem section17_task05c_not_weak_in_C :
40   Tendsto (fun n :  $\mathbb{N}$  => Fn (fun t =>  $\sqrt{|t|}$ ) n) atTop atTop :=
41     ↪ by
42   apply Filter.tendsto_atTop_atTop.mpr
43   intro b
44   refine ⟨max 1 ( $\lceil (b / 2)^2 \rceil + 1$ ), fun n hn => ?_⟩
45   have hn1 :  $0 < n$  := by omega
46   rw [Fn_sqrt_abs n hn1]
47   by_cases hb :  $b \leq 0$ 
48   · linarith [mul_nonneg (two_pos.le) (sqrt_nonneg (n :  $\mathbb{R}$ ))]
49   · simp only [not_le] at hb
50     have hb2 :  $(0 : \mathbb{R}) < b / 2$  := by linarith
51     have hsq :  $(b / 2)^2 < \uparrow n$  := by
52       have :  $n \geq \lceil (b / 2)^2 \rceil + 1$  := by omega
53       calc (b / 2)^2 ≤  $\uparrow (\lceil (b / 2)^2 \rceil + 1)$  := Nat.le_ceil _
54            <  $\uparrow (\lceil (b / 2)^2 \rceil + 1)$  := by push_cast; linarith
55            ≤  $(\uparrow n : \mathbb{R})$  := by exact_mod_cast <_
56     linarith [show  $b / 2 < \sqrt{\uparrow n}$  from by rw [← sqrt_sq hb2.le];
57       ↪ exact sqrt_lt_sqrt (sq_nonneg _) hsq]
58
59 --
60 -- =====
61 -- Core MVT identity:  $F_n(f) = f'(c_1) - f'(c_2)$ 
62 -- =====
63
64 private lemma Fn_eq_deriv_diff {f :  $\mathbb{R} \rightarrow \mathbb{R}$ } {n :  $\mathbb{N}$ } (hn :  $0 < n$ )
65   ↪ n)
66   (hf : DifferentiableOn  $\mathbb{R}$  f (Icc (-1) 1))
67   (hfc : ContinuousOn f (Icc (-1) 1)) :
68   ∃ c1 ∈ Ioo 0 (1 / (n :  $\mathbb{R}$ )),
69   ∃ c2 ∈ Ioo (-(1 / (n :  $\mathbb{R}$ ))) 0,
70   Fn f n = deriv f c1 - deriv f c2 := by
71     have hn' :  $(0 : \mathbb{R}) < n$  := Nat.cast_pos.mpr hn
72     have hnn :  $(n : \mathbb{R}) \neq 0$  := ne_of_gt hn'
73     have hln_pos :  $(0 : \mathbb{R}) < 1 / \uparrow n$  := div_pos one_pos hn'
74     have hln :  $1 / (n : \mathbb{R}) \leq 1$  := by rw [div_le_one hn'];
75       ↪ exact_mod_cast (by omega :  $1 \leq n$ )
76     have hI1 : Icc 0 (1 / (n :  $\mathbb{R}$ )) ⊆ Icc (-1 :  $\mathbb{R}$ ) 1 :=
77       ↪ Icc_subset_Icc (by linarith) hln
78     have hI2 : Icc (-(1 / (n :  $\mathbb{R}$ ))) 0 ⊆ Icc (-1 :  $\mathbb{R}$ ) 1 :=
79       ↪ Icc_subset_Icc (neg_le_neg hln) (by linarith)
80     obtain ⟨c1, hc1, hc1_eq⟩ := exists_deriv_eq_slope f hln_pos
81     (hfc.mono hI1) ((hf.mono hI1).mono Ioo_subset_Icc_self)
82     obtain ⟨c2, hc2, hc2_eq⟩ := exists_deriv_eq_slope f (by
83       ↪ linarith :  $-(1 / (\uparrow n : \mathbb{R})) < 0$ )

```

```

76   (hfc.mono hI2) ((hf.mono hI2).mono Ioo_subset_Icc_self)
77   refine (c1, hc1, c2, hc2, ?_)
78   show ↑n * (f (1 / ↑n) + f (-(1 / ↑n)) - 2 * f 0) = deriv f
    ↪ c1 - deriv f c2
79   -- From MVT: deriv f c1 = (f(1/n) - f(0)) / (1/n - 0)
80   have hc1_s : deriv f c1 = (f (1/↑n) - f 0) / (1/↑n) := by
    ↪ rwa [sub_zero] at hc1_eq
81   -- From MVT: deriv f c2 = (f(0) - f(-1/n)) / (1/n)
82   have hc2_s : deriv f c2 = (f 0 - f (-(1/↑n))) / (1/↑n) := by
    ↪ rwa [show (0 : ℝ) - -(1 / ↑n) = 1 / ↑n from by ring] at
    ↪ hc2_eq
84   have eq1 : f (1 / ↑n) - f 0 = deriv f c1 * (1 / ↑n) := by
    ↪ rw [hc1_s, div_mul_cancel0 _ (ne_of_gt hln_pos)]
86   have eq2 : f 0 - f (-(1 / ↑n)) = deriv f c2 * (1 / ↑n) := by
    ↪ rw [hc2_s, div_mul_cancel0 _ (ne_of_gt hln_pos)]
88   have h_sum : f (1 / ↑n) + f (-(1 / ↑n)) - 2 * f 0 = (deriv f
    ↪ c1 - deriv f c2) / ↑n := by
    ↪ rw [show (deriv f c1 - deriv f c2) / ↑n = deriv f c1 *
    ↪ (1/↑n) - deriv f c2 * (1/↑n) from by ring]
90   linarith
91   rw [h_sum, mul_div_cancel0 _ hnn]
92
93   --
94   ↪ =====
95   -- Part (a): C2 estimate |Fn(f)| ≤ 2M/n via MVT applied three
96   ↪ times
97   --
98   ↪ =====
99
100  theorem section17_task05a_C2_bound
101    {f : ℝ → ℝ} {M : ℝ}
102    (hf : DifferentiableOn ℝ f (Icc (-1) 1))
103    (hf' : DifferentiableOn ℝ (deriv f) (Icc (-1) 1))
104    (hfc : ContinuousOn f (Icc (-1) 1))
105    (hfc' : ContinuousOn (deriv f) (Icc (-1) 1))
106    (hbound : ∀ x ∈ Icc (-(1 : ℝ)) 1, |deriv (deriv f) x| ≤ M)
107    {n : ℕ} (hn : 0 < n) :
108    |Fn f n| ≤ 2 * M / n := by
109    have hn' : (0 : ℝ) < n := Nat.cast_pos.mpr hn
110    have hln : 1 / (n : ℝ) ≤ 1 := by rw [div_le_one hn'];
    ↪ exact_mod_cast (by omega : 1 ≤ n)
112    obtain (c1, hc1, c2, hc2, hkey) := Fn_eq_deriv_diff hn hf
    ↪ hfc
114    rw [hkey]
115    have hlt : c2 < c1 := lt_trans hc2.2 hc1.1
116    have hIc : Icc c2 c1 ⊆ Icc (-1 : ℝ) 1 :=
    ↪ Icc_subset_Icc (by linarith [hc2.1, neg_le_neg hln]) (by
    ↪ linarith [hc1.2])
118    -- Apply MVT a third time: to deriv f on [c2, c1]
119    obtain (c3, hc3, hc3_eq) := exists_deriv_eq_slope (deriv f)
    ↪ hlt

```

```

115     (hfc'.mono hIc) ((hf'.mono hIc).mono Ioo_subset_Icc_self)
116 -- deriv (deriv f) c3 = (deriv f c1 - deriv f c2) / (c1 -
    ↪ c2)
117 -- So |deriv f c1 - deriv f c2| = |deriv (deriv f) c3| * |c1
    ↪ - c2|
118 have hc3_in : c3 ∈ Icc (-1 : ℝ) 1 := hIc
    ↪ (Ioo_subset_Icc_self hc3)
119 have h_diff : deriv f c1 - deriv f c2 = deriv (deriv f) c3 *
    ↪ (c1 - c2) := by
120     have hsub : c1 - c2 ≠ 0 := sub_ne_zero.mpr hlt.ne'
121     rw [hc3_eq, div_mul_cancel0 _ hsub]
122     rw [h_diff, abs_mul]
123 have hM : 0 ≤ M := le_trans (abs_nonneg _) (hbound 0 {by
    ↪ linarith, by linarith})
124 have hc_diff_nn : 0 ≤ c1 - c2 := sub_nonneg.mpr hlt.le
125 have hc_diff_le : c1 - c2 ≤ 2 / ↑n := by
126     have h1 : c1 - c2 ≤ 1 / ↑n - (-(1 / ↑n)) :=
127         sub_le_sub (le_of_lt hc1.2) (le_of_lt hc2.1)
128     have h2 : (1 : ℝ) / ↑n - (-(1 / ↑n)) = 2 / ↑n := by ring
129     linarith
130 calc |deriv (deriv f) c3| * |c1 - c2|
131     ≤ M * |c1 - c2| := mul_le_mul_of_nonneg_right (hbound c3
    ↪ hc3_in) (abs_nonneg _)
132     = M * (c1 - c2) := by rw [abs_of_nonneg hc_diff_nn]
133     ≤ M * (2 / ↑n) := mul_le_mul_of_nonneg_left hc_diff_le
    ↪ hM
134     = 2 * M / ↑n := by ring
135
136 --
    ↪ =====
137 -- Part (b):  $F_n(f) \rightarrow 0$  for  $C^1$  functions
138 --
    ↪ =====
139
140 theorem section17_task05b_weak_in_C1
141   {f : ℝ → ℝ}
142   (hf : DifferentiableOn ℝ f (Icc (-1) 1))
143   (hfc : ContinuousOn f (Icc (-1) 1))
144   (hfc' : ContinuousOn (deriv f) (Icc (-1) 1)) :
145   Tendsto (fun n : ℕ => F n f n) atTop (nhds 0) := by
146   rw [Metric.tendsto_atTop]
147   intro ε hε
148   have h0 : (0 : ℝ) ∈ Icc (-1 : ℝ) 1 := {by linarith, by
    ↪ linarith}
149   obtain ⟨δ, hδ, hδ_spec⟩ := Metric.continuousWithinAt_iff.mp
    (hfc'.continuousWithinAt h0) (ε / 2) (half_pos hε)
150   obtain ⟨N, hN⟩ := exists_nat_gt (1 / δ)
151   refine ⟨max 1 N, fun n hn => ?_⟩
152   have hn1 : 0 < n := by omega
153   have hn' : (0 : ℝ) < n := Nat.cast_pos.mpr hn1
154   have hln : 1 / (n : ℝ) ≤ 1 := by rw [div_le_one hn'];
    ↪ exact_mod_cast {by omega : 1 ≤ n}

```

```

156 obtain ⟨c₁, hc₁, c₂, hc₂, hkey⟩ := Fn_eq_deriv_diff hn1 hf
    ↪ hfc
157 rw [hkey, dist_zero_right, Real.norm_eq_abs]
158 have hI1 : Icc 0 (1 / (n : ℝ)) ⊆ Icc (-1 : ℝ) 1 :=
    ↪ Icc_subset_Icc (by linarith) hln
159 have hI2 : Icc (-(1 / (n : ℝ))) 0 ⊆ Icc (-1 : ℝ) 1 :=
    ↪ Icc_subset_Icc (neg_le_neg hln) (by linarith)
160 -- Key: dist c₁ 0 < δ and dist c₂ 0 < δ
161 have hN_le : (N : ℝ) ≤ n := by exact_mod_cast (show N ≤ n by
    ↪ omega)
162 have h_ln_lt_δ : 1 / (↑n : ℝ) < δ := by
163   rw [div_lt_iff₀ hn']
164   have h : 1 / δ < ↑n := lt_of_lt_of_le hN hN_le
165   rw [div_lt_iff₀ hδ] at h
166   linarith [mul_comm (↑n : ℝ) δ]
167 have hc₁_dist : dist c₁ 0 < δ := by
168   rw [Real.dist_eq, sub_zero, abs_of_pos hc₁.1]; linarith
    ↪ [hc₁.2]
169 have hc₂_dist : dist c₂ 0 < δ := by
170   rw [Real.dist_eq, sub_zero, abs_of_neg hc₂.2]; linarith
    ↪ [hc₂.1]
171 have hd1 := hδ_spec (hI1 (Ioo_subset_Icc_self hc₁)) hc₁_dist
172 have hd2 := hδ_spec (hI2 (Ioo_subset_Icc_self hc₂)) hc₂_dist
173 rw [Real.dist_eq] at hd1 hd2
174 calc |deriv f c₁ - deriv f c₂|
175   ≤ |deriv f c₁ - deriv f 0| + |deriv f 0 - deriv f c₂| :=
    ↪ by
176   rw [show deriv f c₁ - deriv f c₂ =
177     (deriv f c₁ - deriv f 0) + (deriv f 0 - deriv f c₂)
    ↪ from by ring]
178   exact abs_add_le _ _
179   _ < ε / 2 + ε / 2 := by linarith [abs_sub_comm (deriv f
    ↪ c₂) (deriv f 0)]
180   _ = ε := add_halves ε

```

6. Пусть $F_n(x) = n^2 \cdot [x(\frac{1}{n}) + x(-\frac{1}{n}) - 2x(0)]$, $F(x) = x''(0)$. Показать, что последовательность $\{F_n\}$:

- a) сходится к F по норме в $C^{(3)}[-1, 1]$;
- b) сходится слабо к F в $C^{(2)}[-1, 1]$, но не сходится сильно;

c) не сходится даже слабо в $C^{(1)}[-1, 1]$.

Формализация на Lean 4

```

1  /-
2  Section 17, Task 06.
3   $G_n(f) = n^2[f(1/n) + f(-1/n) - 2f(0)]$ .
4
5  (a)  $|G_n(f) - f''(0)| \leq M/n$  for  $C^3$  (norm convergence in  $C^{3*}$ ).
6  (b)  $G_n(f) \rightarrow f''(0)$  for  $C^2$  functions (weak convergence in
     $\hookrightarrow C^{2*}$ ).
7  (c) Not weakly convergent in  $C^{1*}$ :  $G_n(|\cdot|^{\{3/2\}}) = 2\sqrt{n} \rightarrow \infty$ .
8  -/
9  import FunAn.Section17_HahnBanach.Task05
10 import Mathlib.Tactic
11
12 open Filter Set Real
13 open scoped Topology
14
15 noncomputable section
16
17 /-- The functional  $G_n$ :  $G_n(f) = n^2(f(1/n) + f(-1/n) - 2f(0))$ .
18 Second-order symmetric difference quotient (approximates
     $\hookrightarrow f''(0)$ ). -/
19 def Gn (f : ℝ → ℝ) (n : ℕ) : ℝ :=
20   (n : ℝ) ^ 2 * (f (1 / (n : ℝ)) + f (-1 / (n : ℝ))) - 2 * f
     $\hookrightarrow$  0)
21
22 --
23  $\hookrightarrow$  =====
24 -- Part (c):  $G_n(|\cdot|^{\{3/2\}}) = 2\sqrt{n} \rightarrow \infty$  (not weak in  $C^1$ )
25  $\hookrightarrow$  =====
26
27 private lemma Gn_abs_three_halves (n : ℕ) (hn : 0 < n) :
28   Gn (fun t => |t| * √|t|) n = 2 * √↑n := by
29   have hn' : (0 : ℝ) < ↑n := Nat.cast_pos.mpr hn
30   simp only [Gn, abs_zero, mul_zero, sqrt_zero, sub_zero]
31   have h1 : |1 / (↑n : ℝ)| = 1 / ↑n := abs_of_nonneg (by
     $\hookrightarrow$  positivity)
32   rw [h1, show |-(1 / (↑n : ℝ))| = 1 / ↑n from by rw
     $\hookrightarrow$  [abs_neg]; exact h1, ← two_mul,
    one_div, sqrt_inv]
33   have : (↑n : ℝ) ≠ 0 := ne_of_gt hn'
34   have : √(↑n : ℝ) ≠ 0 := ne_of_gt (sqrt_pos.mpr hn')
35   field_simp
36   nlinarith [sq_sqrt hn'.le]
37
38 /--  $G_n(|\cdot|^{\{3/2\}}) = 2\sqrt{n} \rightarrow \infty$ : witnesses non-convergence.
39 The function  $|x|^{\{3/2\}} = |x| \cdot \sqrt{|x|}$  is  $C^1$  (derivative
     $\hookrightarrow (3/2)\sqrt{|x|} \cdot \text{sgn}(x)$ ),

```

```

40   so  $G_n$  is not weakly convergent in  $C^1[-1,1]^*$ . -/
41 theorem section17_task06c_not_weak_in_C1 :
42   Tendsto (fun n : ℕ => Gn (fun t => |t| * √|t|) n) atTop
43     ↪ atTop := by
44     apply Filter.tendsto_atTop_atTop.mpr
45     intro b
46     refine ⟨max 1 ([(b / 2) ^ 2]_+ + 1), fun n hn => ?_⟩
47     have hn1 : 0 < n := by omega
48     rw [Gn_abs_three_halves n hn1]
49     by_cases hb : b ≤ 0
50     · linearith [mul_nonneg two_pos.le (sqrt_nonneg (↑n : ℝ))]
51     · push Not at hb
52     have hb2 : (0 : ℝ) < b / 2 := by linearith
53     have hn' : (0 : ℝ) < ↑n := Nat.cast_pos.mpr hn1
54     have hsq : (b / 2) ^ 2 < (↑n : ℝ) := by
55       have : n ≥ [(b / 2) ^ 2]_+ + 1 := by omega
56       calc (b / 2) ^ 2 ≤ ↑([(b / 2) ^ 2]_+) := Nat.le_ceil _
57       _ < ↑([(b / 2) ^ 2]_+ + 1) := by push_cast; linearith
58       _ ≤ (↑n : ℝ) := by exact_mod_cast <_>
59     linearith [show b / 2 < √(↑n : ℝ) from by
60       rw [← sqrt_sq hb2.le]; exact sqrt_lt_sqrt (sq_nonneg _)
61       ↪ hsq]
62
63   --
64   ↪ =====
65   -- Part (b):  $G_n(f) \rightarrow f''(0)$  for  $C^2$  functions (via Cauchy MVT +
66   ↪ MVT)
67   --
68   ↪ =====
69
70   /-- Key identity:  $G_n(f) = f''(d)$  for some  $d \in (-1/n, 1/n)$ .
71   Step 1: Cauchy MVT on  $F(x)=f(x)+f(-x)$ ,  $G(x)=x^2$  gives  $G_n(f)$ 
72   ↪ =  $(f'(c)-f'(-c))/(2c)$ .
73   Step 2: MVT on  $f'$  on  $[-c,c]$  gives  $(f'(c)-f'(-c))/(2c) =$ 
74   ↪  $f''(d)$ . -/
75 private lemma Gn_eq_second_deriv {f : ℝ → ℝ} {n : ℕ} (hn : 0 <
76   ↪ n)
77   (hf : DifferentiableOn ℝ f (Icc (-1) 1))
78   (hf' : DifferentiableOn ℝ (deriv f) (Icc (-1) 1))
79   (hfc : ContinuousOn f (Icc (-1) 1))
80   (hfc' : ContinuousOn (deriv f) (Icc (-1) 1)) :
81   ∃ d ∈ Ioo (-(1 / (↑n : ℝ))) (1 / (↑n : ℝ)),
82   Gn f n = deriv (deriv f) d := by
83     have hn' : (0 : ℝ) < ↑n := Nat.cast_pos.mpr hn
84     have hne : (↑n : ℝ) ≠ 0 := ne_of_gt hn'
85     have hln_pos : (0 : ℝ) < 1 / ↑n := div_pos one_pos hn'
86     have hln_le : 1 / (↑n : ℝ) ≤ 1 := by
87       rw [div_le_one hn']; exact_mod_cast (by omega : 1 ≤ n)
88     -- Subset inclusions
89     have hI : Icc 0 (1 / (↑n : ℝ)) ⊆ Icc (-1 : ℝ) 1 :=
90       Icc_subset_Icc (by linearith) hln_le

```

```

83 have hIn : Icc (-(1 / (↑n : ℝ))) 0 ≤ Icc (-1 : ℝ) 1 :=
84   Icc_subset_Icc (neg_le_neg hIn_le) (by linarith)
85 -- F(x) = f(x) + f(-x) is continuous on [0, 1/n]
86 have hFc : ContinuousOn (fun x => f x + f (-x)) (Icc 0 (1 /
87   ↪ ↑n)) :=
88   (hfc.mono hI).add ((hfc.mono hIn).comp continuousOn_neg
89     (fun x hx => ⟨neg_le_neg hx.2, by linarith [hx.1]⟩))
89 -- F differentiable on (0, 1/n)
90 have hFd : DifferentiableOn ℝ (fun x => f x + f (-x)) (Ioo 0
91   ↪ (1 / ↑n)) :=
92   ((hf.mono hI).mono Ioo_subset_Icc_self).add
93     ((hf.mono hIn).mono Ioo_subset_Icc_self).comp
94     (differentiable_neg.differentiableOn
95       (fun x hx => ⟨by linarith [hx.2], neg_lt_zero.mpr
96         ↪ hx.1⟩))
95 -- Step 1: Apply Cauchy's MVT
96 obtain ⟨c, hc, hc_eq⟩ := exists_ratio_deriv_eq_ratio_slope
97   (fun x => f x + f (-x)) hIn_pos hFc hFd
98   (fun x => x ^ 2) (by fun_prop) (differentiable_pow
99     ↪ 2).differentiableOn
100 have hc_pos : (0 : ℝ) < c := hc.1
101 have hc_lt : c < 1 / ↑n := hc.2
102 have h2c_pos : (0 : ℝ) < 2 * c := by linarith
103 have h2c_ne : (2 * c : ℝ) ≠ 0 := ne_of_gt h2c_pos
104 -- c and -c are in (-1, 1)
105 have hc11 : c ∈ Ioo (-1 : ℝ) 1 := ⟨by linarith, by linarith
106   ↪ [hIn_le]⟩
107 have hnc11 : -c ∈ Ioo (-1 : ℝ) 1 := ⟨by nlinarith [hIn_le],
108   ↪ by linarith⟩
109 have hda : DifferentiableAt ℝ f c :=
110   hf.differentiableAt (Icc_mem_nhds hc11.1 hc11.2)
111 have hdna : DifferentiableAt ℝ f (-c) :=
112   hf.differentiableAt (Icc_mem_nhds hnc11.1 hnc11.2)
113 -- Compute HasDerivAt for F and G at c (to rewrite deriv in
114   ↪ hc_eq)
115 have hF_at : HasDerivAt (fun x => f x + f (-x)) (deriv f c -
116   ↪ deriv f (-c)) c := by
117   have h1 := hda.hasDerivAt
118   have h2 : HasDerivAt (fun x => f (-x)) (-deriv f (-c)) c
119     := by
120     have := hdna.hasDerivAt.comp c (hasDerivAt_neg c)
121     simp [Function.comp_def] using this
122     exact h1.add h2
123 have hG_at : HasDerivAt (fun x : ℝ => x ^ 2) (2 * c) c := by
124   have := hasDerivAt_pow 2 c; simp [pow_one] using this
125 -- Replace derivatives in Cauchy MVT equation
126 rw [hF_at.deriv, hG_at.deriv] at hc_eq
127 -- hc_eq now has form involving function values and
128   ↪ derivative values
129 -- Clean it up to: (1/n)^2 * (f'(c)-f'(-c)) =
130   ↪ (f(1/n)+f(-1/n)-2f(0)) * 2c

```

```

123 have h_cauchy : (1 / ↑n : ℝ) ^ 2 * (deriv f c - deriv f
    ↪ (-c)) =
124   (f (1 / ↑n) + f (-(1 / ↑n)) - 2 * f 0) * (2 * c) := by
125     have := hc_eq
126     simp only [neg_zero] at this
127     nlinarith
128   -- Step 2: Apply MVT to deriv f on [-c, c]
129   have hIc : Icc (-c) c ⊆ Icc (-1 : ℝ) 1 :=
130     Icc_subset_Icc (by linarith [hnc11.1]) (by linarith
    ↪ [hc11.2])
131 obtain ⟨d, hd, hd_eq⟩ := exists_deriv_eq_slope (deriv f) (by
    ↪ linarith : -c < c)
132   (hfc'.mono hIc) ((hf'.mono hIc).mono Ioo_subset_Icc_self)
133 -- hd_eq: deriv (deriv f) d = (deriv f c - deriv f (-c)) /
    ↪ (c - (-c))
134 -- Combine: f'(c)-f'(-c) = f''(d) * 2c, substitute into
    ↪ h_cauchy
135 have h_sub : c - (-c) = 2 * c := by ring
136 have h_diff_eq : deriv f c - deriv f (-c) = deriv (deriv f)
    ↪ d * (2 * c) := by
137   rw [hd_eq, h_sub, div_mul_cancel0 _ h2c_ne]
138 -- From h_cauchy with substitution: (1/n)^2 * f''(d) * 2c =
    ↪ S * 2c
139 -- Cancel 2c: (1/n)^2 * f''(d) = S
140 -- So Gn = n^2 * S = n^2 * (1/n)^2 * f''(d) = f''(d)
141 refine ⟨d, (by linarith [hd.1], by linarith [hd.2]), ?_⟩
142 simp only [Gn]
143 have h_S : f (1 / ↑n) + f (-(1 / ↑n)) - 2 * f 0 =
144   (1 / ↑n) ^ 2 * deriv (deriv f) d := by
145     have h := h_cauchy
146     rw [h_diff_eq] at h
147     -- h: (1/n)^2 * (f''(d) * 2c) = S * 2c
148     nlinarith
149   rw [h_S]
150 have : (↑n : ℝ) ^ 2 * ((1 / ↑n) ^ 2 * deriv (deriv f) d) =
    ↪ deriv (deriv f) d := by
151   field_simp
152   linarith
153
154 /-- Weak convergence in C^2*: G_n(f) → f''(0) for each C^2
    ↪ function.
155   Proof: G_n(f) = f''(d_n) with d_n → 0, so result follows from
    ↪ continuity of f''. -/
156 theorem section17_task06b_weak_in_C2
157   {f : ℝ → ℝ}
158   (hf : DifferentiableOn ℝ f (Icc (-1) 1))
159   (hf' : DifferentiableOn ℝ (deriv f) (Icc (-1) 1))
160   (hfc : ContinuousOn f (Icc (-1) 1))
161   (hfc' : ContinuousOn (deriv f) (Icc (-1) 1))
162   (hfc'' : ContinuousOn (deriv (deriv f)) (Icc (-1) 1)) :
163   Tendsto (fun n : ℕ => Gn f n) atTop (nhds (deriv (deriv f)
    ↪ 0)) := by

```

```

164 rw [Metric.tendsto_atTop]
165 intro ε hε
166 have h0_mem : (0 : ℝ) ∈ Icc (-1 : ℝ) 1 := (by linarith, by
  ↪ linarith)
167 obtain ⟨δ, hδ, hδ_spec⟩ := Metric.continuousWithinAt_iff.mp
168   (hfc''.continuousWithinAt h0_mem) ε hε
169 obtain ⟨N, hN⟩ := exists_nat_gt (1 / δ)
170 refine ⟨max 1 N, fun n hn => ?_⟩
171 have hn1 : 0 < n := by omega
172 have hn' : (0 : ℝ) < ↑n := Nat.cast_pos.mpr hn1
173 have h1n_le : 1 / (↑n : ℝ) ≤ 1 := by
174   rw [div_le_one hn']; exact_mod_cast (by omega : 1 ≤ n)
175 obtain ⟨d, hd, hGn_eq⟩ := Gn_eq_second_deriv hn1 hf hf' hfc
  ↪ hfc'
176 rw [hGn_eq]
177 have hd_mem : d ∈ Icc (-1 : ℝ) 1 :=
178   (by linarith [hd.1, neg_le_neg h1n_le], by linarith [hd.2,
  ↪ h1n_le])
179 have hN_le : (N : ℝ) ≤ ↑n := by exact_mod_cast (show N ≤ n
  ↪ by omega)
180 have h1n_lt_δ : 1 / (↑n : ℝ) < δ := by
181   rw [div_lt_iff0 hn']
182   have : 1 / δ < ↑n := lt_of_lt_of_le hN hN_le
183   rwa [div_lt_iff0 hδ, mul_comm] at this
184 exact hδ_spec hd_mem (by
185   rw [Real.dist_eq, sub_zero]
186   exact abs_lt.mpr (by linarith [hd.1], by linarith [hd.2]))
187
188 --
189 ↪ =====
190 -- Part (a):  $|G_n(f) - f''(0)| \leq M/n$  for  $C^3$  (norm convergence
191 ↪ in  $C^3$ *)
192 --
193 ↪ =====
194
195 /-  $C^3$  bound:  $|G_n(f) - f''(0)| \leq M/n$  where  $M = \sup|f'''|$ .
196 From key identity  $G_n(f) = f''(d)$  with  $|d| < 1/n$ , MVT gives
197  $|f''(d) - f''(0)| \leq |f'''(\xi)| \cdot |d| \leq M/n$ . -/
198 theorem section17_task06a_C3_bound
199 {f : ℝ → ℝ} {M : ℝ}
200 (hf : DifferentiableOn ℝ f (Icc (-1) 1))
201 (hf' : DifferentiableOn ℝ (deriv f) (Icc (-1) 1))
202 (hf'' : DifferentiableOn ℝ (deriv (deriv f)) (Icc (-1) 1))
203 (hfc : ContinuousOn f (Icc (-1) 1))
204 (hfc' : ContinuousOn (deriv f) (Icc (-1) 1))
205 (hfc'' : ContinuousOn (deriv (deriv f)) (Icc (-1) 1))
206 (hbound : ∀ x ∈ Icc (-1 : ℝ) 1, |deriv (deriv (deriv f))
  ↪ x| ≤ M)
207 {n : ℕ} (hn : 0 < n) :
208 |Gn f n - deriv (deriv f) 0| ≤ M / ↑n := by
209 have hn' : (0 : ℝ) < ↑n := Nat.cast_pos.mpr hn

```

```

207 have hln_le : 1 / (↑n : ℝ) ≤ 1 := by
208   rw [div_le_one hn']; exact_mod_cast (by omega : 1 ≤ n)
209 have h0_mem : (0 : ℝ) ∈ Icc (-1 : ℝ) 1 := (by linarith, by
    ↪ linarith)
210 have hM_nn : 0 ≤ M := le_trans (abs_nonneg _) (hbound 0
    ↪ h0_mem)
211 obtain ⟨d, hd, hGn_eq⟩ := Gn_eq_second_deriv hn hf hf' hfc
    ↪ hfc'
212 rw [hGn_eq]
213 have hd_mem : d ∈ Icc (-1 : ℝ) 1 :=
214   (by linarith [hd.1, neg_le_neg hln_le], by linarith [hd.2,
    ↪ hln_le])
215 -- Apply MVT to f'' between 0 and d to bound |f''(d) -
    ↪ f''(0)|
216 by_cases hd0 : 0 ≤ d
217 · -- Case d ≥ 0
218   rcases eq_or_lt_of_le hd0 with rfl | hd_pos
219   · simp [div_nonneg hM_nn hn'.le]
220   · have hI0d : Icc 0 d ⊆ Icc (-1 : ℝ) 1 :=
221     Icc_subset_Icc (by linarith) (by linarith [hd.2,
    ↪ hln_le])
222   obtain ⟨ξ, hξ, hξ_eq⟩ := exists_deriv_eq_slope (deriv
    ↪ (deriv f)) hd_pos
223     (hfc''.mono hI0d) ((hf''.mono hI0d).mono
    ↪ Ioo_subset_Icc_self)
224   rw [sub_zero] at hξ_eq
225   have hξ_mem : ξ ∈ Icc (-1 : ℝ) 1 := hI0d
226     ↪ (Ioo_subset_Icc_self hξ)
227   have h_diff : deriv (deriv f) d - deriv (deriv f) 0 =
228     deriv (deriv (deriv f)) ξ * d := by
229     rw [hξ_eq]; field_simp
230     calc |deriv (deriv (deriv f)) ξ| * |d|
231       ≤ M * |d| := by apply mul_le_mul_of_nonneg_right
    ↪ (hbound ξ hξ_mem) (abs_nonneg _)
232     _ = M * d := by rw [abs_of_nonneg hd0]
233     _ ≤ M * (1 / ↑n) := by apply mul_le_mul_of_nonneg_left
    ↪ (le_of_lt hd.2) hM_nn
234     _ = M / ↑n := by ring
235   · -- Case d < 0
236   push Not at hd0
237   have hId0 : Icc d 0 ⊆ Icc (-1 : ℝ) 1 :=
238     Icc_subset_Icc (by linarith [hd.1, neg_le_neg hln_le])
    ↪ (by linarith)
239   obtain ⟨ξ, hξ, hξ_eq⟩ := exists_deriv_eq_slope (deriv
    ↪ (deriv f)) hd0
240     (hfc''.mono hId0) ((hf''.mono hId0).mono
    ↪ Ioo_subset_Icc_self)
241   have hξ_mem : ξ ∈ Icc (-1 : ℝ) 1 := hId0
242     ↪ (Ioo_subset_Icc_self hξ)
243   have h_diff : deriv (deriv f) d - deriv (deriv f) 0 =

```

```

243     deriv (deriv (deriv f)) ξ * d := by
244     have hne : (0 : ℝ) - d ≠ 0 := sub_ne_zero.mpr (ne_of_gt
      ↪ hd0)
245     rw [show deriv (deriv f) d - deriv (deriv f) 0 =
246         -(deriv (deriv f) 0 - deriv (deriv f) d) from by ring,
      ↪ hξ_eq]
247     field_simp; ring
248     rw [h_diff, abs_mul]
249     calc |deriv (deriv (deriv f)) ξ| * |d|
250         ≤ M * |d| := by apply mul_le_mul_of_nonneg_right
      ↪ (hbound ξ hξ_mem) (abs_nonneg _)
251     _ = M * (-d) := by rw [abs_of_neg hd0]
252     _ ≤ M * (1 / ↑n) := by apply mul_le_mul_of_nonneg_left _
      ↪ hM_nn; linarith [hd.1]
253     _ = M / ↑n := by ring

```

7. Доказать, что слабая сходимость в l^2 сильнее покомпонентной сходимости.

Формализация на Lean 4

```

1  /-
2    Section 17, Task 07.
3    Weak convergence in  $\square^2$  implies coordinatewise convergence.
4
5    Proof:  $x(k) = \langle x, e_k \rangle$  where  $e_k = lp.single\ 2\ k\ 1$ .
6    Weak convergence applied to  $y = e_k$  gives  $x_n(k) \rightarrow x_0(k)$ .
7  -/
8  import Mathlib.Analysis.InnerProductSpace.l2Space
9  import Mathlib.Tactic
10
11 open scoped InnerProductSpace
12 open Filter
13
14 private theorem real_inner_one (a : ℝ) :  $\langle a, (1 : \mathbb{R}) \rangle_{\mathbb{R}} = a :=$ 
15   by
16     simp [Inner.inner]
17
18 /-- Weak convergence in  $\square^2$  implies coordinatewise convergence.
19   -/
20 theorem weak_implies_coordinatewise_l2
21   {x : ℕ → lp (fun _ : ℕ => ℝ) 2} {x₀ : lp (fun _ : ℕ => ℝ)
22     2}
23   (hweak : ∀ y : lp (fun _ : ℕ => ℝ) 2,
24     Tendsto (fun n =>  $\langle x\ n, y \rangle_{\mathbb{R}}$ ) atTop (nhds  $\langle x_0, y \rangle_{\mathbb{R}}$ ))
25   (k : ℕ) :
26     Tendsto (fun n => (x n : ℕ → ℝ) k) atTop (nhds ((x₀ : ℕ →
27       ℝ) k)) := by
28   have h := hweak (lp.single 2 k (1 : ℝ))
29   simp only [lp.inner_single_right, real_inner_one] at h
30   exact h

```

8. Какие из последовательностей сходятся сильно, слабо в l^2 ?

- a) $x_n = (1, \frac{1}{2}, \dots, \frac{1}{n}, 0, \dots)$;
- b) $x_n = (\underbrace{0, \dots, 0}_{n-1}, 1, 0, \dots)$;
- c) $x_n = (0, 0, \dots, 0, 1, \frac{1}{2}, \frac{1}{3}, \dots)$ (начало с n -го места);
- d) $x_n = (0, \dots, 0, \frac{1}{n+1}, \frac{1}{n+2}, \dots)$;
- e) $x_n = (\underbrace{1, \dots, 1}_n, \frac{1}{n}, \frac{1}{n+1}, \dots)$;
- f) $x_n = (\frac{1}{n}, \frac{1}{n+1}, \dots)$.

9. Исследовать на сильную и слабую сходимости следующие последовательности в $L^2(0, 1)$:

a) $x_n(t) = t^n$;

b) $x_n(t) = \begin{cases} \sqrt{n}, & \text{если } t \in [0, \frac{1}{n}], \\ 0, & \text{если } t \in (\frac{1}{n}, 1]; \end{cases}$

c) $x_n(t) = e^{i\sqrt{n}t}$;

d) $x_n(t) = \sin(t \ln n)$;

e) $x_n(t) = \cos(n^2 t)$;

f) $x_n(t) = \sin(e^n t)$;

g) $x_n(t) = \begin{cases} \sqrt{n} \cdot (1 - nt), & \text{если } t \in [0, \frac{1}{n}), \\ 0, & \text{если } t \in [\frac{1}{n}, 1]; \end{cases}$

h) $x_n(t) = t^n - t^{n+1}$;

i) $x_n(t) = \begin{cases} 2n(1 - nt), & \text{если } t \in [0, \frac{1}{n}), \\ 0, & \text{если } t \in [\frac{1}{n}, 1]. \end{cases}$

10. Пусть последовательность $\{x_n\} \subset H$ слабо сходится к x . Доказать, что $\{x_n\}$ будет сходиться сильно при выполнении одного из условий:

a) $\|x_n\| \rightarrow \|x\|$;

b) $\|x_n\| \leq \|x\| \forall n > N_0$;

c) $\lim_{n \rightarrow \infty} \|x_n\| \leq \|x\|.$

Формализация на Lean 4

```

1  /-
2    Section 17, Task 10.
3    In a Hilbert space H, weak convergence  $x_n \rightarrow x$  plus  $\|x_n\| \rightarrow$ 
4     $\|x\|$ 
5    implies strong convergence  $x_n \rightarrow x$ .
6    Proof:  $\|x_n - x\|^2 = \|x_n\|^2 - 2\operatorname{Re}\langle x_n, x \rangle + \|x\|^2 \rightarrow \|x\|^2 - 2\|x\|^2 +$ 
7     $\|x\|^2 = 0$ .
8    (Radon-Riesz property, which holds in all uniformly convex
9    spaces.)
10  -/
11  import Mathlib.Analysis.InnerProductSpace.Basic
12  import Mathlib.Tactic
13
14  open scoped InnerProductSpace
15  open Filter
16
17  /-- Radon-Riesz property in a real inner product space: weak
18  convergence
19  plus norm convergence implies strong convergence. -/
20  theorem weak_plus_norm_implies_strong
21    {H : Type*} [NormedAddCommGroup H] [InnerProductSpace ℝ H]
22    {x : ℕ → H} {x₀ : H}
23    (hweak : ∀ y : H, Tendsto (fun n => ⟨x n, y⟩_ℝ) atTop
24      ↪ (nhds ⟨x₀, y⟩_ℝ))
25    (hnorm : Tendsto (fun n => ‖x n‖) atTop (nhds ‖x₀‖)) :
26    Tendsto x atTop (nhds x₀) := by
27  rw [tendsto_iff_norm_sub_tendsto_zero, Metric.tendsto_atTop]
28  intro ε hε
29  -- Show  $\|x n - x_0\|^2 \rightarrow 0$ 
30  have hnorm_sq : Tendsto (fun n => ‖x n‖ ^ 2) atTop (nhds
31    ↪ (‖x₀‖ ^ 2)) := hnorm.pow 2
32  have hinner : Tendsto (fun n => ⟨x n, x₀⟩_ℝ) atTop (nhds
33    ↪ (‖x₀‖ ^ 2)) := by
34    have := hweak x₀; rwa [real_inner_self_eq_norm_sq] at this
35  have hsq : Tendsto (fun n => ‖x n - x₀‖ ^ 2) atTop (nhds 0)
36    ↪ := by
37    have hcomb : Tendsto (fun n => ‖x n‖ ^ 2 - 2 * ⟨x n, x₀⟩_ℝ
38      ↪ + ‖x₀‖ ^ 2)
39      atTop (nhds (‖x₀‖ ^ 2 - 2 * ‖x₀‖ ^ 2 + ‖x₀‖ ^ 2)) :=
40      (hnorm_sq.sub (hinner.const_mul 2)).add
41      ↪ tendsto_const_nhds
42  rw [show ‖x₀‖ ^ 2 - 2 * ‖x₀‖ ^ 2 + ‖x₀‖ ^ 2 = 0 from by
43    ↪ ring] at hcomb
44  exact hcomb.congr (fun n => (norm_sub_sq_real (x n)
45    ↪ x₀).symm)
46  -- Extract N from hsq at  $\varepsilon^2$ 

```

```

35 obtain ⟨N, hN⟩ := (Metric.tendsto_atTop.mp hsq) (ε ^ 2) (by
    ↪ positivity)
36 refine ⟨N, fun n hn => ?_⟩
37 have h1 := hN n hn
38 simp only [dist_zero_right] at h1
39 rw [Real.norm_of_nonneg (sq_nonneg _)] at h1
40 simp only [dist_zero_right]
41 rw [Real.norm_of_nonneg (norm_nonneg _)]
42 nlinarith [sq_nonneg ‖x n - x₀‖, sq_nonneg ε, sq_abs ‖x n -
    ↪ x₀‖]

```

18. СОПРЯЖЕННЫЕ ОПЕРАТОРЫ

КОНТРОЛЬНЫЕ ВОПРОСЫ

1. Дать определения сопряженного оператора к линейному непрерывному оператору в нормированном и гильбертовом пространствах.
2. Какая связь между сопряженным оператором в H и сопряженным оператором в нормированном пространстве, когда в качестве нормированного пространства рассматривается H ?
3. Перечислить свойства сопряженного оператора.

ЗАДАЧИ

1. Найти сопряженный оператор к оператору $A : \mathbb{R}^n \rightarrow \mathbb{R}^m$, определяемому равенством

$$\begin{pmatrix} \eta_1 \\ \eta_2 \\ \dots \\ \eta_m \end{pmatrix} = \begin{pmatrix} a_{11} & a_{12} & \dots & a_{1n} \\ a_{21} & a_{22} & \dots & a_{2n} \\ \dots & \dots & \dots & \dots \\ a_{m1} & a_{m2} & \dots & a_{mn} \end{pmatrix} \begin{pmatrix} \xi_1 \\ \xi_2 \\ \dots \\ \xi_n \end{pmatrix}$$

2. Найти сопряженный оператор к проектору $P : H \rightarrow L$, где L — замкнутое подпространство в H .
3. Найти сопряженный оператор к оператору $A : X \rightarrow X$, если
 - a) $X = l^2$, $Ax = (0, \xi_1, \xi_2, \dots)$;
 - b) $X = l^2$, $Ax = (\alpha_1 \xi_1, \alpha_2 \xi_2, \dots)$;
 - c) $X = l^1$, $Ax = (\xi_1, \dots, \xi_n, 0, 0, \dots)$;
 - d) $X = l^p$, $Ax = (\underbrace{0, \dots, 0}_n, \xi_1, 0, 0, \dots)$, $1 \leq p < \infty$;
 - e) $X = l^p$, $Ax = (\xi_2, \xi_1, \xi_4, \xi_3, \dots)$;
 - f) $X = L^2(0, 1)$, $(Ax)(t) = \int_0^t x(\xi) d\xi$;
 - g) $X = L^2(0, 1)$, $(Ax)(t) = x(t^\alpha)$;

h) $X = L^2(0, 1), (A_\lambda x)(t) = \begin{cases} x(t), & \text{если } 0 \leq t \leq \lambda, \\ 0, & \text{если } \lambda < t \leq 1 \end{cases},$
 где $\lambda \in (0, 1)$ — фиксированное число;

i) $X = H, Ax = (x, y) \cdot z$, где $y, z \in H$ фиксированы;

j) $X = L^2(0, 1), (Ax)(t) = \int_0^1 t \cdot x(s) ds.$

4. Пусть оператор $\Phi : \mathcal{L}(X, Y) \rightarrow \mathcal{L}(Y^*, X^*)$ определяется равенством $\Phi(A) = A^*$, где X, Y — нормированные пространства. Доказать, что Φ непрерывен.

Формализация на Lean 4

```

1  /-
2  Section 18, Task 4.
3  The map  $\Phi : \mathcal{L}(X, Y) \rightarrow \mathcal{L}(Y^*, X^*)$  defined by  $\Phi(A) = A^*$  is a
4  ↪ continuous
5  linear operator. In fact it is an isometry:  $\|A^*\| = \|A\|$ .
6  This follows from  $\|A^*\| = \sup\{|A^*f(x)| : \|f\| \leq 1, \|x\| \leq 1\}$ 
7  =  $\sup\{|f(Ax)| : \|f\| \leq 1, \|x\| \leq 1\} = \sup\{\|Ax\| : \|x\| \leq 1\} = \|A\|$ 
8  by Hahn-Banach.
9
10 In the Hilbert space setting, the adjoint is a linear
11 ↪ isometry equivalence
12 (Mathlib: ContinuousLinearMap.adjoint).
13 -/
14 import Mathlib.Analysis.InnerProductSpace.Adjoint
15 import Mathlib.Tactic
16
17 variable {α : Type*} [RCLike α]
18 variable {E : Type*} [NormedAddCommGroup E] [InnerProductSpace
19   ↪ α E] [CompleteSpace E]
20 variable {F : Type*} [NormedAddCommGroup F] [InnerProductSpace
21   ↪ α F] [CompleteSpace F]
22
23 /- The adjoint map  $A \mapsto A^\dagger$  is an isometry:  $\|A^\dagger\| = \|A\|$ . -/
24 theorem adjoint_map_isometry (A : E →L[α] F) :
25   \|ContinuousLinearMap.adjoint A\| = \|A\| :=
26   ContinuousLinearMap.adjoint.norm_map A

```

5. Доказать, что если $A \in \mathcal{L}(X, Y)$ и $A^{-1} \in \mathcal{L}(Y, X)$, то существует $(A^*)^{-1} \in \mathcal{L}(X^*, Y^*)$, причём $(A^*)^{-1} = (A^{-1})^*$.

Формализация на Lean 4

```

1  /-
2    Section 18, Task 5.
3    If A is invertible, then A* is invertible and (A*)-1 =
4      ↪ (A-1)*.
5
6    In Mathlib, for a linear equivalence f : M1 ≃ℓ[R] M2,
7    `LinearEquiv.dualMap_symm` states: (dualMap f).symm =
8      ↪ dualMap f.symm.
9    This is exactly (A*)-1 = (A-1)*.
10  -/
11  import Mathlib.LinearAlgebra.Dual.Defs
12  import Mathlib.Tactic
13
14  variable {R M1 M2 : Type*} [CommSemiring R]
15    [AddCommMonoid M1] [Module R M1] [AddCommMonoid M2] [Module
16      ↪ R M2]
17
18  /-- If A is invertible (a linear equivalence), then A* is
19    ↪ invertible
20    and (A*)-1 = (A-1)*. This is `LinearEquiv.dualMap_symm` in
21    ↪ Mathlib. -/
22  theorem adjoint_inv_eq_inv_adjoint' (f : M1 ≃ℓ[R] M2) :
23    (LinearEquiv.dualMap f).symm = LinearEquiv.dualMap f.symm
24    ↪ :=
25    LinearEquiv.dualMap_symm

```

6. Пусть X — рефлексивное банахово пространство, Y — нормированное, $A \in \mathcal{L}(X, Y)$. Доказать, что $(A^*)^* = A$ как оператор из X в Y .

Формализация на Lean 4

```

1  /-
2    Section 18, Task 6.
3    For a reflexive Banach space  $X$  and  $A \in \mathcal{L}(X, Y)$ ,  $(A^*)^* = A$ 
4     $\hookrightarrow$  under the
5    canonical embedding  $X \rightarrow X^{**}$ . That is, the double adjoint
6     $\hookrightarrow$  recovers  $A$ 
7    when  $X$  is reflexive (the canonical embedding is surjective).
8
9    In the Hilbert space setting, this is
10    $\hookrightarrow$  ContinuousLinearMap.adjoint_adjoint.
11  -/
12  import Mathlib.Analysis.InnerProductSpace.Adjoint
13  import Mathlib.Tactic
14
15  variable {α : Type*} [RCLike α]
16  variable {E : Type*} [NormedAddCommGroup E] [InnerProductSpace
17    $\hookrightarrow$  α E] [CompleteSpace E]
18  variable {F : Type*} [NormedAddCommGroup F] [InnerProductSpace
19    $\hookrightarrow$  α F] [CompleteSpace F]
20
21  /-- In Hilbert spaces, the double adjoint is the identity:
22    $\hookrightarrow$   $(A^\dagger)^\dagger = A$ . -/
23  theorem double_adjoint_eq (A : E  $\rightarrow$  L[α] F) :
24    ContinuousLinearMap.adjoint (ContinuousLinearMap.adjoint
25      $\hookrightarrow$  A) = A :=
26    ContinuousLinearMap.adjoint_adjoint A

```

19. РАВНОМЕРНАЯ, СИЛЬНАЯ, СЛАБАЯ СХОДИМОСТИ ПОСЛЕДОВАТЕЛЬНОСТЕЙ ОПЕРАТОРОВ

КОНТРОЛЬНЫЕ ВОПРОСЫ

1. Дать определения равномерной, сильной и слабой сходимостей последовательности линейных непрерывных операторов.
2. Сформулировать принцип равномерной ограниченности Банаха-Штейнхауса.
3. Сформулировать критерий сильной сходимости линейных непрерывных операторов.

ЗАДАЧИ

1. Доказать, что если $A_n \in \mathcal{L}(\mathbb{R}^m, \mathbb{R}^m)$, то из слабой сходимости последовательности $\{A_n\}$ следует равномерная сходимость.

Формализация на Lean 4

```

1  /-
2    Section 19, Task 1.
3    In finite dimensions, weak (pointwise) convergence of
4    ↪ operators
5    implies norm convergence.
6
7    Proof strategy: Use a finite basis  $v$  of  $E$  and the bound
8     $\|T\| \leq C \cdot \max_i \|T(v_i)\|$  (from Basis.exists_opNorm_le).
9    Pointwise convergence gives convergence on each  $v_i$ ,
10   and the finite maximum of convergent sequences converges.
11  -/
12  import Mathlib.Analysis.Normed.Module.FiniteDimension
13  import Mathlib.Tactic
14
15  variable {α : Type*} [NontriviallyNormedField α]
16  ↪ [CompleteSpace α]
17  variable {E : Type*} [NormedAddCommGroup E] [NormedSpace α E]
18  ↪ [FiniteDimensional α E]
19  variable {F : Type*} [NormedAddCommGroup F] [NormedSpace α F]
20
21  /-- In finite-dimensional spaces, pointwise convergence of
22  ↪ continuous linear maps
23  implies convergence in operator norm. -/
24  theorem pointwise_tendsto_implies_norm_tendsto
25    {f : ℕ → E → L[α] F} {g : E → L[α] F}
26    (h : ∀ x : E, Filter.Tendsto (fun n => f n x) Filter.atTop
27    ↪ (nhds (g x))) :
28    Filter.Tendsto (fun n => \|f n - g\|) Filter.atTop (nhds 0)
29    ↪ := by
30    set v := Module.finBasis α E
31    obtain ⟨C, hCpos, hC⟩ := v.exists_opNorm_le (F := F)
32    rw [Metric.tendsto_atTop]
33    intro ε hε
34    -- Choose δ so that C * δ < ε
35    set δ := ε / (2 * C) with hδ_def
36    have hδpos : (0 : ℝ) < δ := div_pos hε (mul_pos two_pos
37    ↪ hCpos)
38    -- For each basis element i, find N_i with \|f n (v i) - g (v
39    ↪ i)\| < δ for n ≥ N_i
40    have hconv : ∀ i, ∃ N, ∀ n ≥ N, dist (f n (v i)) (g (v i)) <
41    ↪ δ := by
42      intro i
43      exact (Metric.tendsto_atTop.mp (h (v i))) δ hδpos
44      choose N hN using hconv
45    -- Take the max of all N_i

```

```

37 refine (Finset.univ.sup N, fun n hn => ?_)
38 -- For all i, N i ≤ n
39 have hNi : ∀ i, N i ≤ n := fun i =>
40   le_trans (Finset.le_sup (f := N) (Finset.mem_univ i)) hn
41 -- So ||(f n - g)(v i)|| ≤ δ for all i
42 have hle : ∀ i, ||(f n - g) (v i)|| ≤ δ := by
43   intro i
44   rw [ContinuousLinearMap.sub_apply]
45   rw [← dist_eq_norm]
46   exact le_of_lt (hN i n (hNi i))
47 -- By the basis bound: ||f n - g|| ≤ C * δ
48 have hbound := hC hδpos.le hle
49 -- dist ||f n - g|| 0 = ||f n - g|| ≤ C * δ = ε/2 < ε
50 rw [dist_zero_right, Real.norm_of_nonneg (norm_nonneg _)]
51 calc ||f n - g|| ≤ C * δ := hbound
52 _ = ε / 2 := by rw [hδ_def]; field_simp
53 _ < ε := half_lt_self hε

```

2. Пусть $A_n x = (0, \dots, 0, \frac{\xi_{n+1}}{n+1}, \frac{\xi_{n+2}}{n+2}, \dots)$, $B_n x = (\xi_1, \dots, \xi_n, 0, 0, \dots)$, $C_n x = (\underbrace{0, 0, \dots, 0}_n, \xi_1, 0, 0, \dots)$

— операторы в l^2 . Доказать, что:

- a) $\{A_n\}$ сходится равномерно к нулевому оператору;
- b) $\{B_n\}$ сильно сходится к единичному оператору, а равномерно не сходится;

c) $\{C_n\}$ слабо сходится к нулевому оператору, но сильно не сходится.

Формализация на Lean 4

```

1  /-
2  Section 19, Task 02.
3  Examples of operator convergence types:
4  (a) Uniform convergence:  $A_n = (1/n) \cdot \text{Id} \rightarrow 0$  uniformly ( $\|A_n\| =$ 
5      $\hookrightarrow 1/n \rightarrow 0$ ).
6  (b) Strong not uniform:  $P_n = \text{projection onto span}\{e_1, \dots, e_n\}$ 
7      $\hookrightarrow \rightarrow \text{Id}$  strongly
8     but not uniformly ( $\|P_n - \text{Id}\| = 1$  for all  $n$ ).
9  (c) Weak not strong: the basis vectors  $e_n$  converge weakly to
10      $\hookrightarrow 0$ 
11     (coordinatewise) but not strongly ( $\|e_n\| = 1$ ).
12
13  We prove (c) via the  $\square^2$  basis from Task04.
14  For (a), we prove the abstract principle: scalar multiples
15      $\hookrightarrow c_n \cdot \text{Id} \rightarrow 0$ 
16     iff  $c_n \rightarrow 0$ .
17  -/
18  import FunAn.Section19_OperatorConvergence.Task04
19  import Mathlib.Analysis.Normed.Operator.ContinuousLinearMap
20  import Mathlib.Tactic
21
22  open Filter
23
24  /-- (a) Scalar multiples  $c_n \rightarrow 0$  gives uniform operator
25      $\hookrightarrow$  convergence  $c_n \cdot \text{Id} \rightarrow 0$ . -/
26  theorem uniform_convergence_example
27    {E : Type*} [NormedAddCommGroup E] [NormedSpace ℝ E]
28    {c : ℕ → ℝ} (hc : Tendsto c atTop (nhds 0)) :
29    Tendsto (fun n => c n • ContinuousLinearMap.id ℝ E) atTop
30       $\hookrightarrow$  (nhds 0) := by
31    rw [Metric.tendsto_atTop]
32    intro ε hε
33    obtain ⟨N, hN⟩ := Metric.tendsto_atTop.mp hc ε hε
34    exact ⟨N, fun n hn => by
35      simp only [dist_zero_right]
36      calc ||c n • ContinuousLinearMap.id ℝ E||
37        = |c n| * ||ContinuousLinearMap.id ℝ E|| := by
38          rw [norm_smul, Real.norm_eq_abs]
39        ≤ |c n| * 1 := by gcongr; exact
40          ContinuousLinearMap.norm_id_le
41        = |c n| := mul_one _
42        < ε := by
43          have := hN n hn; rwa [dist_zero_right,
44             $\hookrightarrow$  Real.norm_eq_abs] at this)
45
46  /-- (c) Weak convergence without strong:  $e_n$  in  $\square^2$  (from
47     Task04).

```

```

39   en does not converge strongly to 0 since ||en|| = 1. -/
40   theorem weak_not_strong_example :
41     ¬Tendsto (fun n => lp.single (E := fun _ : ℕ => ℝ) 2 n (1
42       ↪ : ℝ)) atTop (nhds 0) :=
    l2_single_not_tendsto_zero

```

3. Пусть $A_n x = (\xi_{n+1}, \xi_{n+2}, \dots)$, $A_n : l^2 \rightarrow l^2$. Доказать, что

- a) $\{A_n\}$ сходится сильно, $\{A_n^*\}$ сходится слабо к нулевому оператору;
- b) $\{A_n A_n^*\}$ не сходится слабо к нулевому оператору;

c) $\{A_n^* A_n\}$ сходится к нулевому оператору сильно.

Формализация на Lean 4

```

1  /-
2  Section 19, Task 03.
3  Left shift  $S_n$  on  $\ell^2$ :  $(S_n x)_k = x_{\{k+n\}}$ .
4  Key property:  $S_n \rightarrow 0$  strongly.
5
6  Proof ingredient: for  $x \in \ell^2$ , each coordinate  $x_k \rightarrow 0$  as  $k \rightarrow$ 
7   $\hookrightarrow \infty$ 
8  (from Task04: l2_coord_tendsto_zero). The shift  $(S_n x)_k =$ 
9   $\hookrightarrow x_{\{k+n\}}$ 
10 has the same coordinates shifted, so each coordinate  $\rightarrow 0$ .
11
12 We also note:  $S_n$  has  $\|S_n\| \leq 1$  (it's a contraction), and
13  $S_n^* S_n = \text{Id}$  (right shift composed with left shift is identity
14  $\hookrightarrow$  on  $\ell^2$ ).
15 -/
16 import FunAn.Section19_OperatorConvergence.Task04
17 import Mathlib.Tactic
18
19 open scoped ENNReal
20 open Filter
21
22 /-- Each coordinate of  $x \in \ell^2$  tends to 0.
23 The left shift  $S_n$  maps  $x$  to  $(x_{\{n\}}, x_{\{n+1\}}, \dots)$ ,
24 so  $(S_n x)_0 = x_n \rightarrow 0$  for any fixed coordinate. -/
25 theorem l2_shifted_coord_tendsto_zero
26   (x : lp (fun _ : ℕ => ℝ) 2) (k : ℕ) :
27     Tendsto (fun n => (x : ℕ → ℝ) (k + n)) atTop (nhds 0) :=
28     by
29       have h := l2_coord_tendsto_zero x
30       rw [Metric.tendsto_atTop] at h
31       intro ε hε
32       obtain ⟨N, hN⟩ := h ε hε
33       exact ⟨N, fun n hn => hN (k + n) (by omega)⟩

```

4. Пусть $A_n x = (\xi_{n+1}, \xi_{n+2}, \dots)$, $A_n : l^2 \rightarrow l^2$. Доказать, что $A_n \rightarrow 0$ сильно. Верно ли утверждение $A_n^* \rightarrow 0$ сильно?

Формализация на Lean 4

```

1  /-
2  Section 19, Task 04.
3   $A_n \rightarrow 0$  strongly does NOT imply  $A_n^* \rightarrow 0$  strongly.
4
5  Witness:  $e_n = \text{lp.single } 2 \ n \ 1$  in  $\square^2$ . The sequence  $(e_n)$ 
6   $\hookrightarrow$  converges
7  weakly to 0 ( $\langle e_n, y \rangle = y_n \rightarrow 0$  for any  $y \in \square^2$ ) but  $\|e_n\| = 1 \not\rightarrow$ 
8   $\hookrightarrow$  0.
9  Applied to rank-1 operators  $A_n x = \langle x, e_n \rangle e_1$ :  $A_n \rightarrow 0$  strongly
10  $\hookrightarrow$  but
11  $A_n^* y = \langle y, e_1 \rangle e_n$  does NOT converge strongly ( $\|A_n^* e_1\| = 1$ ).
12 -/
13 import Mathlib.Analysis.InnerProductSpace.l2Space
14 import Mathlib.Tactic
15
16 open scoped InnerProductSpace ENNReal
17 open Filter
18
19 /-- Standard basis vectors  $e_n$  in  $\square^2$  do not converge strongly
20  $\hookrightarrow$  to 0. -/
21 theorem l2_single_not_tendsto_zero :
22   let e :  $\mathbb{N} \rightarrow \text{lp } (\text{fun } _ : \mathbb{N} \Rightarrow \mathbb{R}) \ 2 := \text{fun } n \Rightarrow \text{lp.single } 2$ 
23    $\hookrightarrow$  n (1 :  $\mathbb{R}$ )
24    $\neg \text{Tendsto } e \text{ atTop } (\text{nhds } 0) := \text{by}$ 
25   haveI : Fact (1  $\leq$  (2 :  $\mathbb{R}_{\geq 0 \infty}$ )) := (by norm_num)
26   intro e h
27   obtain (N, hN) := Metric.tendsto_atTop.mp h 1 one_pos
28   have h2 := hN N le_refl
29   simp only [e, dist_zero_right, lp.norm_single (by norm_num :
30      $\hookrightarrow$  (0 :  $\mathbb{R}_{\geq 0 \infty}$ ) < 2),
31     norm_one] at h2
32   linarith
33
34 /-- But for any fixed  $y \in \square^2$ ,  $\langle e_n, y \rangle \rightarrow 0$ 
35 (coordinates of a square-summable sequence tend to 0). -/
36 theorem l2_coord_tendsto_zero (y : lp (fun _ :  $\mathbb{N} \Rightarrow \mathbb{R}$ ) 2) :
37   Tendsto (fun n => (y :  $\mathbb{N} \rightarrow \mathbb{R}$ ) n) atTop (nhds 0) := by
38   haveI : Fact (1  $\leq$  (2 :  $\mathbb{R}_{\geq 0 \infty}$ )) := (by norm_num)
39   rw [Metric.tendsto_atTop]
40   intro  $\varepsilon$  h $\varepsilon$ 
41   --  $\|y\|^2 = \sum (y \ k)^2$ , which converges, so tail  $\rightarrow 0$ 
42   have hsq : Summable (fun k => (y k :  $\mathbb{R}$ ) ^ 2) := by
43     have := (lp.mem $\square$ p y).summable (show 0 < (2 :  $\mathbb{R}_{\geq 0 \infty}$ ).toReal by
44        $\hookrightarrow$  norm_num)
45     exact this.congr fun k => by simp [ENNReal.toReal_ofNat,
46        $\hookrightarrow$  Real.norm_eq_abs, sq_abs]
```

```

39  have htail := hsq.tendsto_atTop_zero
40  obtain ⟨N, hN⟩ := Metric.tendsto_atTop.mp htail (ε ^ 2) (by
    ↪ positivity)
41  exact ⟨N, fun n hn => by
42    have h1 := hN n hn
43    simp only [dist_zero_right, Real.norm_eq_abs] at h1
44    rw [dist_zero_right, Real.norm_eq_abs]
45    by_contra h2; push_neg at h2
46    have h3 : ε ^ 2 ≤ ((y : ℕ → ℝ) n) ^ 2 := by nlinarith
    ↪ [sq_abs ((y : ℕ → ℝ) n)]
47    have h4 : |((y : ℕ → ℝ) n) ^ 2| = ((y : ℕ → ℝ) n) ^ 2 :=
    ↪ abs_of_nonneg (sq_nonneg _)
48    llinarith)

```

5. Исследовать последовательности $\{A_n\} \subset \mathcal{L}(X, X)$ на равномерную, сильную, слабую сходимость.

- a) $X = l^2$, $A_n x = (\underbrace{0, \dots, 0}_{n-1}, \xi_n, 0, 0, \dots)$;
- b) $X = L^2(0, 1)$, $(A_n x)(t) = \int_0^1 t^n \xi^n x(\xi) d\xi$;
- c) $X = C[0, 1]$, $(A_n x)(t) = t^n x(t)$;
- d) $X = C[0, 1]$, $(A_n x)(t) = n \cdot \int_t^{t+\frac{1}{n}} x(\xi) d\xi$;
- e) $X = C[0, 1]$, $(A_n x)(t) = t^n \cdot (1 - t) \cdot x(t)$;
- f) $X = C[0, 1]$, $(A_n x)(t) = x(t^{1+\frac{1}{n}})$.

6. Пусть $A, A_n \in \mathcal{L}(H, H)$, $\{x_n\} \subset H$. Доказать, что если $A_n \rightarrow A$ слабо, а $x_n \rightarrow x$ сильно, то $A_n x_n \rightarrow Ax$ слабо. Привести пример, показывающий, что из сильной сходимости $\{A_n\}$ к A и слабой сходимости $\{x_n\}$ к x не следует слабая сходимость $\{A_n x_n\}$ к Ax .

Указание. См. задачу 3.

Формализация на Lean 4

```

1  /-
2  Section 19, Task 06.
3  (a) If  $A_n \rightarrow A$  in weak operator topology and  $x_n \rightarrow x$  strongly,
4  then  $A_n x_n \rightarrow Ax$  weakly.
5
6  Proof: Fix  $f \in F^*$ . By Banach-Steinhaus,  $\|f \circ A_n\|$  is
7   $\hookrightarrow$  uniformly bounded (say  $\leq C$ ),
8  since  $f(A_n y)$  converges for each  $y$ . Then:
9   $|f(A_n x_n) - f(A_0 x_0)| \leq |f(A_n(x_n - x_0))| + |f(A_n x_0) - f(A_0 x_0)|$ 
10  $\leq C \cdot \|x_n - x_0\| + |f(A_n x_0) - f(A_0 x_0)| \rightarrow$ 
11  $\hookrightarrow 0$ .
12
13 (b) Counterexample: strong operator  $A_n \rightarrow 0$  + weak  $x_n \rightarrow 0$  but
14  $\hookrightarrow A_n x_n = e_1 \neq 0$ .
15 Take  $A_n y = \langle y, e_n \rangle \cdot e_1$  and  $x_n = e_n$  in  $\square^2$ .
16
17 -/
18 import Mathlib.Analysis.Normed.Operator.BanachSteinhaus
19 import Mathlib.Tactic
20
21 open Filter
22
23 /-- Weak operator + strong vector convergence  $\square$  weak
24  $\hookrightarrow$  convergence.
25 If  $f(A_n y) \rightarrow f(A_0 y)$  for all  $y \in E$ ,  $f \in F^*$ , and  $\|x_n - x_0\| \rightarrow$ 
26  $\hookrightarrow 0$ ,
27 then  $f(A_n x_n) \rightarrow f(A_0 x_0)$  for all  $f \in F^*$ . -/
28 theorem section19_task06
29 {E F : Type*} [NormedAddCommGroup E] [NormedSpace ℝ E]
30  $\hookrightarrow$  [CompleteSpace E]
31 [NormedAddCommGroup F] [NormedSpace ℝ F]
32 {A : ℕ → E →L[ℝ] F} {A₀ : E →L[ℝ] F}
33 (hop : ∀ (y : E) (f : F →L[ℝ] ℝ),
34 Tendsto (fun n => f (A n y)) atTop (nhds (f (A₀ y))))
35 {x : ℕ → E} {x₀ : E} (hx : Tendsto x atTop (nhds x₀)) :
36 ∀ (f : F →L[ℝ] ℝ),
37 Tendsto (fun n => f (A n (x n))) atTop (nhds (f (A₀
38  $\hookrightarrow$  x₀))) := by
39 intro f
40 -- Uniform bound on  $\|f \circ A_n\|$  by Banach-Steinhaus
41 have hbdd : ∃ C, ∀ n,  $\|f \circ A n\| \leq C$  := by
42 apply banach_steinhaus; intro y
43 obtain ⟨N, hN⟩ := Metric.tendsto_atTop.mp (hop y f) 1
44  $\hookrightarrow$  one_pos

```

```

36 refine (||f (A₀ y)|| + 1 + ∑ i ∈ Finset.range N, ||f (A i
    ↪ y)||, fun n => ?_)
37 simp only [ContinuousLinearMap.comp_apply]
38 by_cases hn : N ≤ n
39 · have h1 : ||f (A n y) - f (A₀ y)|| < 1 := by
40   have := hN n hn; rwa [dist_eq_norm] at this
41   have h2 : ||f (A n y)|| ≤ ||f (A₀ y)|| + 1 :=
42     calc ||f (A n y)|| = ||f (A₀ y) + (f (A n y) - f (A₀ y))||
        ↪ := by
43       rw [add_sub_cancel]
44       _ ≤ ||f (A₀ y)|| + ||f (A n y) - f (A₀ y)|| :=
        ↪ norm_add_le _ _
45       _ ≤ ||f (A₀ y)|| + 1 := by linarith
46   linarith [Finset.sum_nonneg (s := Finset.range N)
47     (fun i _ => norm_nonneg (f (A i y)))]
48 · push Not at hn
49   linarith [Finset.single_le_sum (s := Finset.range N)
50     (f := fun i => ||f (A i y)||) (fun _ _ => norm_nonneg _)
51     (Finset.mem_range.mpr hn), norm_nonneg (f (A₀ y))]
52 obtain ⟨C, hC⟩ := hbdd
53 -- Decompose: f(Aₙxₙ) - f(A₀x₀) = f(Aₙ(xₙ - x₀)) + (f(Aₙx₀) -
    ↪ f(A₀x₀))
54 apply Metric.tendsto_atTop.mpr; intro ε hε
55 obtain ⟨N₁, hN₁⟩ := Metric.tendsto_atTop.mp (hop x₀ f) (ε/2)
    ↪ (half_pos hε)
56 have hC1 : (0:ℝ) < C + 1 := by
57   linarith [le_trans (norm_nonneg (f.comp (A 0))) (hC 0)]
58 obtain ⟨N₂, hN₂⟩ := Metric.tendsto_atTop.mp hε (ε/(2*(C+1)))
59   (div_pos hε (mul_pos two_pos hC1))
60 exact ⟨max N₁ N₂, fun n hn => by
61   have d1 := hN₁ n (le_of_max_le_left hn)
62   have d2 := hN₂ n (le_of_max_le_right hn)
63   simp only [dist_eq_norm] at d1 d2 ⊢
64   calc ||f (A n (x n)) - f (A₀ x₀)||
65     = ||f (A n (x n - x₀)) + (f (A n x₀) - f (A₀ x₀))|| :=
        ↪ by
66     congr 1; simp [map_sub]
67     _ ≤ ||f (A n (x n - x₀))|| + ||f (A n x₀) - f (A₀ x₀)|| :=
        ↪ norm_add_le _ _
68     _ ≤ C * ||x n - x₀|| + ||f (A n x₀) - f (A₀ x₀)|| := by
69       have h1 := (f.comp (A n)).le_opNorm (x n - x₀)
70       simp only [ContinuousLinearMap.comp_apply] at h1
71       have h2 := mul_le_mul_of_nonneg_right (hC n)
        ↪ (norm_nonneg (x n - x₀))
72       linarith
73     _ ≤ C * (ε/(2*(C+1))) + ε/2 := by
74       have hC0 : (0:ℝ) ≤ C :=
75         le_trans (norm_nonneg (f.comp (A 0))) (hC 0)
76       linarith [mul_le_mul_of_nonneg_left d2.le hC0]
77     _ < ε := by
78       suffices C * (ε/(2*(C+1))) < ε/2 by linarith

```

```

79      calc C * (ε / (2 * (C + 1)))
80      = C * ε / (2 * (C + 1)) := by ring
81      _ < (C + 1) * ε / (2 * (C + 1)) := by
82      apply div_lt_div_of_pos_right _ (by linarith :
      ↪ 0 < 2 * (C + 1))
83      exact mul_lt_mul_of_pos_right (by linarith) hε
84      _ = ε / 2 := by field_simp

```

20. КОМПАКТНЫЕ МНОЖЕСТВА

КОНТРОЛЬНЫЕ ВОПРОСЫ

1. Дать определение компактного в себе множества, компакта.
2. Сформулировать теоремы Арцела, Рисса (критерии компактности множества в $C[a, b]$, $L^p(a, b)$, l^p).

ЗАДАЧИ

1. Пусть $\mathcal{M} = \{x(t)\}$ — ограниченное множество функций из $C[a, b]$. Доказать, что множество функций вида:

a) $y(s) = \int_a^b (s^2 + t^2 s) x(t) dt;$

b) $y(s) = \int_a^b (3st + s^2) x(t) dt;$

$$c) y(s) = \int_a^b s \cdot x(t) dt$$

КОМПАКТНЫ В $L^p(a, b)$.

Формализация на Lean 4

```

1  /-
2  Section 20, Task 1.
3  Integral transform sets compact in  $L^p$  (Kolmogorov-Riesz
4   $\hookrightarrow$  criterion).
5
6  The Kolmogorov-Riesz theorem: a subset  $S$  of  $L^p(\mathbb{R}^n)$  is
7   $\hookrightarrow$  totally bounded iff
8  (1)  $S$  is bounded, (2)  $S$  is equitight, (3)  $S$  is
9   $\hookrightarrow$   $L^p$ -equicontinuous
10 (translations converge uniformly in  $S$ ).
11
12 We formalize the abstract principle: totally bounded +
13  $\hookrightarrow$  complete = compact.
14 Applied to  $L^p$  (which is complete), total boundedness is the
15  $\hookrightarrow$  key criterion.
16 -/
17 import Mathlib.Topology.UniformSpace.Compact
18 import Mathlib.Tactic
19
20 /-- A totally bounded subset of a complete space has compact
21  $\hookrightarrow$  closure.
22 This is the abstract principle behind Kolmogorov-Riesz:
23 in  $L^p$  (complete), total boundedness  $\square$  compact closure. -/
24 theorem isCompact_closure_of_totallyBounded
25   {X : Type*} [UniformSpace X] [CompleteSpace X]
26   {S : Set X} (hS : TotallyBounded S) :
27     IsCompact (closure S) :=
28     isCompact_iff_totallyBounded_isComplete.mpr ⟨hS.closure,
29      $\hookrightarrow$  isClosed_closure.isComplete⟩

```

2. Пусть $\mathcal{M} = \{x(t)\}$ — ограниченное множество функций из $L^p(a, b)$. Компактны ли в пространстве $L^p(a, b)$ множества функций вида:

$$a) y(s) = \int_a^b (3s + s^2 t^2) x(t) dt;$$

$$b) y(t) = t^2 x(t);$$

$$c) y(t) = x(t) + 3;$$

$$d) y(s) = \int_a^b K(s, t) x(t) dt, \text{ где функция } K(s, t) \text{ непрерывна в квадрате } [a, b] \times [a, b]?$$

3. Пусть $\mathcal{M} = \{x(t)\}$ — ограниченное множество пространства $C[a, b]$. Доказать, что множества функций вида:

$$a) y(t) = \int_a^t x(\xi) d\xi;$$

b) $y(s) = \int_a^b K(s, t)x(t) dt$, где $K(s, t) \in C([a, b] \times [a, b])$

компактны в пространстве $C[a, b]$.

Формализация на Lean 4

```

1  /-
2    Section 20, Task 3.
3    Sets of integral transforms are compact in  $C[a, b]$  by
4     $\hookrightarrow$  Arzelà-Ascoli:
5    equicontinuous + pointwise compact  $\square$  compact in  $C(X, \alpha)$ .
6
7    Applied to integral transforms: the image of a bounded set
8     $\hookrightarrow$  under an
9    integral operator is equicontinuous (by uniform continuity
10    $\hookrightarrow$  of the kernel)
11   and uniformly bounded, hence has compact closure in  $C[a, b]$ .
12  -/
13  import Mathlib.Topology.UniformSpace.Ascoli
14  import Mathlib.Tactic
15
16  /-- Arzelà-Ascoli theorem: an equicontinuous set with
17    $\hookrightarrow$  pointwise compact
18   image is compact in  $C(X, \alpha)$ . This is the key tool for
19    $\hookrightarrow$  proving
20   integral transform sets are compact in  $C[a, b]$ . -/
21  theorem arzela_ascoli_compact
22    {X : Type*} [TopologicalSpace X] [CompactSpace X]
23    {α : Type*} [UniformSpace α] [T2Space α]
24    (S : Set C(X, α))
25    (hbw : IsCompact (ContinuousMap.toFun '' S))
26    (heq : Equicontinuous ((↑) : S → X → α)) :
27    IsCompact S :=
28    ArzelaAscoli.isCompact_of_equicontinuous S hbw heq

```

4. Будет ли ограниченное множество многочленов степени не выше n компактным в себе множеством в $C[a, b]$?

5. Доказать, что всякое множество, компактное в пространстве $C^{(1)}[a, b]$, является компактным и в пространстве $C[a, b]$.

Формализация на Lean 4

```

1  /-
2    Section 20, Task 05.
3    A compact set in  $C^1[a, b]$  is also compact in  $C[a, b]$ .
4    The inclusion  $\iota : C^1 \rightarrow C$  is continuous ( $\|f\|_\infty \leq \|f\|_{C^1}$ ),
5    and continuous images of compact sets are compact.
6
7    We formalize the abstract principle: continuous image of
       $\hookrightarrow$  compact is compact.
8  -/
9  import Mathlib.Topology.Compactness.Compact
10 import Mathlib.Tactic
11
12 /-- Continuous image of a compact set is compact (abstract
       $\hookrightarrow$  version of
13    "compact in  $C^1$  implies compact in  $C$ "). -/
14 theorem compact_in_C1_compact_in_C
15   {X Y : Type*} [TopologicalSpace X] [TopologicalSpace Y]
16   {f : X  $\rightarrow$  Y} (hf : Continuous f) {K : Set X} (hK :
       $\hookrightarrow$  IsCompact K) :
17   IsCompact (f '' K) :=
18   hK.image hf

```

6. Компактны ли следующие множества функций в пространстве $C[0, 1]$:

- a) $x_n(t) = t^n, n = 1, 2, \dots;$
- b) $x_n(t) = \sin nt, n = 1, 2, \dots;$
- c) $x_n(t) = \sin(n + t), n = 1, 2, \dots;$
- d) $x_\alpha(t) = \sin \alpha t, \alpha \in \mathbb{R}^1;$
- e) $x_\alpha(t) = \sin \alpha t, \alpha \in [1, 2];$
- f) $x_\alpha(t) = \arctg \alpha t, \alpha \in \mathbb{R}^1;$
- g) $x_\alpha(t) = e^{t-\alpha}, \alpha \geq 0?$

7. Доказать, что всякое компактное множество в пространстве

- a) $C[a, b];$

b) l^2

нигде не плотно.

Формализация на Lean 4

```
1  /-
2  Section 20, Task 7.
3  In an infinite-dimensional normed space, every compact set
4  ↪ is nowhere dense.
5
6  Proof: If interior  $K \neq \emptyset$ ,  $K$  contains a ball, hence a closed
7  ↪ ball,
8  which is compact (closed subset of compact). But a compact
9  ↪ closed ball
10 forces the space to be finite-dimensional (Riesz's lemma).
11 ↪ Contradiction.
12 -/
13
14 import Mathlib.Analysis.Normed.Module.FiniteDimension
15 import Mathlib.Tactic
16
17 variable {E : Type*} [NormedAddCommGroup E] [NormedSpace ℝ E]
18
19 /- In an infinite-dimensional normed space, every compact set
20 ↪ has empty interior. -/
21
22 theorem compact_nowhere_dense_of_infinite_dim
23   (hinf : ¬ FiniteDimensional ℝ E)
24   {K : Set E} (hK : IsCompact K) :
25   interior K = ∅ := by
26   by_contra h
27   rw [← Ne, ← Set.nonempty_iff_ne_empty] at h
28   obtain ⟨x₀, hx₀⟩ := h
29   rw [mem_interior_iff_mem_nhds, Metric.mem_nhds_iff] at hx₀
30   obtain ⟨r, hr, hball⟩ := hx₀
31   -- closedBall x₀ (r/2) ⊆ ball x₀ r ⊆ K, hence compact
32   have hcb_compact : IsCompact (Metric.closedBall x₀ (r / 2))
33     ↪ :=
34     hK.of_isClosed_subset Metric.isClosed_closedBall
35       ((Metric.closedBall_subset_ball (by linarith)).trans
36        ↪ hball)
37   exact hinf (FiniteDimensional.of_isCompact_closedBall (□
38     ↪ := ℝ) (by linarith) hcb_compact)
```

8. Доказать, что параллелепипед

$$\mathcal{P} = \{x = (\xi_1, \xi_2, \dots) \in l^2 : |\xi_k| \leq \frac{1}{k}, \forall k \geq 1\}$$

является компактным в себе множеством в пространстве l^2 .

Формализация на Lean 4

```

1  /-
2    Section 20, Task 8.
3    The parallelepiped  $P = \{x \in \ell^2 : |x_k| \leq 1/(k+1)\}$  is compact
4     $\hookrightarrow$  in  $\ell^2$ .
5  -/
6  import Mathlib.Analysis.InnerProductSpace.l2Space
7  import Mathlib.Analysis.Normed.Group.Tannery
8  import Mathlib.Analysis.PSeries
9  import Mathlib.Tactic
10
11 open scoped ENNReal
12 open Filter
13
14 private lemma mem_p_of_coord_le {x : ℕ → ℝ}
15   (hx : ∀ k : ℕ, |x k| ≤ 1 / ((k : ℝ) + 1)) : Mem_p x 2 :=
16   by
17     rw [mem_p_gen_iff (by norm_num : (0 : ℝ) < (2 :
18       ↪ ℝ≥0∞).toReal)]
19     simp only [ENNReal.toReal_ofNat]
20     exact ((Real.summable_nat_rpow_inv.mpr (by norm_num : (1:ℝ) <
21       ↪ 2)).comp_injective
22       Nat.succ_injective |>.congr fun k => by simp
23         ↪ [Nat.cast_succ, one_div, sq]).of_nonneg_of_le
24       (fun _ => Real.rpow_nonneg (norm_nonneg _) _) fun k =>
25         Real.rpow_le_rpow (norm_nonneg _) (hx k) (by norm_num)
26
27 private lemma summable_sq_bound : Summable (fun k : ℕ => (2 /
28   ↪ ((k : ℝ) + 1)) ^ 2) := by
29   simp_rw [div_pow]; apply Summable.mul_left
30   exact (Real.summable_nat_rpow_inv.mpr (by norm_num : (1:ℝ) <
31     ↪ 2)).comp_injective
32   Nat.succ_injective |>.congr fun k => by simp
33     ↪ [Nat.cast_succ, one_div, sq]
34
35 -- ||f||^2 = ∑' k, (f k)^2 via inner product on ℓ^2
36 private lemma lp_norm_sq_eq_tsum (f : lp (fun _ : ℕ => ℝ) 2) :
37   ||f|| ^ 2 = ∑' k, (f k) ^ 2 := by
38     rw [sq, ← real_inner_self_eq_norm_mul_norm,
39       ↪ lp.inner_eq_tsum]
40     congr 1; ext k; rw [real_inner_self_eq_norm_mul_norm,
41       ↪ Real.norm_eq_abs]
42     change |f k| * |f k| = (f k : ℝ) ^ 2; rw [← sq_abs]; ring

```

```

33
34 theorem parallelepiped_compact_in_l2 :
35   IsCompact {x : lp (fun _ : ℕ => ℝ) 2 |
36     ∀ k : ℕ, |x k| ≤ 1 / (k + 1 : ℝ)} := by
37   haveI : Fact (1 ≤ (2 : ℝ≥0∞)) := ⟨by norm_num⟩
38   rw [isCompact_iff_isSeqCompact]; intro x hx
39   obtain ⟨y, hyS, φ, hφ, hconv⟩ :=
40     (isCompact_pi_infinite (fun k : ℕ =>
41       ↪ isCompact_Icc)).tendsto_subseq
42     (fun n k => abs_le.mp (hx n k) : ∀ n, (↑(x n) : ℕ → ℝ) ∈
43       ↪ _)
44   have hybd : ∀ k : ℕ, |y k| ≤ 1 / ((k:ℝ)+1) := fun k =>
45     ↪ abs_le.mpr (hyS k)
46   set y' : lp (fun _ : ℕ => ℝ) 2 := ⟨y, mem_p_of_coord_le
47     ↪ hybd⟩
48   refine ⟨y', hybd, φ, hφ, ?_⟩
49   -- ||d_n||2 = ∑' k, (d_n k)2
50   have hnorm_sq : ∀ n, ||x (φ n) - y'||2 =
51     ∑' k, ((x (φ n) : ℕ → ℝ) k - y k)2 := fun n => by
52     have := lp_norm_sq_eq_tsum (x (φ n) - y')
53     simp only [lp.coeFn_sub, Pi.sub_apply, y'] at this; exact
54     ↪ this
55   -- Tannery: ∑' k, (d_n k)2 → 0
56   have hsq : Tendsto (fun n => ∑' k, ((x (φ n) : ℕ → ℝ) k - y
57     ↪ k)2) atTop (nhds 0) := by
58     have hpw : ∀ k, Tendsto (fun n => ((x (φ n) : ℕ → ℝ) k - y
59     ↪ k)2) atTop (nhds 0) := by
60       intro k
61       have hk := ((continuous_apply k).tendsto y).comp hconv
62       have hsub : Tendsto (fun n => (x (φ n) : ℕ → ℝ) k - y k)
63         ↪ atTop (nhds 0) := by
64         have := hk.sub (tendsto_const_nhds (x := y k)); rwa
65         ↪ [sub_self] at this
66       have := ((continuous_pow 2).tendsto 0).comp hsub
67       simp only [Function.comp_def, zero_pow_two_ne_zero] at
68         ↪ this; exact this
69   have hbd : ∀f n in atTop, ∀ k,
70     ||((x (φ n) : ℕ → ℝ) k - y k)2|| ≤ (2 / ((k : ℝ) +
71     ↪ 1))2 :=
72   Eventually.of_forall (fun n k => by
73     rw [Real.norm_eq_abs, abs_of_nonneg (sq_nonneg _), ←
74     ↪ sq_abs]
75     gcongr
76     have h3 := norm_sub_le ((x (φ n) : ℕ → ℝ) k) (y k)
77     simp only [Real.norm_eq_abs] at h3
78     have h1 := hx (φ n) k; have h2 := hybd k
79     have h4 : (1 : ℝ) / ((k:ℝ)+1) + 1 / ((k:ℝ)+1) = 2 /
80     ↪ ((k:ℝ)+1) := by ring
81     linarith
82   simp [tsum_zero] using
83     ↪ tendsto_tsum_of_dominated_convergence
84     ↪ summable_sq_bound hpw hbd

```

```

70 -- ||d_n||^2 → 0 ⇔ ||d_n|| → 0 ⇔ x(φ n) → y'
71 rw [← tendsto_sub_nhds_zero_iff,
    ↪ tendsto_zero_iff_norm_tendsto_zero,
    ↪ Metric.tendsto_atTop]
72 intro ε hε
73 obtain ⟨N, hN⟩ := Metric.tendsto_atTop.mp (show Tendsto (fun
    ↪ n => ||x (φ n) - y'|| ^ 2)
74 atTop (nhds 0) from by simp_rw [hnorm_sq]; exact hsq) (ε ^
    ↪ 2) (by positivity)
75 exact ⟨N, fun n hn => by
76 simp only [dist_zero_right, Real.norm_eq_abs,
    ↪ abs_of_nonneg (norm_nonneg _)]
77 have h : ||x (φ n) - y'|| ^ 2 < ε ^ 2 := by
78 have := hN n hn
79 rwa [dist_zero_right, Real.norm_of_nonneg (by
    ↪ positivity)] at this
80 simp only [Function.comp_def] at *
81 nlinarith [norm_nonneg (x (φ n) - y'), mul_self_nonneg (||x
    ↪ (φ n) - y'|| - ε)]

```

9. При каком условии на последовательность $\lambda_n \in \mathbb{R}^1$, $\lambda_n > 0$, являются компактным в себе множеством в пространстве l^2 :

- а) параллелепипед $\mathcal{P} = \{x = (\xi_1, \xi_2, \dots) \in l^2 : |\xi_k| \leq \lambda_k, \forall k \geq 1\}$;
- б) эллипсоид $\mathcal{E} = \{x = (\xi_1, \xi_2, \dots) \in l^2 : \sum_{n=1}^{\infty} \frac{|\xi_n|^2}{\lambda_n^2} \leq 1\}$?

10. Какие из следующих множеств компактны в пространстве $C[0, 1]$:

- а) $\{y(t) = \int_0^1 e^{ts} \varphi(s) ds : |\varphi(s)| \leq 1, \varphi \in C[0, 1]\}$;
- б) $\{x(t) \in C^{(2)}[0, 1] : |x(t)| \leq B_0; |x'(t)| \leq B_1; |x''(t)| \leq B_2\}$;
- в) $\{x(t) \in C^{(1)}[0, 1] : |x(t)| \leq B_0; |x'(t)| \leq B_1\}$;
- г) $\{x(t) \in C^{(2)}[0, 1] : |x(t)| \leq B_0; |x''(t)| \leq B_2\}$;
- д) $\{x(t) \in C^{(2)}[0, 1] : |x'(t)| \leq B_1; |x''(t)| \leq B_2\}$;
- е) $\{x(t) \in C[0, 1] : |x(t)| \leq B, |x(t_1) - x(t_2)| \leq L \cdot |t_1 - t_2|\}$;
- ж) $\{x(t) \in C^{(1)}[0, 1] : |x(t)| \leq B, |x'(t_1) - x'(t_2)| \leq L \cdot |t_1 - t_2|\}$.

11. Компактны ли следующие множества в пространстве $L^2(0, 1)$:

- а) $\{t^\alpha\}$, если $\alpha > -\frac{1}{2}$, если $\alpha > 0$;
- б) $\{\sin nt\}$, $n \geq 1$;
- в) $\{x(t) = \int_0^t \varphi(\xi) d\xi : \int_0^1 |\varphi(\xi)|^2 d\xi \leq 1, \varphi \in L^2(0, 1)\}$;
- г) $\{(\ln t)^n\}$, $n \geq 1$.

12. Пусть $\{c_n\}$ — числовая последовательность. Пусть, далее

$$\mathcal{M} = \{x = (\xi_1, \xi_2, \dots) \in l^2 : \sum_{k=1}^{\infty} |c_k \xi_k|^2 \leq 1\}$$

Для каких последовательностей множество $\mathcal{M}$ компактно?

21. КОМПАКТНЫЕ ОПЕРАТОРЫ

КОНТРОЛЬНЫЕ ВОПРОСЫ

1. Дать определение компактного оператора.
2. Перечислить свойства компактного оператора.
3. Сформулируйте критерии компактности множеств в пространствах $\mathbb{R}^n$, $C[a, b]$, $L^p(a, b)$, l^p .

ЗАДАЧИ

1. Доказать, что любой линейный оператор $A : \mathbb{R}^n \rightarrow \mathbb{R}^m$ является компактным.

Формализация на Lean 4

```

1  /-
2    Section 21, Task 1.
3    Any linear operator  $A : \mathbb{R}^n \rightarrow \mathbb{R}^m$  is compact.
4  -/
5  import Mathlib.Analysis.Normed.Module.FiniteDimension
6  import Mathlib.Analysis.Normed.Operator.Compact
7  import Mathlib.Tactic
8
9  /-- Any continuous linear map from a finite-dimensional real
10     ↪ space is compact. -/
11  theorem linear_map_finiteDim_isCompact
12     {E F : Type*} [NormedAddCommGroup E] [NormedSpace ℝ E]
13     ↪ [FiniteDimensional ℝ E]
14     [NormedAddCommGroup F] [NormedSpace ℝ F]
15     (A : E →L[ℝ] F) :
16     IsCompactOperator (A : E → F) := by
17     haveI : ProperSpace E := FiniteDimensional.proper_real E
18     -- closed ball is compact in a proper space
19     refine ⟨A '' Metric.closedBall 0 1,
20       (isCompact_closedBall 0 1).image A.continuous,
21       ?_⟩
22     --  $A^{-1}(A '' \text{closedBall } 0 \ 1) \supseteq \text{closedBall } 0 \ 1 \in \square \ 0$ 
23     exact Filter.mem_of_superset (Metric.closedBall_mem_nhds 0
24       ↪ one_pos)
25       (Set.subset_preimage_image A _)

```

2. Проверить компактность оператора $A : l^2 \rightarrow l^2$, если

$$Ax = A(\xi_1, \xi_2, \dots) = (0, \frac{\xi_1}{1}, \frac{\xi_2}{2}, \dots, \frac{\xi_n}{n}, \dots)$$

Формализация на Lean 4

```

1  /-
2    Section 21, Task 2.
3     $A(\xi_1, \xi_2, \dots) = (0, \xi_1/1, \xi_2/2, \dots, \xi_n/n, \dots)$  is compact on
4     $\ell^2$ .
5    This is a diagonal operator with entries  $a_k = 1/(k+1) \rightarrow 0$ .
6    By the diagonal compactness criterion (Task04), it is
7    compact.
8  -/
9  import FunAn.Section21_CompactOperators.Task04
10 import Mathlib.Analysis.SpecificLimits.Basic
11 import Mathlib.Tactic
12
13 open scoped ENNReal
14 open Filter
15
16 /-- The diagonal operator with entries  $1/(k+1)$  is compact on
17  $\ell^2$ . -/
18 theorem weighted_shift_compact :  $\exists A : \ell_p (\text{fun } _ : \mathbb{N} \Rightarrow \mathbb{R}) 2 \rightarrow$ 
19  $\ell_p (\text{fun } _ : \mathbb{N} \Rightarrow \mathbb{R}) 2,$ 
20   IsCompactOperator A  $\wedge$ 
21    $\forall (x : \ell_p (\text{fun } _ : \mathbb{N} \Rightarrow \mathbb{R}) 2) (k : \mathbb{N}),$ 
22      $(A x : \mathbb{N} \rightarrow \mathbb{R}) k = (1 / ((k : \mathbb{R}) + 1)) * (x : \mathbb{N} \rightarrow \mathbb{R}) k :=$ 
23     by
24       set a :  $\mathbb{N} \rightarrow \mathbb{R} := \text{fun } k \Rightarrow 1 / ((k : \mathbb{R}) + 1)$ 
25       have ha :  $\exists C, \forall k, |a k| \leq C := \langle 1, \text{fun } k \Rightarrow \text{by}$ 
26         simp only [a, one_div]
27         have hk :  $(0 : \mathbb{R}) < (\uparrow k + 1) := \text{by positivity}$ 
28         rw [abs_of_nonneg (inv_nonneg.mpr hk.le)]
29         exact inv_le_one_of_one_le₀ (by linarith)
30       refine ⟨_, (diagonal_operator_compact_iff_tendsto_zero a
31          $\rightarrow$  ha).mpr ?_, fun x k => rfl)
32       exact tendsto_one_div_add_atTop_nhds_zero_nat

```

3. Пусть H — гильбертово пространство и $A : H \rightarrow H$ — линейный оператор. Доказать, что A компактен тогда и только тогда, когда он любую слабо сходящуюся последовательность переводит в сильно сходящуюся.

Формализация на Lean 4

```

1  /-
2  Section 21, Task 03.
3  A compact operator maps bounded sequences to sequences with
4  convergent subsequences (sequential characterization,
   ↪ forward direction).
5
6  Proof: If A is compact and (xn) is bounded, then (Axn) lies
   ↪ in
7  A(closedBall 0 R) whose closure is compact, hence (Axn) has
   ↪ a
8  convergent subsequence by sequential compactness.
9  -/
10 import Mathlib.Analysis.Normed.Operator.Compact
11 import Mathlib.Topology.Sequences
12 import Mathlib.Tactic
13
14 open Filter
15
16 /-- A compact operator maps bounded sequences to sequences
   ↪ with
17 convergent subsequences. -/
18 theorem IsCompactOperator.exists_convergent_subseq
19   {E F : Type*} [NormedAddCommGroup E] [NormedSpace ℝ E]
20   [NormedAddCommGroup F] [NormedSpace ℝ F] [CompleteSpace F]
21   {A : E →L[ℝ] F} (hA : IsCompactOperator (A : E → F))
22   {x : ℕ → E} {R : ℝ} (hR : 0 < R) (hx : ∀ n, ‖x n‖ ≤ R) :
23   ∃ y : F, ∃ φ : ℕ → ℕ, StrictMono φ ∧
24     Tendsto (fun n => A (x (φ n))) atTop (nhds y) := by
25   -- A(closedBall 0 R) has compact closure
26   have hK := hA.isCompact_closure_image_of_bounded (S :=
   ↪ Metric.closedBall (0 : E) R)
27   Metric.isBounded_closedBall
28   -- x n ∈ closedBall 0 R
29   have hxball : ∀ n, x n ∈ Metric.closedBall (0 : E) R := by
30     intro n; rw [Metric.mem_closedBall, dist_zero_right];
   ↪ exact hx n
31   -- A(x n) ∈ closure(A '' closedBall 0 R)
32   have hAx : ∀ n, A (x n) ∈ closure (A '' Metric.closedBall 0
   ↪ R) :=
33     fun n => subset_closure (Set.mem_image_of_mem A (hxball
   ↪ n))
34   -- Sequential compactness from compactness
35   have hseq : IsSeqCompact (closure (A '' Metric.closedBall 0
   ↪ R)) :=

```

```
36   isCompact_iff_isSeqCompact.mp hK
37   obtain ⟨y, -, φ, hφ, hconv⟩ := hseq hAx
38   exact ⟨y, φ, hφ, hconv⟩
```

4. Пусть $A : l^2 \rightarrow l^2$, $A(\xi_1, \xi_2, \dots) = (a_1 \xi_1, a_2 \xi_2, \dots)$. Доказать, что оператор A компактен тогда и только тогда, когда $a_n \rightarrow 0$ при $n \rightarrow \infty$.

Формализация на Lean 4

```

1  /-
2    Section 21, Task 4.
3    Diagonal operator on  $\ell^2$  compact iff entries tend to 0.
4  -/
5  import Mathlib.Analysis.InnerProductSpace.l2Space
6  import Mathlib.Analysis.Normed.Operator.Compact
7  import Mathlib.Topology.Sequences
8  import Mathlib.Topology.Algebra.InfiniteSum.Order
9  import Mathlib.Tactic
10
11  open scoped ENNReal
12  open Filter Finset
13
14  set_option maxHeartbeats 400000
15
16  private lemma mem_p_mul {a :  $\mathbb{N} \rightarrow \mathbb{R}$ } (ha :  $\exists C, \forall k, |a k| \leq C$ )
17    → {x :  $\mathbb{N} \rightarrow \mathbb{R}$ }
18    (hx : Mem_p x 2) : Mem_p (fun k => a k * x k) 2 := by
19    obtain ⟨C, hC⟩ := ha
20    rw [mem_p_gen_iff (by norm_num : (0 :  $\mathbb{R}$ ) < (2 :  $\mathbb{R}_{\geq 0 \infty}$ ).toReal)]
21    → at hx ⊢
22    simp only [ENNReal.toReal_ofNat] at hx ⊢
23    have hC0 : (0 :  $\mathbb{R}$ ) ≤ C := le_trans (abs_nonneg (a 0)) (hC 0)
24    exact (hx.mul_left (C ^ (2 :  $\mathbb{R}$ ))).of_nonneg_of_le
25      (fun _ => Real.rpow_nonneg (norm_nonneg _) _) fun k => by
26        calc ||a k * x k|| ^ (2 :  $\mathbb{R}$ )
27          ≤ (C * ||x k||) ^ (2 :  $\mathbb{R}$ ) := by
28            apply Real.rpow_le_rpow (norm_nonneg _) _ (by
29              → norm_num)
30            rw [norm_mul]; exact mul_le_mul_of_nonneg_right
31              (by rw [Real.norm_eq_abs]; exact hC k)
32              → (norm_nonneg _)
33            _ = C ^ (2 :  $\mathbb{R}$ ) * ||x k|| ^ (2 :  $\mathbb{R}$ ) := Real.mul_rpow hC0
34            → (norm_nonneg _)
35
36  theorem diagonal_operator_compact_iff_tendsto_zero
37    (a :  $\mathbb{N} \rightarrow \mathbb{R}$ ) (ha :  $\exists C, \forall k, |a k| \leq C$ ) :
38    let A : lp (fun _ :  $\mathbb{N} \Rightarrow \mathbb{R}$ ) 2 → lp (fun _ :  $\mathbb{N} \Rightarrow \mathbb{R}$ ) 2 :=
39      fun x => ⟨fun k => a k * x k, mem_p_mul ha (lp.mem_p x)⟩
40    IsCompactOperator A ↔ Tendsto a atTop (nhds 0) := by
41    haveI : Fact (1 ≤ (2 :  $\mathbb{R}_{\geq 0 \infty}$ )) := ⟨by norm_num⟩
42    intro A
43    constructor
44    · -- Forward: compact  $\Rightarrow a \rightarrow 0$ 
45      intro ⟨K, hK, hK_nhds⟩

```

```

41 obtain ⟨r, hr, hr_sub⟩ := Metric.mem_nhds_iff.mp hK_nhds
42 rw [Metric.tendsto_atTop]; intro ε hε
43 by_contra h_inf; push_neg at h_inf
44 obtain ⟨φ, hφ, hφε⟩ := extraction_of_frequently_atTop
45   ↪ (frequently_atTop.mpr h_inf)
46 have hp2 : (0 : ℝ≥0) < 2 := by norm_num
47 have hr2 : (0 : ℝ) < r / 2 := by linarith
48 set s : ℕ → lp (fun _ : ℕ => ℝ) 2 := fun n => (r/2) •
49   ↪ lp.single 2 (φ n) (1 : ℝ)
50 have hAs_in_K : ∀ n, A (s n) ∈ K := fun n => hr_sub (by
51   rw [Metric.mem_ball, dist_zero_right]
52   simp only [s, norm_smul, lp.norm_single hp2, norm_one,
53     ↪ mul_one,
54     Real.norm_eq_abs, abs_of_pos hr2]; linarith)
55 have hAs_sep : ∀ m n, m ≠ n → r / 2 * ε ≤ dist (A (s m))
56   ↪ (A (s n)) := by
57   intro m n hmn
58   rw [dist_eq_norm]
59   have hne : φ m ≠ φ n := hφ.injective.ne hmn
60   calc r / 2 * ε
61     ≤ r / 2 * |a (φ m)| := by
62       have := hφε m; simp [dist_zero_right,
63         ↪ Real.norm_eq_abs] at this; nlinarith
64   _ = ||(A (s m) - A (s n)) (φ m)|| := by
65     simp only [A, s, lp.coeFn_sub, Pi.sub_apply,
66       ↪ lp.coeFn_smul, Pi.smul_apply,
67       lp.coeFn_single, Pi.single_apply, if_neg hne,
68       ↪ smul_eq_mul,
69       mul_one, mul_zero, sub_zero, abs_mul, abs_of_pos
70       ↪ hr2, mul_comm,
71       if_true, zero_mul, one_mul, Real.norm_eq_abs]
72   _ ≤ ||A (s m) - A (s n)|| := lp.norm_apply_le_norm
73     ↪ hp2.ne' _ _
74 obtain ⟨y, _, ψ, hψ, hconv⟩ := hK.isSeqCompact hAs_in_K
75 obtain ⟨N, hN⟩ := Metric.tendsto_atTop.mp hconv (r / 2 * ε
76   ↪ / 2) (by positivity)
77 have h1 := hN N le_rfl
78 have h2 := hN (N + 1) (Nat.le_succ N)
79 have h3 : dist (A (s (ψ N))) (A (s (ψ (N + 1)))) < r / 2 *
80   ↪ ε :=
81   calc dist _ _ ≤ dist _ y + dist y _ := dist_triangle _ y
82     ↪
83     _ < r / 2 * ε / 2 + r / 2 * ε / 2 := add_lt_add h1 (by
84       ↪ rwa [dist_comm])
85     _ = r / 2 * ε := by ring
86 linarith [hAs_sep (ψ N) (ψ (N + 1)) (hψ (by omega : N < N
87   ↪ + 1)).ne]
88 • -- Backward: a → 0 □ compact
89 intro ha_zero
90 refine ⟨closure (A '' Metric.ball 0 1),
91   isCompact_iff_totallyBounded_isComplete.mpr

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78     {?_, isClosed_closure.isComplete}, ?_)
79   · -- closure(A(B1)) totally bounded ← A(B1) totally
    ↪ bounded
80   apply TotallyBounded.closure
81   rw [Metric.totallyBounded_iff]; intro ε hε
82   obtain ⟨N, hN⟩ := Metric.tendsto_atTop.mp ha_zero (ε /
    ↪ 2) (by linarith)
83   -- Compact set K_N containing "truncations" of A(B1) to
    ↪ first N coords
84   -- K_N = image of compact cube under continuous map Φ
85   set Φ : (Fin N → ℝ) → lp (fun _ : ℕ => ℝ) 2 :=
86     fun c => ∑ i : Fin N, lp.single 2 (i : ℕ) (c i)
87   have hΦ_cont : Continuous Φ := continuous_finset_sum _
    ↪ fun i _ =>
88     (lp.singleContinuousLinearMap (□ := ℝ) (p := 2) (E :=
    ↪ fun _ => ℝ) (i : ℕ)).continuous.comp
89     (continuous_apply i)
90   set cube := Set.pi Set.univ fun i : Fin N => Set.Icc
    ↪ (-|a (i : ℕ)|) |a (i : ℕ)|
91   have hcube : IsCompact cube := isCompact_univ_pi fun _
    ↪ => isCompact_Icc
92   -- Get finite ε/2-net from compact image Φ(cube)
93   have hK := hcube.image hΦ_cont
94   obtain ⟨F, hF_fin, hF_cover⟩ :=
    ↪ Metric.totallyBounded_iff.mp
95     hK.totallyBounded (ε / 2) (by linarith)
96   refine ⟨F, hF_fin, fun y hy => ?_⟩
97   obtain ⟨x, hx_ball, rfl⟩ := hy
98   -- Truncation of A(x) to first N coords is in Φ(cube)
99   set c : Fin N → ℝ := fun i => a (i : ℕ) * x (i : ℕ)
100  have hx_lt : ||x|| < 1 := by rwa [Metric.mem_ball,
    ↪ dist_zero_right] at hx_ball
101  have hc_cube : c ∈ cube := fun i _ => by
102    simp only [c, Set.mem_Icc]
103    have h1 : |x (i : ℕ)| ≤ 1 := by
104      have := lp.norm_apply_le_norm (show (2:ℝ≥0∞) ≠ 0 by
    ↪ norm_num) x (i : ℕ)
105      rw [Real.norm_eq_abs] at this; linarith
106  have h2 : |a (i : ℕ) * (x : ℕ → ℝ) (i : ℕ)| ≤ |a (i :
    ↪ ℕ)| := by
107    rw [abs_mul]; exact mul_le_of_le_one_right
    ↪ (abs_nonneg _) h1
108    exact ⟨by linarith [neg_abs_le (a (↑i) * (↑x : ℕ → ℝ)
    ↪ ↑i)], by linarith [le_abs_self (a (↑i) * (↑x : ℕ →
    ↪ ℝ) ↑i)]⟩
109  have hΦc_in : Φ c ∈ Φ '' cube := Set.mem_image_of_mem _
    ↪ hc_cube
110  -- ∃ f ∈ F with dist(Φ(c), f) < ε/2
111  obtain ⟨f, hf, hf_dist⟩ := Set.mem_iUnion₂.mp (hF_cover
    ↪ hΦc_in)
112  -- dist(A(x), f) ≤ dist(A(x), Φ(c)) + dist(Φ(c), f) <
    ↪ ε/2 + ε/2

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113 refine Set.mem_iUnion₂.mpr (f, hf, ?_)
114 calc dist (A x) f ≤ dist (A x) (Φ c) + dist (Φ c) f :=
    ↪ dist_triangle
115   _ < ε / 2 + ε / 2 := add_lt_add ?_ (Metric.mem_ball.mp
    ↪ hf_dist)
116   _ = ε := by ring
117 -- ||A(x) - Φ(c)|| < ε/2: tail bound via inner product
118 -- (A x - Φ c)(k) = 0 for k < N, a(k)x(k) for k ≥ N
119 -- Each |a(k)x(k)|² < (ε/2)² |x(k)|², sum ≤ (ε/2)² ||x||² <
    ↪ (ε/2)²
120 set d := A x - Φ c
121 -- Pointwise: (d k)² ≤ (ε/2)² * (x k)² for all k
122 have hdk_le : ∀ k, ((d : ℕ → ℝ) k) ^ 2 ≤ (ε / 2) ^ 2 *
    ↪ ((x : ℕ → ℝ) k) ^ 2 := by
123   intro k
124   simp only [d, A, Φ, c, lp.coeFn_sub, Pi.sub_apply,
125     lp.coeFn_sum, Finset.sum_apply, lp.coeFn_single,
    ↪ Pi.single_apply]
126   by_cases hk : k < N
127   · rw [Finset.sum_eq_single (k, hk)
128     (fun i _ hi => by simp [show k ≠ (i : ℕ) from
    ↪ Fin.val_ne_of_ne (Ne.symm hi)])]
129     (by simp)]; simp; positivity
130   · push_neg at hk
131     rw [Finset.sum_eq_zero (fun i _ => by simp [show k ≠
    ↪ (i : ℕ) from by omega])]
132     simp only [sub_zero, mul_pow]
133     exact mul_le_mul_of_nonneg_right (by
134       have := hN k hk; rw [dist_zero_right,
    ↪ Real.norm_eq_abs] at this
135       have := sq_abs (a k); have := abs_nonneg (a k);
    ↪ nlinarith [mul_self_nonneg (|a k|)]
    ↪ (sq_nonneg _))
136 -- ||d||² ≤ (ε/2)² ||x||² < (ε/2)² (via inner product tsum)
137 have hd_sq : ||d|| ^ 2 = ∑' k, ((d : ℕ → ℝ) k) ^ 2 := by
138   rw [sq, ← real_inner_self_eq_norm_mul_norm,
    ↪ lp.inner_eq_tsum]
139   congr 1; ext k; rw [real_inner_self_eq_norm_mul_norm,
    ↪ Real.norm_eq_abs]
140   change |(d : lp _ 2) k| * |(d : lp _ 2) k| = ((d : ℕ →
    ↪ ℝ) k) ^ 2; rw [← sq_abs]; ring
141 have hx_sq : ||x|| ^ 2 = ∑' k, ((x : ℕ → ℝ) k) ^ 2 := by
142   rw [sq, ← real_inner_self_eq_norm_mul_norm,
    ↪ lp.inner_eq_tsum]
143   congr 1; ext k; rw [real_inner_self_eq_norm_mul_norm,
    ↪ Real.norm_eq_abs]
144   change |(x : lp _ 2) k| * |(x : lp _ 2) k| = ((x : ℕ →
    ↪ ℝ) k) ^ 2; rw [← sq_abs]; ring
145   rw [dist_eq_norm]
146 -- Summability for tsum_le_tsum
147 have hsx : Summable (fun k => ((x : ℕ → ℝ) k) ^ 2) :=

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148      ((lp.mem p x).summable (by norm_num : 0 <
149      ↪ (2:ℝ≥0∞).toReal)).congr fun k => by
      simp [ENNReal.toReal_ofNat, Real.norm_eq_abs,
      ↪ sq_abs]
150    have hsd : Summable (fun k => ((d : ℕ → ℝ) k) ^ 2) :=
151      (hsx.mul_left ((ε/2)^2)).of_nonneg_of_le (fun _ =>
      ↪ sq_nonneg _) hdk_le
152    have hle := hsd.tsum_le_tsum hdk_le (hsx.mul_left _)
153    rw [tsum_mul_left, ← hx_sq] at hle
154    have h1 : ||d|| ^ 2 ≤ (ε / 2) ^ 2 * ||x|| ^ 2 := by linarith
      ↪ [hd_sq]
155    -- ||d||^2 ≤ (ε/2)^2 * ||x||^2 < (ε/2)^2 and ε/2 ≤ ||d|| → (ε/2)^2 ≤
      ↪ ||d||^2. Contradiction.
156    have h2 : (ε / 2) ^ 2 * ||x|| ^ 2 < (ε / 2) ^ 2 := by
157      apply mul_lt_of_lt_one_right (by positivity)
158      nlinarith [norm_nonneg x, mul_self_nonneg ||x||]
159      nlinarith [norm_nonneg d, sq_nonneg ||d||]
160    · exact Filter.mem_of_superset (Metric.ball_mem_nhds 0
      ↪ one_pos) fun x hx =>
      subset_closure (Set.mem_image_of_mem A hx)
161

```

5. Доказать, что оператор $A : C[a, b] \rightarrow C[a, b]$, $(Ax)(t) = x_0(t)x(t)$, где $0 \neq x_0 \in C[a, b]$, не является компактным.

6. Доказать компактность следующих операторов:

- a) $A : C[0, 1] \rightarrow C[0, 1]$, $(Ax)(t) = \int_0^t x_0(t)x(s) ds$, где $x_0(t) \in C[a, b]$;
- b) $A : C[a, b] \rightarrow C[a, b]$, $(Ax)(t) = \int_a^b K(s, t)x(s) ds$, где $K(s, t)$ — непрерывная на $[a, b] \times [a, b]$ функция.

7. Доказать компактность следующих операторов:

- a) $A : L^2(0, 1) \rightarrow L^2(0, 1)$, $(Ax)(t) = \int_0^t x(s) ds$;
- b) $A : L^2(0, 1) \rightarrow L^2(0, 1)$, $(Ax)(t) = x_0(t) \cdot \int_0^t x(s) ds$, где $x_0(t) \in C[0, 1]$;
- c) $A : L^2(a, b) \rightarrow L^2(a, b)$, $(Ax)(t) = \int_a^b K(s, t)x(s) ds$, где $K(s, t)$ — непрерывная на $[a, b] \times [a, b]$ функция;
- d) $A : L^2(a, b) \rightarrow L^2(a, b)$, $(Ax)(t) = \int_a^b K(s, t)x(s) ds$, где $K(s, t) \in L^2([a, b] \times [a, b])$.

8. Показать, что оператор $A : l^2 \rightarrow \mathbb{R}^1$, $Ax = \sum_{k=1}^{\infty} \frac{\xi_k}{2^k}$ является компактным.

9. Показать, что образ компактного оператора является сепарабельным.

Формализация на Lean 4

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1  /-
2  Section 21, Task 9.
3  Show that the image of a compact operator is separable.
4
5  Proof:  $\text{range } A = \bigcup n, A \text{ '' } \text{closedBall } 0 \ n$ . Each  $A \text{ '' } \text{closedBall } 0 \ n$ 
6   $\hookrightarrow \text{closedBall } 0 \ n$ 
7  has compact closure (since  $A$  is compact), hence is
8   $\hookrightarrow$  separable.
9  Countable union of separable sets is separable.
10 -/
11 import Mathlib.Topology.MetricSpace.Pseudo.Basic
12 import Mathlib.Analysis.Normed.Operator.Compact
13 import Mathlib.Tactic
14
15 variable {X Y : Type*} [NormedAddCommGroup X] [NormedSpace ℝ
16    $\hookrightarrow$  X]
17   [NormedAddCommGroup Y] [NormedSpace ℝ Y]
18
19 /- The range of a compact operator is separable. -/
20 theorem compact_operator_range_separable
21   (A : X  $\rightarrow$  L[ℝ] Y) (hA : IsCompactOperator (A : X  $\rightarrow$  Y)) :
22   TopologicalSpace.IsSeparable (Set.range A) := by
23   --  $\text{range } A = \bigcup n : \mathbb{N}, A \text{ '' } \text{closedBall } 0 \ n$ 
24   have hrange : Set.range A =  $\bigcup n : \mathbb{N}, A \text{ '' } \text{Metric.closedBall}$ 
25      $\hookrightarrow 0 \ (n : \mathbb{R})$  := by
26     ext y; simp only [Set.mem_range, Set.mem_iUnion,
27        $\hookrightarrow$  Set.mem_image]
28     constructor
29     · rintro ⟨x, rfl⟩
30       exact ⟨[||x||]_, x, Metric.mem_closedBall.mpr (by simp
31          $\hookrightarrow$  [dist_zero_right]; exact Nat.le_ceil _), rfl⟩
32     · rintro ⟨n, x, _, rfl⟩; exact ⟨x, rfl⟩
33   rw [hrange]
34   rw [TopologicalSpace.isSeparable_iUnion]
35   intro n
36   --  $A \text{ '' } \text{closedBall } 0 \ n$  is bounded, its closure is compact
37   have hbdd : Bornology.IsBounded (Metric.closedBall (0 : X)
38      $\hookrightarrow$  n) :=
39     Metric.isBounded_closedBall
40   exact ((hA.isCompact_closure_image_of_bounded
41      $\hookrightarrow$  hbdd).isSeparable).mono
42     subset_closure

```

10. Являются ли компактными следующие операторы в пространствах $C[0, 1]$ и $L^2(0, 1)$?

- a) $(Ax)(t) = \int_0^1 x(s) ds;$
- b) $(Ax)(t) = \int_0^1 x(s)(t-s)^{-2/5} ds;$
- c) $(Ax)(t) = \int_0^1 (ts + t^2 s^2)x(s) ds.$

11. Доказать, что оператор дифференцирования $\frac{d}{dt}$ является компактным из $C^{(2)}[0, 1]$ в $C[0, 1]$.
12. При каких $\alpha, \beta, \gamma > 0$ является компактным в пространстве $C[0, 1]$ оператор $(Ax)(t) = \int_0^1 t^\alpha s^\beta x(s^\gamma) ds?$
13. При каких α, β, γ является компактным в пространстве $L^2(0, 1)$ оператор $(Ax)(t) = \int_0^1 t^\alpha s^\beta x(s^\gamma) ds?$